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REPORT
CAPSTONE PROJECT (CO4337)

BK–MIND: A MULTIMODAL DATA UNDERSTANDING SYSTEM FOR HCMUT LECTURES

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Declaration of Authorship

We hereby declare that the entire content of this report is the result of research, analysis, evaluation, and implementation conducted by ourselves under the guidance of **Dr. Nguyen An Khuong** and **Mr. Nguyen Trang Sy Lam**. The results achieved in this report are authentic and have never been published in any form before. All content presented in this report has been collected, selected, analyzed, and evaluated with rigor throughout the research process. All materials, data, and content referenced from other sources have been properly cited and listed in accordance with regulations in the bibliography.

During the implementation of this report, we have used certain modern support tools, including analysis software, simulation tools, and artificial intelligence (AI) tools, solely for technical support purposes. Specifically, Large Language Model (LLM)-based tools were employed to assist with language refinement, summarization of complex concepts, and drafting support to improve clarity and presentation quality. These tools served as assistive technologies for writing refinement and did not generate the core academic content, scientific reasoning, or technical implementations.

All substantive academic content, scientific reasoning, model design, implementation architecture, result analysis, and conclusions have been directly conducted and authored by ourselves. The use of AI tools in no way substitutes for or replaces the intellectual work, critical thinking, and original contributions that form the foundation of this research. We affirm full responsibility for the accuracy and integrity of all academic claims and technical solutions presented. This disclosure is made in full compliance with contemporary standards for responsible AI usage in academic work and to maintain the highest standards of academic integrity.

While each team member specialized in different aspects of the project, the team ensured continuous knowledge sharing and collaboration throughout the research process. All members actively participated in discussions, code reviews, and cross-domain learning to maintain a comprehensive understanding of the entire system. Details about each member's contribution is mentioned in the appendix.

We take full responsibility for this statement of commitment as well as the entire content of this report. If any academic misconduct or improper copying is detected, we accept all disciplinary actions according to the regulations of the Faculty Board and the University Board of Ho Chi Minh City University of Technology - Vietnam National University.

Ho Chi Minh City, May 2026

Abstract

Educational materials at Ho Chi Minh City University of Technology (HCMUT), including lecture videos, audio recordings, slides, and documents—are expanding rapidly, yet conventional text-based tools fail to capture their multimodal nature. These methods often rely on transcripts or extracted text, resulting in recognition errors, hallucinations, and loss of visual and auditory context.

This project proposes a web-based application powered by a Multimodal Retrieval-Augmented Generation (MRAG) pipeline to assist students in studying and exam preparation. The system integrates textual transcripts, Optical Character Recognition (OCR)-based slide and document extraction, visual information for diagrams and equations, and audio features that capture emphasis and speaker prosody. Retrieval combines sparse and dense methods with cross-modal verification, and retrieved evidence is grounded in a large language model (LLM) for accurate and explainable responses.

The prototype includes a backend built with FastAPI that interfaces with a vector database for multimodal vector storage and indexing, and a frontend built with React for responsive search and summarization. Key functionalities include multimodal content retrieval, automatic lecture summarization, and personalized study roadmap generation. The system aims to enhance learning efficiency, reduce information loss, and provide a robust, adaptive educational assistant tailored to the HCMUT academic environment.

Chapter 1

Introduction

1.1 Motivations

Students learning on their own often struggle to find and organize course materials. These materials—videos, notes, slides, PDFs—are scattered across different websites and platforms, and gathering them all in one place takes too much time and effort. Additionally, students have to manually connect information from different sources—like watching a video lecture and then finding related notes. This requires a lot of mental effort and makes learning harder [1, 2].

This information management challenge exists at institutional levels as well. At HCMUT, the volume of educational content continues to expand across multiple formats including lecture videos, audio recordings, presentation slides, and text documents. Traditional text-based processing methods for this content present limitations. These methods typically rely on transcript generation, which introduces transcription errors and removes important contextual information present in visual components such as diagrams and mathematical notation, as well as auditory elements including speaker emphasis and intonation patterns.

Recent developments in Multimodal Large Language Models (LLMs)—AI systems that can understand text, images, and audio—have improved dramatically. Tools like Google's Gemini and NotebookLM demonstrate these LLMs can process and reason across different types of information [1]. These advances suggest the feasibility of developing educational systems that can process lecture materials across multiple modalities without information loss. We investigated using an AI retrieval system (called RAG—Retrieval-Augmented Generation, which we explain in Chapter 3) that can search through videos, images, and text together. The proposed system aims to address both institutional content management needs at HCMUT and the broader requirements of self-directed multimodal learning.

[1] https://blog.google/technology/ai/notebooklm-google-ai/

1.2 Objectives

The primary objective of this project is to design and implement a web-based system that enables students to search, summarize, and study lecture materials through a MRAG pipeline. The proposed system aims to:

- Integrate textual, visual, and audio information from lecture videos, slides, and documents into a unified retrieval system.
- Convert slides and lecture videos into searchable text (like transcribing a video to find when the professor said a specific topic).
- Employ a hybrid retrieval mechanism that combines sparse and dense embeddings with cross-modal verification for reliable and accurate information retrieval.
- Use an AI to confirm that answers are supported by the documents it found.
- Provide personalized study tools, including lecture summarization and study roadmap generation, to improve review efficiency and exam readiness.

Through these objectives, the system aims to enhance accessibility and understanding of complex lecture content, serving as an intelligent assistant that bridges multimodal information with adaptive learning.

1.3 Scope

This project focuses on the development of a prototype system that processes lecture-related multimodal data and supports retrieval, summarization, and study roadmap generation. The scope includes:

- Data ingestion from multiple sources such as videos, slides, and documents.
- Extraction of textual and visual information through OCR and ASR techniques.
- Build a search system that finds relevant materials using different search methods that work together.
- Deployment of a RAG-based reasoning layer powered by an LLM.
- Design of a frontend web interface that allows students to search for relevant content and receive summaries or study guidance.

The project does not aim to develop new models from scratch but rather focuses on integrating existing multimodal models and retrieval techniques into a cohesive and practical educational application.

1.4 Challenges and Solutions

The development of a multimodal RAG system for educational content presents both theoretical and technical challenges. This section outlines the key challenges and briefly discusses how existing research and our approach address them. Detailed discussions of related work and our methodologies are presented in Chapters 3 and 4, respectively.

Theoretical Challenges

The theoretical challenges stem from the need to understand and integrate multiple concepts:

- **RAG Adaptation to Multimodal Contexts:** Traditional RAG systems focus on text-only retrieval, but educational content spans text, images, audio, and video. Research has explored multimodal retrieval through models like M2HF and systems like NotebookLM. Our approach employs a dual-stream indexing architecture that processes text and visual content through separate pathways, enabling parallel retrieval across modalities.
- **Retrieval Techniques and Reranking:** Selecting appropriate retrieval methods requires balancing recall, precision, and efficiency. Related work demonstrates that hybrid sparse-dense retrieval outperforms single methods [3], while cross-attentional rerankers capture subtle distinctions. We evaluate multiple retrieval strategies and employ hybrid retrieval combined with reranking for optimal performance.
- **Vision-Language Models:** Extracting semantic meaning from visual content requires sophisticated vision-language alignment. Models like PolyViLT and ColPali extend retrieval paradigms to visual domains. We integrate late-interaction visual models that preserve spatial and semantic nuances, enabling retrieval based on visual content even without descriptive text.
- **Challenges and Solutions** – To build this system, we had to solve several technical problems. Understanding these problems helps explain how we designed our solution.
- **Modality Fusion:** Effectively combining information from different modalities requires choosing appropriate fusion strategies. Research explores early fusion, late fusion, and hybrid approaches. Our architecture implements a hybrid strategy that maintains separate indices during indexing and performs parallel retrieval with result aggregation during query execution.

Technical Challenges

The technical challenges involve implementing and optimizing these concepts into a robust system:

- **Integration of Heterogeneous Modules:** Coordinating diverse processing modules (OCR, ASR, visual embedding, text embedding) requires careful orchestration. We design a unified pipeline that handles different input formats and produces standardized artifacts for consistent downstream processing.
- **Data Synchronization:** Maintaining consistency between text chunks, visual pages, and metadata is essential for accurate citation. We implement comprehensive metadata schemas that enable precise provenance tracking and citation generation.
- **Technology Selection:** Selecting suitable models and frameworks requires balancing performance, cost, and computational constraints. Through systematic experimentation, we evaluate multiple configurations to determine optimal choices for our specific use case.
- **Efficient Cross-Modal Retrieval:** Managing parallel retrieval while minimizing latency requires efficient indexing structures. We implement parallel retrieval with optimized indexing and just-in-time processing to balance efficiency and accuracy.
- **System Deployment:** Deploying with considerations for scalability, privacy, and institutional requirements presents operational challenges. We adopt a local-first architecture that ensures data privacy while supporting modular scaling and containerized deployment.

Addressing these challenges through careful integration of established techniques and systematic experimentation enables the development of a functional MRAG system that supports real-world educational use cases. The detailed methodologies, experimental results, and architectural decisions are presented in subsequent chapters.

1.5 Structure of the Report

The remainder of this report is organized as follows:

- **Chapter 1: Introduction** – Provides background, motivations, objectives, scope, challenges and solutions, and the structure of the report.
- **Chapter 2: Preliminaries** – Presents foundational knowledge, key theories, and tools relevant to multimodal retrieval and LLMs.
- **Chapter 3: Related Work** – Reviews existing literature and prior systems related to MRAG, OCR, ASR, and visual-language integration.
- **Chapter 4: Proposed Solution** – Describes the proposed MRAG architecture, system design, and methodologies for integrating multimodal retrieval with generation.
- **Chapter 5: Implementation** – Details the technical setup, multi-tenant cloud-native architecture, database schemas, and deployment strategy.

- **Chapter 6: System Validation: From Components to Scale** – Defines the metrics, datasets, procedures, and results for validating system performance across multimodal retrieval, summarization, and learning outcomes at both component and full-system scales.
- **Chapter 7: Conclusion** – Summarizes achievements, key learnings from development, and outlines long-term development strategies and future enhancements.
- **Chapter 8: Appendices** – Provides supplementary materials including system interfaces, database schemas, API specifications, and team contributions.

Chapter 2

Preliminaries

2.1 Terminology and Conventions

To ensure clarity and consistency throughout this report, the following standardized terminology and capitalization conventions are utilized:

Table 2.1: Standardized Technical Terminology.

Term	Capitalization	Description / Usage
Vector Database	Capitalized	The high-dimensional similarity search engine (e.g., Qdrant) used for dense retrieval.
BM25	All Caps	The probabilistic lexical ranking function used for keyword-based sparse retrieval.
RAG	All Caps	Retrieval-Augmented Generation; the core system architecture.
MRAG	All Caps	Multimodal Retrieval-Augmented Generation; the specialized extension of RAG for handling non-textual data.
LLM	All Caps	Large Language Model (e.g., GPT-4o-mini, Claude 3.5 Sonnet).
Multimodal	Sentence Case	Systems or processes involving multiple data modalities (text, image, audio, video).
ASR	All Caps	Automatic Speech Recognition; the technology used to transcribe audio/video lecture content.
OCR	All Caps	Optical Character Recognition; the process of extracting text from rasterized document images.

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Standardized Technical Terminology (continued)

Term	Capitalization	Description / Usage
VLM	All Caps	Vision-Language Model; models that process both visual and textual inputs (e.g., ColQwen, SmolVLM).
Docling	Title Case	The specialized document parsing library used for structural extraction.
ColQwen	Title Case	The vision-language model architecture used for visual document retrieval.
Hybrid Search	Capitalized	The combination of sparse (BM25) and dense (Vector) retrieval pathways.
RRF / DBSF	All Caps	Reciprocal Rank Fusion and Distribution-Based Score Fusion (fusion algorithms).
FAISS	All Caps	Facebook AI Similarity Search; a library for fast similarity search and clustering of embeddings.
HNSW	All Caps	Hierarchical Navigable Small World; an efficient similarity search algorithm used in vector indices.
nDCG	Notation	Normalized Discounted Cumulative Gain; an information retrieval metric for evaluating ranking quality.
MaxSim	Notation	Maximum Similarity; the token-level scoring mechanism used in late-interaction models like ColBERT.
ColBERT	Title Case	Contextualized Late Interaction over BERT; a dense retrieval architecture using token-level interactions.
AWS	All Caps	Amazon Web Services; the cloud platform used for deployment and infrastructure.
S3	All Caps	Amazon Simple Storage Service; object storage service for document artifacts and processing outputs.
ECS	All Caps	Amazon Elastic Container Service; container orchestration service for running the BK-MInD application.
SageMaker	Title Case	AWS SageMaker; machine learning service used for model hosting and inference scaling.
WAF	All Caps	Web Application Firewall; AWS WAF provides protection against application-level attacks.

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Standardized Technical Terminology (continued)

Term	Capitalization	Description / Usage
JWT	All Caps	JSON Web Token; cryptographic token used for authenticating API requests.
TLS	All Caps	Transport Layer Security; encryption protocol for securing data in transit.
SSE	All Caps	Server-Side Encryption; encryption method for protecting data at rest in Amazon S3.
API	All Caps	Application Programming Interface; the interface through which frontend and backend systems communicate.
REST	All Caps	Representational State Transfer; architectural style for the backend API design.
ASGI	All Caps	Asynchronous Server Gateway Interface; the asynchronous interface for Python web application servers (e.g., Uvicorn).
SRT	All Caps	SubRip Text; subtitle file format used for storing transcription outputs.
VTT	All Caps	Web Video Text Tracks; subtitle file format used for web-based video player integration.
DPI	All Caps	Dots Per Inch; resolution parameter used when rasterizing PDF pages to images.
CI/CD	All Caps	Continuous Integration / Continuous Deployment; automated pipeline for testing and deployment.

2.2 Automatic Speech Recognition

Automatic Speech Recognition (ASR) converts spoken language into text. It has evolved from statistical pipelines to end-to-end neural models and is widely used in transcription, captioning, accessibility, and voice interfaces [4, 5]. Recent work emphasizes robustness, low-resource languages, and large-scale pretraining [6, 7]. A typical ASR pipeline is illustrated in Figure 2.1[1].

[1] https://www.researchgate.net/figure/ASR-application-pipeline_fig1_355070202

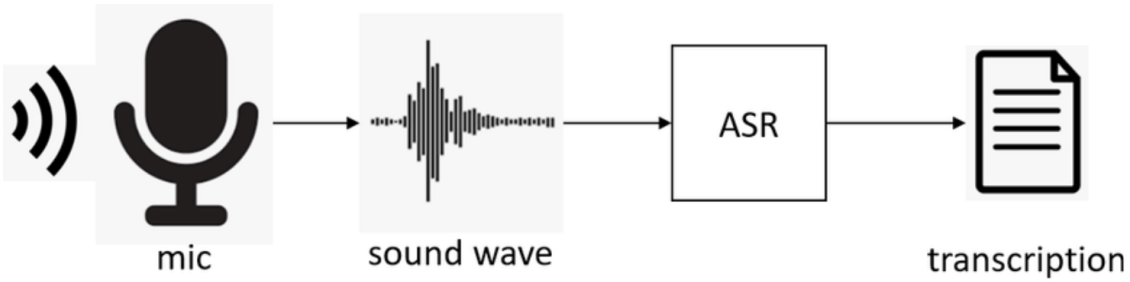

Figure 2.1: ASR pipeline.

ASR is the task of converting an acoustic signal into a sequence of words. In simple terms, the system tries to find the sequence of words (W) that is most likely given the audio (X). It looks for: "What words would most likely produce this audio sound?" Formally, the goal of an ASR system is to determine the most probable word sequence $\hat{W}$ given an input acoustic observation X

$$\hat{W} = \arg\max_{W} P(W|X).$$

Using Bayes' theorem, this can be rewritten as

$$\hat{W} = \arg\max_{W} P(X|W)P(W)$$

where

- $P(X|W)$ is the **acoustic model**, representing the probability of observing the audio features given a word sequence;
- $P(W)$ is the **language model**, representing the prior probability of a word sequence in the target language;
- $\hat{W}$ is the final predicted transcription.

The decoder combines the acoustic and language models to search for the optimal word sequence that maximizes this probability.

Early ASR systems used Gaussian Mixture Models with Hidden Markov Models (GMM–HMMs) for acoustic modeling and temporal alignment [8, 9]. Later DNN–HMM hybrids replaced GMM acoustic models with neural networks and became common in production systems, supported by toolkits such as Kaldi [10] and benchmarks such as Switchboard and WSJ [11, 12].

End-to-end ASR directly maps audio to text with a single neural model. The main paradigms are CTC, which removes the need for frame-level alignments [4]; attention-based encoder–decoder models such as LAS, which model output dependencies through attention [5]; and transducer models such as RNN-T, which support streaming and low-latency inference [13, 14].

End-to-End ASR using Connectionist Temporal Classification (CTC) is a technique that helps the model learn to match audio to text without being told exactly where each letter occurs in the audio. It's like finding the transcript without needing word-by-word timestamps. CTC models the probability of a label sequence Y by summing over all possible alignments π that collapse to Y

$$P(Y|X) = \sum_{\pi \in \mathcal{B}^{-1}(Y)} P(\pi|X)$$

where $\mathcal{B}$ is the mapping that removes blanks and repeated labels from the alignment path.

Transformers further improved ASR through global self-attention. Conformer augments Transformers with convolutional modules to capture local acoustic patterns while preserving global context, achieving strong results on benchmarks such as LibriSpeech [15, 16].

In attention-based ASR models, the system generates the output transcription autoregressively, producing one token at a time conditioned on previous outputs and the encoded acoustic features. The probability of generating a transcription sequence Y given the input speech features X is formulated as

$$P(Y|X) = \prod_{t=1}^{T} P(y_t \mid y_{1:t-1}, X)$$

where

- X is the input acoustic feature sequence extracted from the speech signal;
- $Y = (y_1, y_2, \ldots, y_T)$ is the output sequence of tokens (e.g., characters, subwords, or words);
- y_t is the token predicted at timestep t;
- $y_{1:t-1}$ denotes all previously generated tokens before timestep t;
- T is the length of the output sequence;
- $P(y_t \mid y_{1:t-1}, X)$ is the conditional probability of generating token y_t given the input features and prior predictions.

The decoder uses an attention mechanism to dynamically focus on relevant parts of the encoded speech representation at each timestep, enabling the model to learn soft alignments between acoustic frames and output tokens without explicit alignment labels.

Self-supervised pretraining is another major direction. Models such as wav2vec 2.0 learn from unlabeled audio and can be fine-tuned with limited labels, while Whisper shows that scaling data and model size improves robustness and multilingual transfer.

ASR evaluation commonly uses LibriSpeech [16], Switchboard [11], TED-LIUM [17], and Common Voice [18]. The main metrics are Word Error Rate (WER) and Character Error Rate (CER).

These metrics measure the difference between the reference transcription and the hypothesis transcription produced by the system. Word Error Rate (WER) is computed as

$$\text{WER} = \frac{S + D + I}{N}$$

where

- S is the number of word substitutions;
- D is the number of word deletions;
- I is the number of word insertions;
- N is the total number of words in the reference transcript.

Character Error Rate (CER) applies the same formulation at the character level:

$$\text{CER} = \frac{S + D + I}{N} \tag{2.1}$$

where S, D, and I are the number of character substitutions, deletions, and insertions, respectively, and N is the total number of reference characters.

Lower WER and CER values indicate better recognition performance, with WER $= 0$ and CER $= 0$ representing perfect transcriptions.

Practical ASR still faces challenges in noise, accents, domain shifts, streaming latency, and on-device constraints. Current research therefore focuses on multilingual and code-switching recognition, efficient streaming architectures, model compression, diarization, alignment, and confidence estimation for safer deployment.

2.3 Optical Character Recognition

Optical Character Recognition (OCR) extracts text from scanned documents and images. It has progressed from rule-based and template-matching methods to deep learning systems that support digitization, automation, and document understanding across many domains [19, 20].

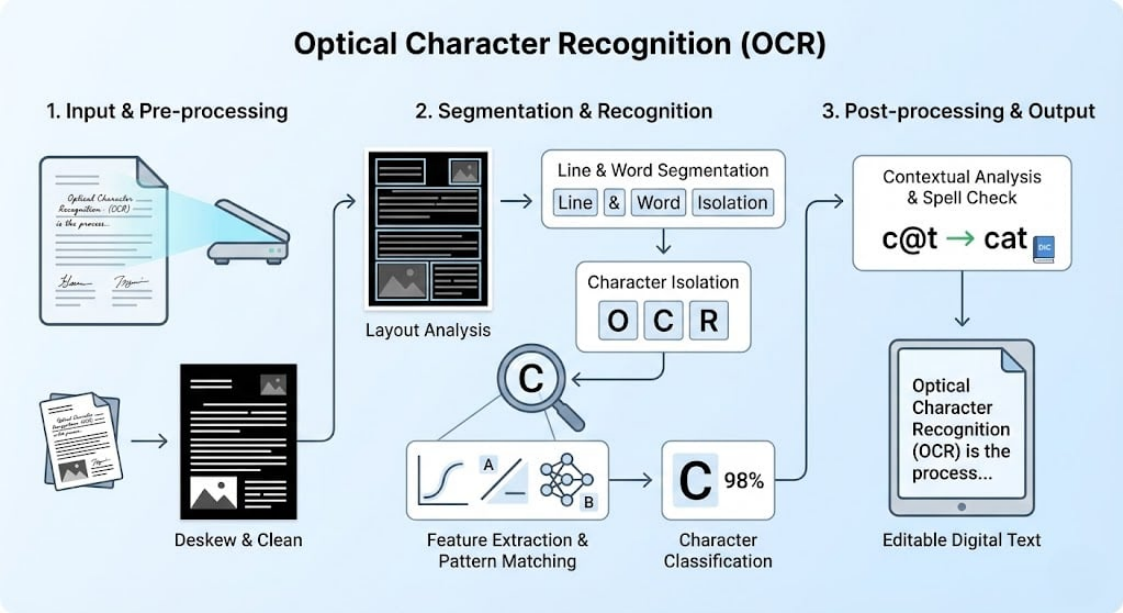

Figure 2.2: OCR pipeline.

Early OCR systems relied on mechanical recognition and template matching, which worked only in constrained settings and were sensitive to fonts, noise, and document quality [21]. Commercial use expanded in banking and postal services, but robustness remained limited [22].

Conventional OCR pipelines used segmentation, feature extraction, and classification. Hand-crafted features such as zoning, projection profiles, and contours were paired with classifiers such as SVMs and HMMs [23]. Handwriting remained harder because of style variation, cursiveness, and ligatures, motivating more flexible statistical approaches [24].

Deep learning transformed OCR by replacing hand-crafted features with learned visual and sequence representations. CNNs learn spatial features directly from images [25], while RNN/LSTM models and CTC support sequence recognition without explicit character segmentation [4, 26]. More recently, transformer-based systems such as TrOCR have improved recognition across scripts and visual conditions [27].

With advances in deep learning, end-to-end models such as CRNN and Transformer-based architectures directly learn visual-to-text mappings from raw images [28–30]. These models estimate the output text sequence by maximizing the conditional probability given the input image

$$\hat{Y} = \arg\max_{Y} P(Y|X)$$

where X denotes the input image and Y is the predicted text sequence.

Although OCR has improved significantly, handwritten documents, low-resolution images, multilingual text, domain-specific noise, and complex page layouts remain difficult. OCR is commonly evaluated using Character Error Rate (CER) and Word Error Rate

(WER), following the same edit-distance formulation defined in Section 2.2, Equations 2.1 and 2.2. Lower CER and WER indicate better recognition performance.

Scene text recognition (STR) extends OCR to text in natural images, where backgrounds, distortion, and lighting are uncontrolled. Modern STR often combines object detection models such as Faster R-CNN or YOLO with recognition modules to improve text localization and transcription [29].

OCR supports archive digitization [31], check and identity-document processing [32], medical record digitization [33], assistive reading technologies [34], and license plate recognition [35].

Key research directions include self-supervised learning to reduce labeled-data dependence, multimodal OCR that combines visual and linguistic cues, and few-shot methods for underrepresented languages and scripts [27]. Overall, OCR bridges visual and textual information and remains central to document automation and accessibility.

2.4 Transformer

Transformers are a breakthrough AI architecture used in modern language and vision models. Understanding how they work helps explain why our system is effective. Before diving into details, here's the structure:

The Transformer model described in this section is based on *Attention Is All You Need* [36]. It replaces recurrent sequence processing with self-attention, enabling parallel training and effective modeling of long-range dependencies in tasks such as translation, generation, and language understanding.

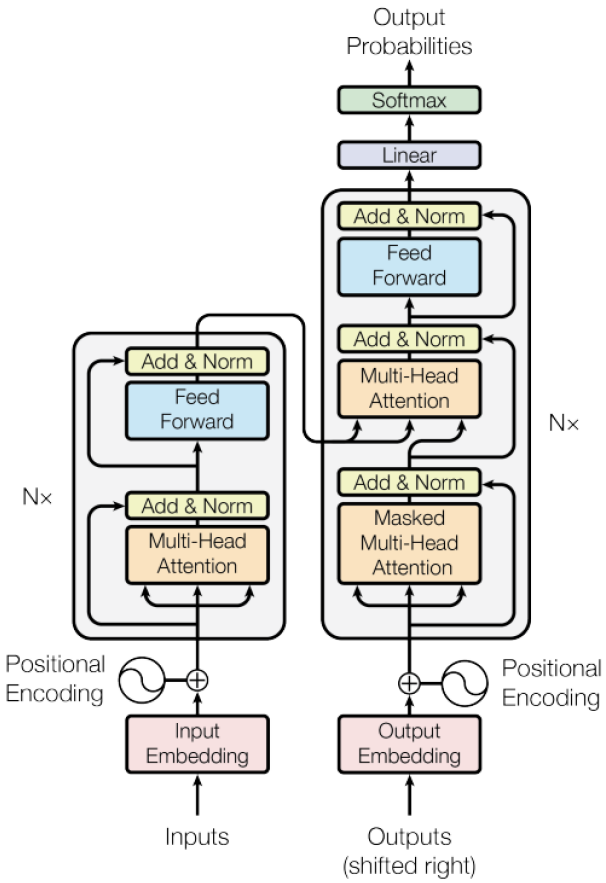

Figure 2.3: Transformer Architecture.

A Transformer consists of an **Encoder** and a **Decoder**, each composed of N identical layers ($N = 6$ in the original implementation).

Each Encoder layer contains two sub-layers:

1. Multi-head self-attention.
2. Position-wise fully connected feed-forward network.

Each sub-layer is wrapped with a residual connection. The output of each sub-layer is

$$\text{LayerNorm}(x + \text{Sublayer}(x)).$$

Each Decoder layer contains three sub-layers:

1. Masked multi-head self-attention, preventing each position from attending to future tokens.
2. Multi-head attention over the Encoder outputs.
3. Position-wise feed-forward network.

As in the Encoder, residual connections and layer normalization are applied to each sub-layer.

Because the Transformer contains no recurrence or convolution, positional information must be injected explicitly. The model adds Positional Encodings to input embeddings at the bottoms of both the Encoder and Decoder stacks. The original Transformer uses sinusoidal encodings defined as

$$\mathrm{PE}(pos, 2i) = \sin\left(\frac{pos}{10000^{2i/d_{\text{model}}}}\right),$$

$$\mathrm{PE}(pos, 2i+1) = \cos\left(\frac{pos}{10000^{2i/d_{\text{model}}}}\right),$$

where pos is the token position and i is the dimension index. Each dimension corresponds to a sinusoid at a different frequency, enabling the model to encode both absolute and relative positional patterns.

Self-attention computes an output for each position by mapping a query vector to a set of key-value pairs.

Similarity between queries and keys is computed via the dot product QK^T, followed by a softmax to obtain attention weights. Because large values can destabilize gradients when the dimensionality d_K is high, the dot products are scaled by $\sqrt{d_K}$

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d_K}}\right)V. \tag{2.2}$$

where

- $Q \in \mathbb{R}^{T_q \times d}$ are query vectors representing what information to look for;
- $K \in \mathbb{R}^{T_k \times d}$ are key vectors representing what information is available;
- $V \in \mathbb{R}^{T_k \times d}$ are value vectors containing the actual information to be extracted;
- d_K (or d) is the dimension of the key/query vectors;
- T_q, T_k are the sequence lengths for queries and keys respectively.

Instead of performing a single attention operation, the Transformer projects queries, keys, and values into h subspaces using learned linear transformations. Attention is computed independently in each head

$$\mathrm{head}_i = \mathrm{Attention}(QW_i^Q,\, KW_i^K,\, VW_i^V).$$

The outputs of all h heads are concatenated and linearly transformed

$$\mathrm{MultiHead}(Q, K, V) = \mathrm{Concat}(\mathrm{head}_1, \ldots, \mathrm{head}_h)W^O.$$

This mechanism allows the model to capture multiple types of relationships simultaneously by attending to diverse representation subspaces.

2.5 Multimodal Fusion and Cross-Attentional Models

Specialized retrieval pipelines often need to combine visual, audio, motion, and ASR-text signals rather than relying on a single encoder. For video retrieval, M2HF uses multi-stream feature extraction, attentional fusion, and multi-level feature/decision fusion to produce richer retrieval signals [37].

Cross-attentional models are especially useful for reranking. There are two main ways AI models compare text: **Independent comparison** (each text separately) or **Joint comparison** (texts together). Joint comparison (called cross-attention) is more accurate but slower. Cross-encoders jointly process the query and document, allowing token-level interactions that capture fine semantic distinctions. This improves precision but is computationally expensive, so cross-attention is usually applied to a smaller candidate set after first-stage retrieval [38, 39][2].

Cross-attention computes attention weights using queries from one sequence and key-value pairs from another. Given query matrix Q, key matrix K, and value matrix V

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

The cross-attention mechanism is illustrated in Figure 2.4[3].

[2] https://www.pinecone.io/learn/series/rag/rerankers/

[3] https://magazine.sebastianraschka.com/p/understanding-multimodal-llms

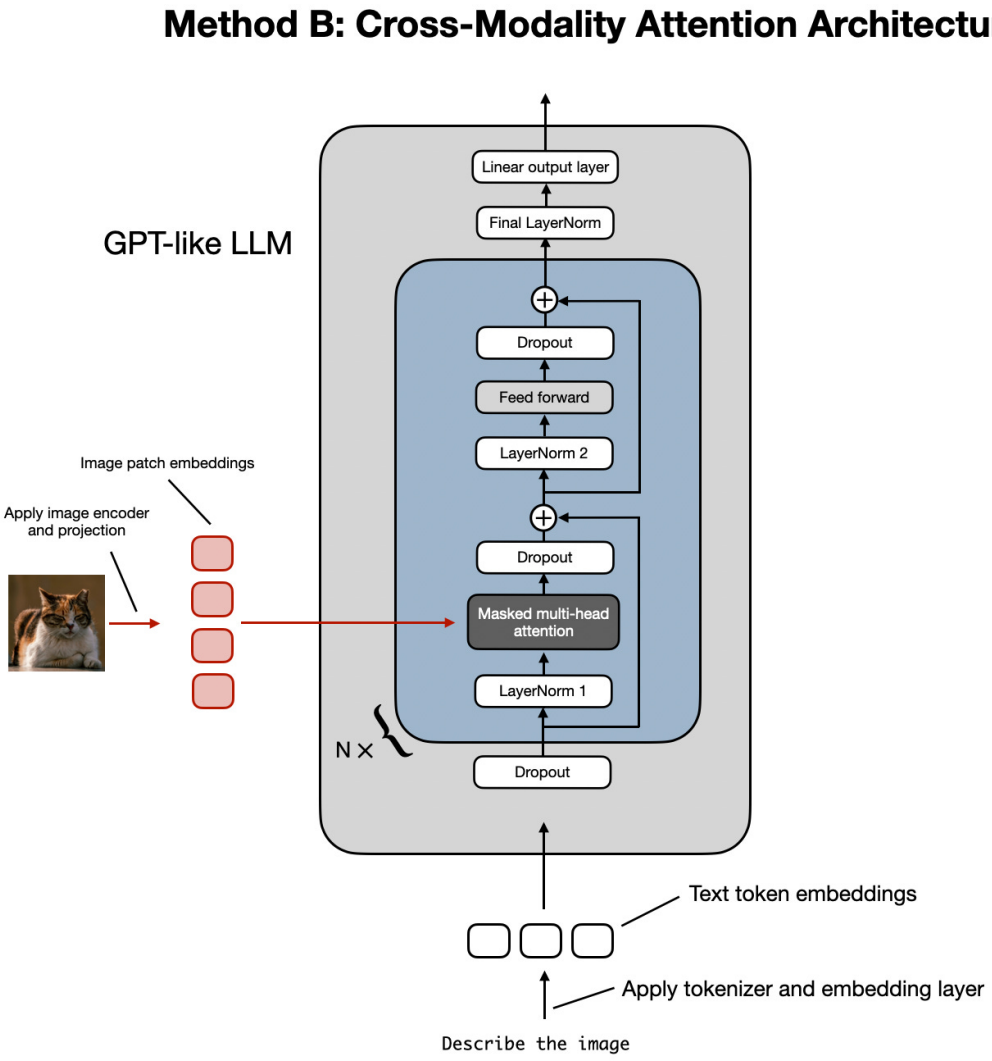

Figure 2.4: Cross-attention.

Cross-attention uses the same attention mechanism defined in Section 2.4 (Equation 2.2), but with a key difference: queries come from one sequence (the target modality, e.g., decoder text embeddings) while keys and values come from a different sequence (the source modality, e.g., encoder speech/image features). This allows the model to attend across different modalities or sequences, enabling information flow from source to target.

Formally, for cross-attention:

- $Q \in \mathbb{R}^{T_q \times d}$ are queries from the target modality (e.g., decoder text embeddings);
- $K, V \in \mathbb{R}^{T_k \times d}$ are keys and values from the source modality (e.g., encoder speech/image features);
- d_k is the dimension of the key vectors;

- T_q, T_k are token lengths for target and source sequences.

Multimodal fusion via dual attention for two modalities A and B, cross-attention is computed independently:

$$H_A = \text{Attention}(Q, K_A, V_A); H_B = \text{Attention}(Q, K_B, V_B).$$

The fused representation is then computed as

$$H = \alpha H_A + (1 - \alpha) H_B$$

where $0 \leq \alpha \leq 1$ is a learnable fusion coefficient.

2.6 Multimodal Embedding

Multimodal embeddings represent heterogeneous inputs in a shared space for cross-modal retrieval. CLIP established a strong baseline by jointly training image and text encoders on large-scale paired data, enabling zero-shot image–text retrieval and serving as a backbone for later systems [40].

To search across different types of information (text, images, audio), we need to represent them in a way the AI system can understand. We use numerical vectors (arrays of numbers) to represent each piece of information.

Each input modality $m \in \{\text{text}, \text{audio}, \text{vision}\}$ is encoded into a latent vector using a modality-specific encoder

$$z_m = f_m(x_m; \theta_m)$$

where x_m is the raw input of modality m, $f_m(\cdot)$ is the corresponding encoder (e.g., Transformer encoder for text, Conformer for audio, ViT for vision), and θ_m denotes its learnable parameters. An example of multimodal embedding is shown in Figure 2.5[4].

[4] https://newsletter.theaiedge.io/p/how-to-build-a-multimodal-rag-pipeline-8d6

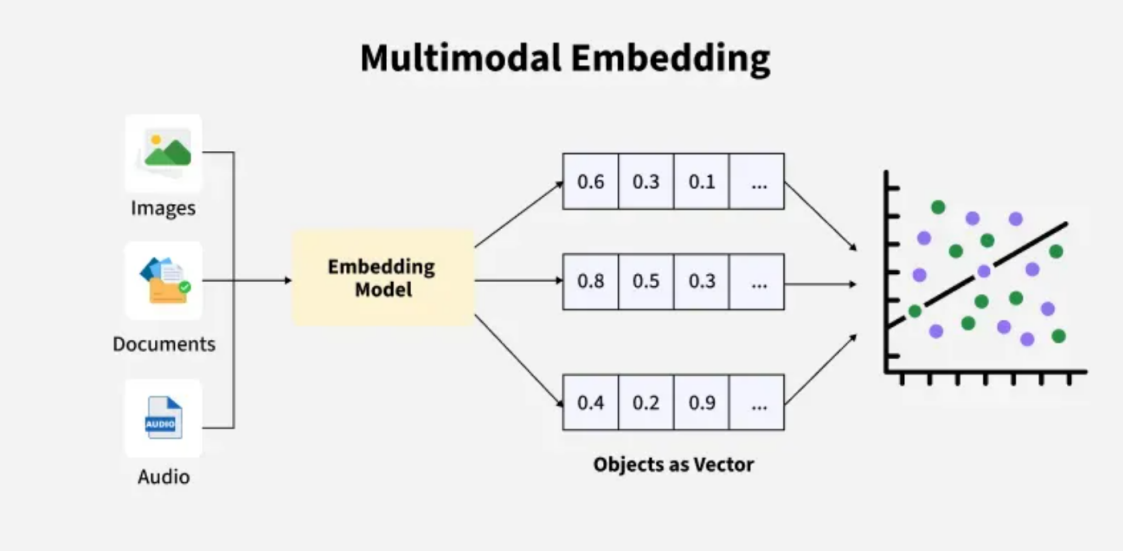

Figure 2.5: Multimodal embedding.

We fuse modality embeddings into a shared representation

$$z = g([z_{\text{text}}; z_{\text{audio}}; z_{\text{vision}}]; \phi)$$

where $[z_{\text{text}}; z_{\text{audio}}; z_{\text{vision}}]$ is the concatenation operator, $g([z_{\text{text}}; z_{\text{audio}}; z_{\text{vision}}])$ is a fusion network (e.g., MLP or cross-attention module), and ϕ denotes fusion parameters.

To align representations across modalities, we use a technique called "contrastive learning." To make sure the AI learns to match similar information across modalities, we use a technique called 'contrastive learning.' It teaches the model to make similar things close together and different things far apart.

$$\mathcal{L}_{\text{contrast}} = -\sum_{i=1}^{N} \log \frac{\exp(\text{sim}(z_i^{(a)}, z_i^{(t)})/\tau)}{\sum_{j=1}^{N} \exp(\text{sim}(z_i^{(a)}, z_j^{(t)})/\tau)}$$

where $z_i^{(a)}$ and $z_i^{(t)}$ are the audio and text embeddings of paired sample i, $\text{sim}(z_i^{(a)}, z_j^{(t)})$ is a similarity function (e.g., cosine similarity), τ is the temperature scaling factor, and N is the batch size.

Recent work repurposes multimodal LLMs as retrievers. MM-Embed fine-tunes a multimodal LLM as a bi-encoder for image-text queries and uses modality-aware hard negatives to reduce modality bias, improving performance on M-BEIR and text retrieval benchmarks [41].

Unified embedding models extend this idea beyond image and text. ImageBind aligns images, video, text, audio, depth, thermal data, and IMU signals in one space without requiring every modality pair to be explicitly annotated [42]. This supports educational retrieval where speech, slides, images, and video should be searchable together.

Domain-specific models target visually rich documents such as slides, tables, diagrams, and formulas. For example, jina-embeddings-v4 supports text-image retrieval, single-vector

embeddings, multi-vector late interaction, and task adapters, with strong results on visually dense retrieval benchmarks such as Jina-VDR [43].

Industry systems also focus on interleaved documents such as PDFs and slides. Nomic's Embed Multimodal 7B embeds mixed text-image document structures for RAG over visually rich files[5].

Overall, the gap between multimodal and single-modal retrieval is narrowing. Multimodal training can improve semantic representations even for text-only tasks, which is valuable in educational settings where queries often combine textual and visual references.

Simplified Overview of Multimodal Embedding Models

The following table provides a concise summary of the landmark multimodal embedding models discussed in this section, highlighting their architectural focus and primary strengths for educational RAG applications.

Table 2.2: Comparison of Multimodal Embedding Models.

Model	Modalities	Key Innovation	Primary Use Case
CLIP	Image, Text	Joint contrastive pre-training on 400M pairs.	Baseline cross-modal alignment.
MM-Embed	Image, Text	Modality-aware hard negatives to reduce text bias.	Balanced multi-domain retrieval.
ImageBind	6 Modalities	Unified embedding space for Text, Image, Audio, etc.	Cross-modal search (e.g., Audio → Image).
Jina-v4	Image, Text	Task-specific LoRA adapters and late-interaction.	Visually rich documents (slides, tables).
Nomic 7B	Interleaved	Native support for interleaved document structures.	Multi-page PDF/Slide RAG.

2.7 RAG

2.7.1 Historical Evolution of RAG

2017–2019: Pre-RAG foundations. Before RAG, open-domain QA commonly used retrieve-then-read pipelines: a sparse retriever fetched passages and a neural reader extracted answers. DrQA established this pattern with TF-IDF/BM25 retrieval and neural reading [44]. ORQA then showed that dense retrievers could be trained from QA supervision [45], while kNN-LMs demonstrated inference-time access to external datastores [46].

2020: Formalizing RAG. REALM introduced differentiable retrieval during

[5] https://www.nomic.ai/blog/posts/nomic-embed-multimodal

language-model pretraining, showing that an LM could consult updateable external knowledge [47]. RAG then combined DPR retrieval with a BART generator and trained the retriever-generator system for knowledge-intensive tasks [48].

2021–2022: Stronger Retrievers, Multi-Document Reasoning, and Lighter Readers. This period improved retrieval, evidence fusion, and query reformulation. RocketQA, RocketQAv2, Condenser, Contriever, and GTR strengthened dense retrieval through hard negatives, distillation, retriever-oriented pretraining, unsupervised contrastive learning, and scaling [49–53]. ColBERTv2 made late-interaction retrieval more practical, while SPLADE showed the value of learned sparse representations and hybrid fusion [54–56]. Multi-document systems such as EMDR, MDR, FiD, KG-FiD, FiD-Light, and RETRO improved multi-hop evidence use and semi-parametric generation [57–62]. Query reformulation methods such as GAR, Self-Ask, ReAct, WebGPT, and HyDE became reusable components for improving first-stage recall and reasoning [63–67].

2023–2024: From fixed pipelines to adaptive, verifiable, and structure-aware systems. Recent RAG systems increasingly decide when, what, and how much to retrieve. FLARE, Adaptive-RAG, SEAKR, and DRAGIN trigger or route retrieval dynamically [68–71]. SELF-RAG, CRAG, and CoV-RAG add critique, filtering, refusal, or verification to reduce unsupported generation [72–75]. Context compression and ranking became central through RankRAG and LLMLingua [76–78], while GraphRAG/GRAG introduced structure-aware retrieval over entities and relationships [79, 80]. LongRAG and self-routing explored how long-context LLMs and targeted retrieval can be combined [81, 82].

2025: Agentic control, standardized modules, faithful decoding, evaluation, and memory. By 2025, Agentic RAG emphasizes planning, tool use, reflection, and multi-agent control over retrieval; HopRAG is an example of graph-guided multi-hop retrieval before generation [83–85]. Surveys increasingly describe RAG as modular: retrieval optimization, context processing, generation optimization, and evaluation/monitoring [86, 87]. Faithfulness methods such as FaithfulRAG, ReCLAIM, CiteFix, and citation benchmarks focus on grounding and attribution [88–92]. Evaluation and memory-oriented systems such as ARES, HippoRAG-2, MemoRAG, Mem0, MIRIX, and MemOS further connect RAG with monitoring, persistent memory, provenance, and continual learning [93–100].

2.7.2 MRAG: The Convergence

Pre-RAG signals for vision–language tasks. Before MRAG was named, factual VQA exposed the limits of image-only reasoning. "Straight to the Facts" explicitly learned KB retrieval for VQA, and OK-VQA required outside knowledge, catalyzing retrieval-style VQA where images must be grounded in world facts [101].

2022–2023: Early MRAG patterns. Early MRAG combined VQA-style retrieval with cross-modal embeddings. TRiG converted images into text, retrieved passages, and answered in natural-language space [102], while entity-focused dense retrieval improved OK-VQA retrieval [103]. CLIP/ALIGN supplied shared image-text spaces, and REVEAL

connected retrieval and generation during visual-language pretraining with multimodal memory [104]. Most systems still retrieved textual evidence such as captions, OCR, or Wikipedia, then fused it into a VL encoder/decoder.

2024: Maturation – better retrieval and fusion for VLMs, and broader modality bindings. VisRAG treats visually rich documents as first-class evidence by combining OCR text with visual region cues [105]. RAVEN frames retrieval-augmented VLM fine-tuning as a general recipe [106], while ImageBind supports one shared embedding space across six modalities, motivating unified MRAG indexes for heterogeneous educational assets [42].

2025: Advanced MRAG – domains, video, documents, and built-in retrieval. Recent MRAG work expands into domain-specific and video/document settings. RS-RAG integrates satellite imagery with world knowledge for remote sensing tasks [107], while unified MRAG systems combine text, image, table, and video evidence in one pipeline [108]. Surveys report that MRAG improves grounding and reduces hallucinations for visually grounded queries compared with text-only RAG [109].

2.7.3 Pipeline

A MRAG pipeline is illustrated in Figure 2.6[6].

[6] https://newsletter.theaiedge.io/p/how-to-build-a-multimodal-rag-pipeline-8d6

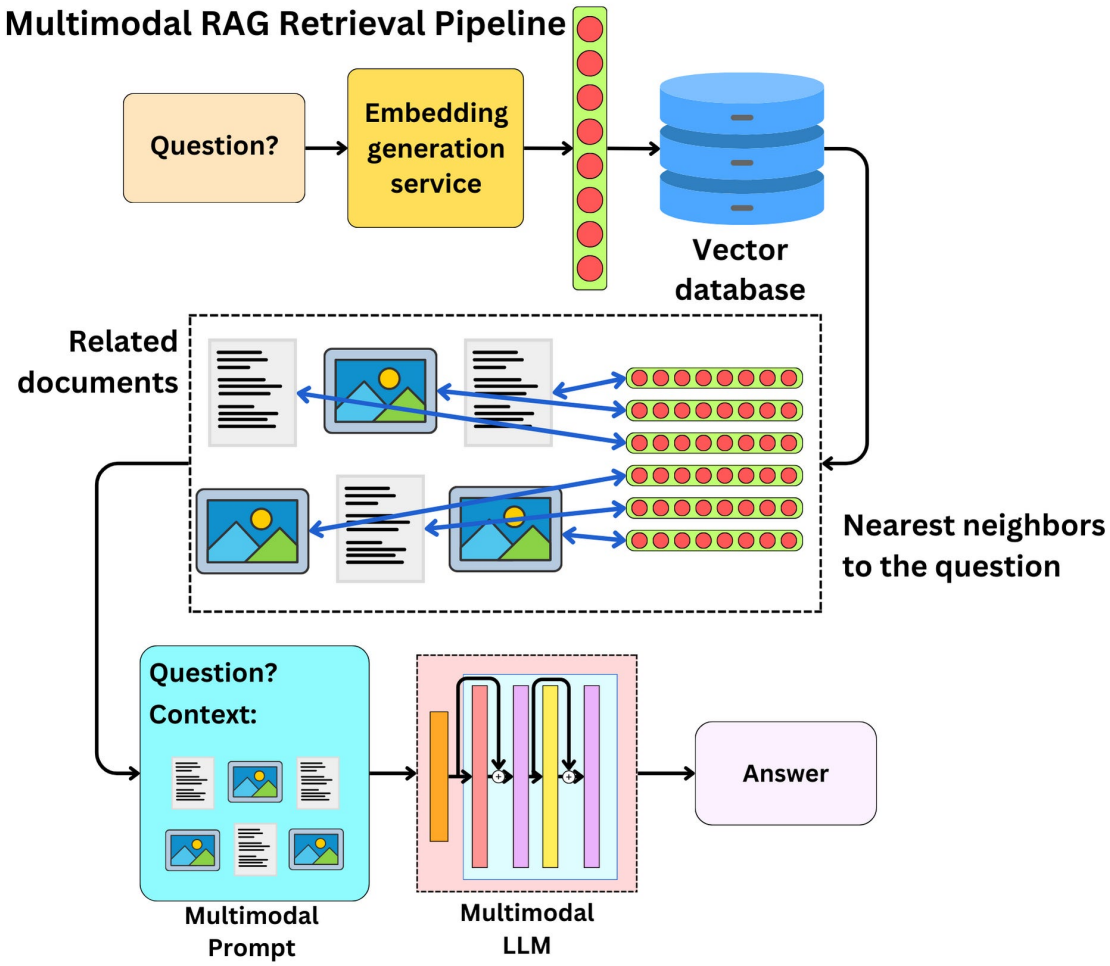

Figure 2.6: MRAG pipeline.

The pipeline combines retrieval with generation to produce grounded answers. A query is encoded into a vector representation, matched against a pre-indexed Vector Database, and used to retrieve relevant documents or passages [48, 110]. Retrieved evidence may then be filtered, summarized, or restructured before being passed to the LLM context [111].

RAG has two core modules: a retriever and a generator. The retriever searches external knowledge using sparse, dense, hybrid, or neural methods [3, 48]; the generator conditions on both the user query and retrieved context to produce the answer [112]. This design improves factual grounding and enables access to domain-specific or updated knowledge without retraining the language model [111, 113].

MRAG extends this pipeline to text, image, video, table, and graph evidence. Recent systems explore modality-aware retrieval, reranking, multi-agent retrieval, and reflective generation to improve answer quality in visually rich and knowledge-intensive tasks [114–116].

2.7.4 Retrieval

Sparse retrieval maps text into high-dimensional lexical vectors [56, 117]. Methods such as TF-IDF, **BM25**, and SPLADE are efficient and interpretable for exact keyword matching, usually implemented with inverted indexes. Their limitation is lexical mismatch: relevant documents may be missed when they use different wording from the query.

Dense retrieval uses neural encoders to map queries and documents into a shared embedding space where semantic similarity is measured by vector proximity [110]. Dual-encoder models such as DPR and SPECTER2 can retrieve semantically related content even when keywords differ [118]. Large-scale dense retrieval typically relies on Approximate Nearest Neighbor Search, including HNSW indexes [119]. However, dense vectors can be weaker on exact term matching and may lose detail in long documents [56].

Hybrid retrieval combines sparse and dense signals to improve robustness in RAG systems [48, 56, 110]. Common strategies include score fusion, merging separate sparse/dense result lists, or building unified hybrid indexes. Key challenges are score calibration, distribution mismatch, and the computational cost of high-dimensional sparse vectors; distribution alignment and adaptive two-stage search can help balance accuracy and speed [120].

Once embeddings have been generated, the retrieval module identifies and returns the most relevant documents for a given query. In a RAG pipeline, this step is essential because it supplies the language model with context that guides accurate response generation. Formally, given a query q in language x, the retriever returns a document d in language y (which may or may not match x) from a corpus D_y. The retrieval function f_n^* may be implemented using dense, lexical, or multi-vector retrieval methods [121]

$$d_y \leftarrow f_n^*(q_x, D_y).$$

Dense Retrieval: A query q is encoded into hidden states H_q using a text encoder. The representation of the query is the normalized hidden state of the special token `[CLS]`

$$e_q = \text{norm}(H_q[0]).$$

Similarly, for a passage p

$$e_p = \text{norm}(H_p[0]).$$

Relevance is computed as the inner product

$$s_{\text{dense}} = \langle e_p, e_q \rangle.$$

Lexical Retrieval: For each term t in the query

$$w_{qt} = \text{ReLU}(W_{\text{lex}}^\top H_q[i]),$$

and similarly for a passage

$$w_{pt} = \text{ReLU}(W_{\text{lex}}^\top H_p[i])$$

where

$$W_{\text{lex}} \in \mathbb{R}^{d\times 1}; \qquad \text{ReLU}(x) = \max(0, x).$$

Multi-vector Retrieval: Both queries and passages are represented using multiple vectors.

$$E_q = \text{norm}(W_{\text{mul}}^{\top} H_q), \qquad E_p = \text{norm}(W_{\text{mul}}^{\top} H_p),$$

where

$$W_{\text{mul}} \in \mathbb{R}^{d\times d},$$

and the normalization function is applied column-wise.

2.7.5 Rerankers

Rerankers, commonly implemented as cross-encoders, refine search results by jointly encoding a query–document pair to compute a relevance score. Unlike bi-encoders, which independently embed queries and documents into fixed-length vectors, rerankers leverage full token-level interactions, enabling more precise semantic matching. They are typically integrated into two-stage retrieval pipelines, where the first-stage retriever generates candidate documents and the reranker refines their ordering to achieve higher accuracy. The reranking process is illustrated in Figure 2.7[7].

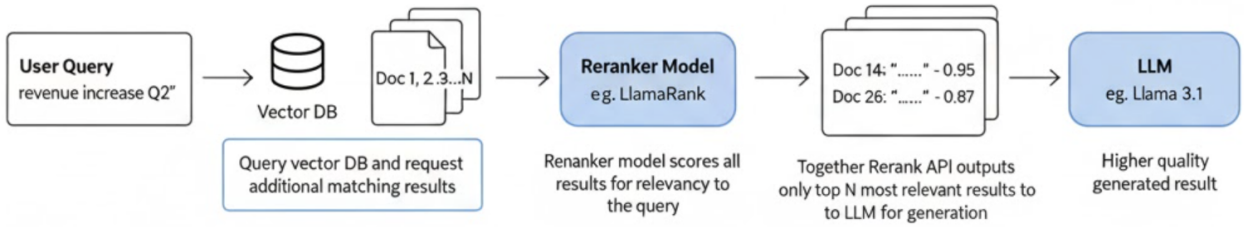

Figure 2.7: Reranker.

Rerankers model token-level cross-attention between the query and document, which improves precision for entity matching, negation, semantic alignment, factual verification, and multi-hop reasoning.

Rerankers are computationally expensive and therefore operate on a narrowed candidate set rather than the full corpus. This architecture is decomposed as follows:

- Stage 1 (Efficient Retrieval): A bi-encoder or sparse retriever (e.g., BM25, DPR, ColBERT) selects the top-k candidates efficiently using vector similarity or lexical matching.
- Stage 2 (Precision Reranking): A cross-encoder transformer takes (q, d) jointly as input and outputs a scalar relevance score used to reorder the candidate list.

[7] https://www.together.ai/blog/together-rerank-api-and-salesforce-llamarank

This separation balances scalability and accuracy: fast retrieval reduces search space, while rerankers optimize precision at the top ranks.

Bi-encoders compute representations independently:

$$\text{doc} \to \mathbf{v}_d \in \mathbb{R}^d, \quad \text{query} \to \mathbf{v}_q \in \mathbb{R}^d$$

Similarity is computed via vector metrics (e.g., dot product), which is efficient but compresses document semantics into a single vector without query-context information.

Rerankers instead compute relevance jointly. This allows the model to consider local alignments and contextual dependencies, improving performance on tasks requiring fine-grained semantic reasoning. However, the computation happens online for each query-document pair and cannot be precomputed.

Bi-encoder indexing is performed offline, so inference requires only query encoding and approximate nearest neighbor search. Rerankers require full transformer inference for every candidate document:

$$q + d \to \text{Transformer} \to \text{score}$$

Because this cost scales with the candidate count, rerankers are typically applied to only 10–1000 candidates rather than the full index.

Impact on Retrieval Quality

Empirically, rerankers improve metrics such as Recall and Mean Reciprocal Rank, especially in:

- open-domain question answering
- legal and biomedical retrieval where precision is critical
- multi-step reasoning and claim verification
- reranking passages in RAG pipelines for LLM-based generation

Token-level attention helps distinguish subtle semantic variations, such as contradiction versus support, that bi-encoders may compress away.

- **Bi-encoders:** Scalable retrieval with precomputed embeddings, but limited by lossy compression.
- **Rerankers:** High precision via joint computation, but expensive at query time.
- **Two-stage retrieval:** Combines scalability and precision, forming the standard architecture for modern retrieval systems.

Neural reranking has evolved from cross-encoders over BM25 candidates to more scalable and generative approaches. ColBERT-v2 preserves token-level late interaction while using compressed embeddings and approximate search for efficiency [122]. MonoT5 frames relevance as a sequence-to-sequence generation task [123], TART uses instruction-tuned reranking for zero-shot transfer [124], and RankGPT applies LLMs to listwise ranking and reasoning across candidate sets [125]. These trends make rerankers both precision filters and semantic evaluators in modern RAG pipelines.

2.7.6 Augmentation

The augmentation process in RAG is illustrated in Figure 2.8[8].

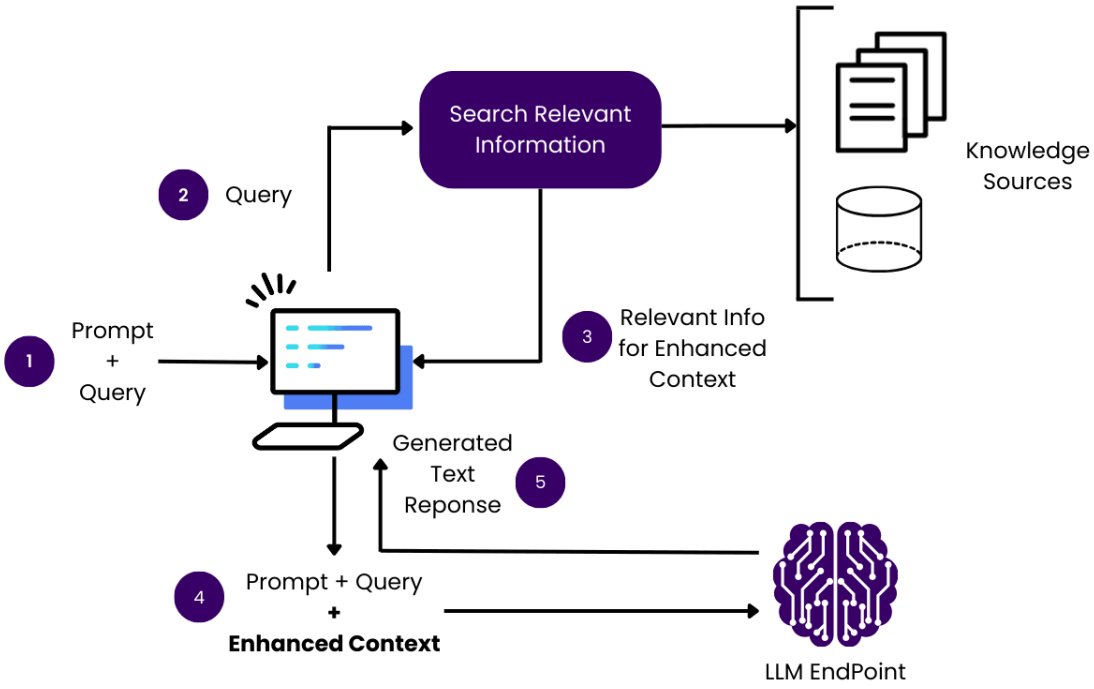

Figure 2.8: Augmentation.

Augmentation in RAG transforms retrieved knowledge before generation. It may summarize, compress, restructure, or expand evidence to improve relevance and reduce noise [126, 127].

Given a retrieved document set $D = \{d_1, \ldots, d_k\}$, augmentation produces a transformed set $D' = \mathcal{A}(D)$ where $\mathcal{A}$ is an augmentation operator such as summarization, chunk merging, entity extraction, or reasoning-based synthesis. Formally:

$$D' = \mathcal{A}(D) = \{a_1, \ldots, a_m\}, \quad m \leq k \text{ or } m \geq k,$$

[8] https://obot.ai/resources/learning-center/retrieval-augmented-generation/

where each augmented element a_i may be compressed, reformatted, or newly generated from retrieval.

To preserve semantic coverage while reducing redundancy, augmentation can be framed as an optimization objective:

$$\max_{D'} \ \mathrm{Rel}(D', q) - \lambda \, \mathrm{Redund}(D'),$$

where Rel measures relevance to query q and Redund quantifies informational overlap.

A common augmentation strategy is active document compression, where generative models summarize retrieved passages while preserving answer-critical semantics [126]. This can be modeled as:

$$a_i = \arg\max_{a} P(a \mid d_i, q),$$

and scored using a utility function:

$$s(a_i, q) = \alpha \cdot \mathrm{Sim}(a_i, q) + \beta \cdot \mathrm{Coverage}(a_i, d_i),$$

where

$$\mathrm{Sim}(a_i, q) = \cos(g(q), h(a_i)), \quad \mathrm{Coverage}(a_i, d_i) = \cos(h(a_i), h(d_i)).$$

Augmentation also enables iterative cycles of retrieval and rewriting. Self-reflection frameworks refine evidence through reasoned critique:

$$D^{(t+1)} = \mathcal{A}\Big(D^{(t)}, y^{(t)}, c^{(t)}\Big),$$

where $y^{(t)}$ is the generated answer and $c^{(t)}$ is a critique model output guiding refinement [127].

For programming tasks, retrieved demonstrations can be reformatted into template-based exemplars:

$$a_i = \mathrm{Format}\Big(\mathrm{Demo}(d_i)\Big),$$

supporting structural learning in applications [128].

Thus, augmentation acts as a knowledge refinement stage that improves grounding, reduces hallucination risk, and increases domain robustness [113].

Once we retrieve relevant documents, we need an AI to read them and write answers in the student's own words. This is where generation comes in.

2.7.7 Generation

The generator is the synthesis module in RAG. It conditions on both the user query and retrieved evidence, improving factual grounding compared with a standalone parametric language model [113].

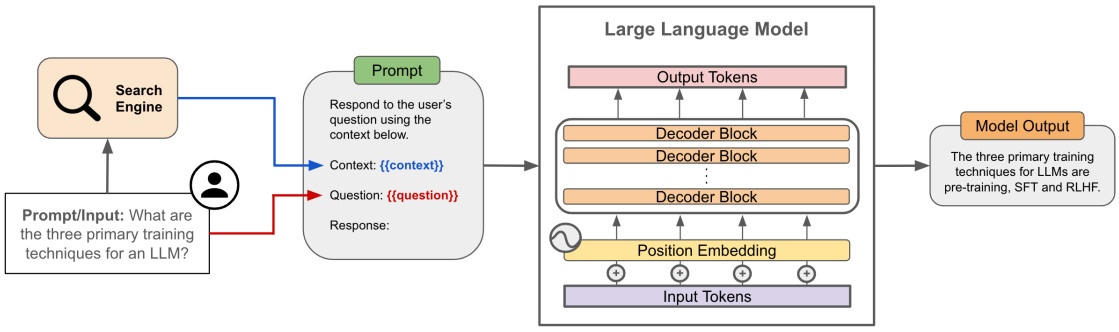

Figure 2.9: Generation.

Formally, given a query q retrieves a document set $D = \{d_1, \ldots, d_k\}$, the generator models the output sequence $y = (y_1, \ldots, y_T)$ as:

$$P(y \mid q, D) = \prod_{t=1}^{T} P(y_t \mid y_{<t}, q, D).$$

Instead of treating retrieved documents as a single concatenated input, marginalization across documents can be applied:

$$P(y_t \mid y_{<t}, q, D) = \sum_{d_i \in D} P(y_t \mid y_{<t}, q, d_i)\, P(d_i \mid q).$$

Each retrieved document is encoded into a latent representation:

$$h_i = f_{\text{enc}}(d_i),$$

and decoding attends to these representations:

$$P(y_t \mid y_{<t}, q, d_i) = \text{softmax}(W \cdot \text{Dec}(y_{<t}, h_i))\,.$$

Modern RAG systems typically use autoregressive or encoder-decoder Transformers, integrating retrieved content through concatenation, cross-attention, or late fusion [126, 127].

- **Concatenation-based generation:** Retrieved passages are appended to the prompt; simple but limited by context length and prone to noisy retrieval.
- **Cross-attention generation:** Retrieved representations act as separate attention sources to improve semantic alignment and reduce positional bias [128].

In cross-attention generation, retrieved representations form a secondary attention memory:

$$\text{Attention}(Q, K_D, V_D) = \text{softmax}\left(\frac{QK_D^\top}{\sqrt{d_k}}\right) V_D,$$

where K_D, V_D are keys and values derived from retrieval encodings.

- **Rerank-and-generate pipelines:** Only top-ranked chunks are passed to the generator after evidence filtration [129].
- **Self-reflective iterative generation:** Retrieval, generation, and critique occur recursively to enhance factual accuracy and reasoning [127].

To further improve faithfulness, contrastive losses can be incorporated to prefer evidence-grounded explanations:

$$\mathcal{L}_{\text{faith}} = -\log \frac{\exp(s(y, D^+))}{\exp(s(y, D^+)) + \exp(s(y, D^-))},$$

where D^+ contains supporting evidence and D^- contains retrieved distractors. The score function $s(y, D)$ measures how well a generated output y is grounded in a document set D, commonly implemented as a similarity function such as:

$$s(y, D) = \cos\left(g(y),\, h(D)\right),$$

where $g(y)$ encodes the generated text and $h(D)$ aggregates document embeddings (e.g., mean pooling or attention-weighted fusion).

2.7.8 Prompt Engineering

Prompt engineering designs input prompts to control LLM behavior during inference. It steers pretrained models without modifying their parameters [130].

Core formats include **zero-shot prompting**, which uses only instructions [131], and **one-shot/few-shot prompting**, which add demonstrations for in-context learning [130]. Reasoning-oriented methods such as **Chain-of-Thought (CoT)** elicit intermediate reasoning for multi-step tasks [132], while self-consistency samples multiple reasoning paths to improve stability [133].

Prompt engineering also supports **structured and reliable outputs**. Schema-based prompts can request JSON, bullet lists, or domain templates, reducing parsing failures in downstream systems[9]. Function calling extends this idea by constraining outputs to well-defined tool or data schemas.

Manual prompt design is heuristic and labor-intensive, so recent work explores **automated prompt optimization**. Candidate prompts can be searched and selected using validation metrics, including instruction phrasing and example ordering. Because exhaustive search is expensive, methods such as TRIPLE formulate prompt selection as a bandit problem to balance exploration and exploitation under limited query budgets [134].

Another line treats prompts as parameters. Prefix-Tuning learns continuous soft prompts without updating model weights [135], while AutoPrompt performs gradient-guided search over discrete trigger tokens [136]. These methods improve control but may reduce interpretability.

[9] OpenAI GPT-4 Technical Report, https://openai.com/research/gpt-4

Iterative editing methods refine human-written prompts. GrIPS applies gradient-free edits such as adding, deleting, or reordering phrases [137]. Instruction Induction and Automatic Prompt Engineer (APE) use LLMs to generate and score candidate instructions [138].

OPRO further frames optimization as an LLM-driven loop: a meta-prompt stores candidate prompts and scores, and the model proposes improved prompts iteratively [139]. Such methods must control overfitting to validation data.

Reinforcement learning treats prompts as actions optimized for rewards such as validation accuracy or human preference. RLPrompt applies policy gradients to discrete prompt tokens and can improve over manual baselines, though search spaces and sparse rewards remain challenging [140].

Overall, prompt engineering has matured into a systematic methodology spanning demonstrations, reasoning scaffolds, structured outputs, and automated optimization. It remains central to reliable, controllable, and domain-aligned LLM behavior.

Chapter 3

Related Work

3.1 Retrieval Methodologies for Educational Content

Educational systems need to search through different types of materials: text notes, slides with images, and videos. The best systems use two search stages: a fast initial search, then a more careful second check to improve accuracy.

3.1.1 Hybrid Sparse-Dense Retrieval for Text

In the first stage, retrieval is broad and efficient, designed to quickly narrow down a large collection to a smaller candidate set. Hybrid sparse–dense retrieval has proven especially effective for text-heavy materials such as lecture transcripts, textbooks, or slide decks. In this approach, a sparse retriever such as BM25 or SPLADE is combined with a dense retriever such as a bi-encoder BERT or SciBERT. Sparse retrieval ensures that exact technical terms, formulas, or names are not overlooked, while dense retrieval provides semantic matching that can capture paraphrases or rewordings of concepts. For instance, a query about a "photosynthesis diagram" would benefit from sparse retrieval if the exact term appears in the document, while a dense retriever could still surface relevant passages describing plant energy processes even when phrased differently. Empirical evidence supports the robustness of the hybrid approach; Luan et al. (2021) demonstrated that combining sparse and dense signals yields superior performance over either method alone, particularly in scientific domains [3]. To scale hybrid retrieval, efficient vector indices such as hierarchical navigable small-world (HNSW) graphs can be extended to support hybrid vectors or to efficiently merge results [141]. Many RAG systems for education adopt this hybrid method on a knowledge base of lessons or FAQs to ensure high recall.

3.1.2 Multimodal Embeddings for Video Retrieval

For video lectures or other multimodal content, dense retrieval based on multimodal embeddings is increasingly common. Models such as M2HF index video segments into

dense vectors that encode visual frames, audio cues, and ASR transcript information [37]. Text queries are mapped into the same embedding space, enabling cross-modal retrieval. This approach is advantageous over relying solely on transcript text, as it allows the retrieval of segments in which the visual modality is decisive—for example, locating "the slide showing the cell cycle" even when the transcript contains only a vague reference. The challenge of index size is addressed by segmenting videos into short windows, each represented by high-dimensional embeddings, and using approximate nearest neighbor search (e.g., FAISS or Annoy) to maintain efficiency. When transcripts or slide text are available, the hybrid sparse–dense approach introduced in Section 3.1.1 can be extended to multimodal contexts: combining OCR-extracted text and captions with visual tags produces more effective retrieval than relying solely on dense visual features, as purely visual features are not easily expressed as sparse signals. In practice, educational video retrieval often relies on maintaining both dense indices for semantic matches and inverted indices for textual signals.

3.1.3 Query Adaptation and Reranking

Before reranking, sometimes the system needs to improve the original question. For example, mathematical queries might be reformatted so the system understands them better. Then, results are reranked for accuracy.

For specialized educational queries, preprocessing or expansion may be required. For example, mathematical queries may be normalized by converting LaTeX into textual equivalents to ensure matches across both formats. LLMs can also be used to generate paraphrases of queries or, more experimentally, to produce synthetic images from text queries that can serve as additional search modalities. Although still exploratory, such query-driven modality augmentation holds promise for educational scenarios, such as automatically generating diagrams from student questions to facilitate diagram-based retrieval.

In the second stage, reranking is applied to the top candidates—typically the top 50 or 100—using slower but more precise methods. Cross-attentional rerankers, often based on transformer models such as BERT, take both the query and the candidate as input and compute a joint relevance score. This approach captures subtle distinctions that bi-encoder methods miss, such as differentiating between "a slide that mentions photosynthesis" and "a slide that explains photosynthesis in detail." While computationally expensive, reranking remains feasible at this stage since only a limited number of candidates are evaluated. Multimodal cross-attentional rerankers further enrich this process by incorporating visual or layout information alongside text.

More recently, LLM-based reranking has emerged as a promising paradigm. Instead of relying on a fixed, pre-trained reranker, a multimodal LLM can be prompted to evaluate the relevance of retrieved passages directly or to synthesize an answer using retrieved evidence. For instance, DynamicRAG uses outputs from an LLM as feedback for reranking in a RAG pipeline, dynamically adjusting which passages to keep [142]. Similarly, in a MRAG setup, more advanced relevancy measures have been proposed to better select which images, captions, or text to feed to the LLM, thereby reducing noise in the context [143]. Such methods blur the line between retrieval and generation, providing both reranking and

contextualized responses, and have been increasingly applied to educational chatbots and tutoring systems.

In practice, educational retrieval pipelines combine these methods to maximize performance. A typical system might retrieve textbook passages through hybrid retrieval, video segments through multimodal dense retrieval, and then rerank all candidates using either a cross-attentional model or a multimodal LLM. This two-stage strategy has become the prevailing standard because it delivers an effective balance between recall in the first stage and precision in the final ranking, thereby ensuring that the retrieved answers are both comprehensive and contextually relevant.

3.2 Educational applications

3.2.1 Multimodal Retrieval

To support evaluation and foster progress in multimodal retrieval for educational applications, researchers have assembled several new datasets that capture the multimodal nature of academic content. Between 2023 and 2025, multiple benchmarks have been introduced to reflect the diversity of lecture materials, textbooks, and visually rich documents. The Lecture Presentations Multimodal (LPM) dataset, released at ICCV 2023, is a notable example [144]. It contains approximately 180 hours of lecture videos delivered by ten lecturers across multiple subjects, aligned with more than 9,000 lecture slides. The dataset provides high-quality annotations linking slide content, including both figures and text, with corresponding speech segments. LPM defines three tasks that are directly relevant to education: figure-to-text retrieval, where a figure from a slide must be matched to the appropriate spoken segment; text-to-figure retrieval, where a segment of lecture speech or transcript is used to identify the relevant illustrative figure; and the generation of slide explanations. Baselines and human performance have been reported, and even fine-tuned vision–language models encountered significant difficulty on these tasks. This dataset has already inspired follow-up work, such as PolyViLT, which seeks to improve cross-modal alignment between slides and speech.

Another dataset is CK-12 Textbook QA (CK12-QA), which, although originally designed for question answering, doubles as a testbed for retrieval-augmented reasoning. Alawwad et al. (2025) propose a multimodal document ranking and retrieval supervision method for CK12-QA, showing that RAG significantly helps when retrieving both text and diagrams from lessons [145]. Their findings revealed that retrieval substantially improves reasoning performance, though challenges remain, particularly in aligning question references with the correct diagram.

Beyond these examples, additional academic lecture video datasets have been introduced. The M^3AV dataset (ACL 2024) comprises 367 hours of university lectures in computer science, mathematics, and medicine, with transcripts and OCR-extracted slide text, including mathematical formulas, aligned with the videos [146]. Although M^3AV emphasizes recognition tasks such as ASR and slide generation, it provides rich multimodal data that

can be repurposed for retrieval research. Earlier resources, such as lecture collections from VideoLectures.NET, contained smaller-scale alignments of subtitles and slides, but LPM and M^{3}AV substantially scale up both the size and the quality of annotations, enabling more rigorous evaluations of lecture segment retrieval.

Another relevant area involves benchmarks for visually rich document retrieval. The Jina-VDR (Visually Rich Document Retrieval) benchmark suite was introduced alongside Jina-Embeddings-v4, covering multimodal and multilingual document types (charts, tables, scanned pages, etc.) [43]. Improved performance on these tasks directly translates into more effective educational resource search, such as retrieving the precise slide containing a particular chart referenced in a query.

General-purpose cross-modal retrieval benchmarks also continue to play a role. Datasets such as MSR-VTT, MSVD, LSMDC, Flickr30k, and COCO provide open-domain evaluation tasks, while BEIR and MTEB offer text-focused retrieval suites. For example, the M^{3}HF model reported strong results on MSR-VTT, MSVD, DiDeMo, and ActivityNet Captions, demonstrating robustness across popular video–text retrieval benchmarks. More recently, the introduction of M-BEIR extended BEIR into multimodal domains, offering a broad evaluation suite that includes both text-only and cross-modal retrieval tasks. This extension has been adopted in the training of models such as MM-Embed, which benefits from exposure to multiple retrieval domains, some of which share similarities with educational materials.

In summary, the ecosystem of benchmarks has rapidly expanded to reflect educational scenarios, including lecture alignment datasets such as LPM, textbook-based reasoning datasets such as CK12-QA, and visually rich document retrieval datasets such as Jina-VDR. Nevertheless, important gaps remain. In particular, a large-scale benchmark for within-video moment retrieval in lecture recordings designed to locate the exact timestamp at which a given concept is explained has yet to be developed. As multimodal applications in education continue to grow, it is likely that the research community will address these gaps by creating more task-specific datasets.

3.2.2 Modality Fusion

A central challenge in multimodal retrieval is determining how to effectively fuse information from different modalities. There are two main ways to combine information from different sources: (1) **Early fusion**: combine them before searching. (2) **Late fusion**: search separately then combine results. Early fusion can pick up on interactions (like what sound is happening where in a video), but it's more complex.

Early fusion, or feature-level fusion, merges modalities prior to computing similarity. This approach yields joint representations that more richly encode cross-modal interactions. For example, the M2HF model for video retrieval combines CLIP-based visual features with audio features to generate audio-guided visual representations that highlight sound-producing regions within frames, and fuses motion features derived from optical flow with visual features to emphasize movement. In this way, a video segment can be embedded as a single vector capturing synchronized cues across modalities, which is particularly useful in lecture settings where spoken language, slide text, and gestures jointly define relevance.

Late fusion, or decision-level fusion, instead computes similarity scores separately for each modality and combines them at the output stage, typically through weighted score averaging or rank fusion. In an educational search context, a query may be matched independently against slide text and slide images, with the results subsequently aggregated. Late fusion offers robustness and modularity, as demonstrated by cases in which a keyword hit in the ASR transcript can elevate a candidate despite weak visual similarity. Many retrieval pipelines adopt late fusion as part of a two-stage process, where candidates are retrieved using one modality and reranked using a cross-modal model.

Hybrid fusion strategies have recently gained attention as they combine the strengths of both approaches. M2HF exemplifies such a design by performing early fusion within video streams (e.g., audio–visual and motion–visual) and then applying text–video retrieval at multiple levels, followed by late fusion of the resulting scores. This multi-level approach significantly improved recall across diverse video retrieval tasks and allowed M2HF to achieve strong performance on benchmarks such as MSR-VTT, MSVD, LSMDC, DiDeMo, and ActivityNet.

A further refinement is cross-attentional fusion, which departs from dual-encoder architectures that generate independent embeddings per modality. Instead, cross-encoder models compute direct attention between modalities and between query and item, enabling deep, token-level fusion. For example, the PolyViLT model ingests aligned slide text, figures, and speech transcripts and applies cross-attention with a multi-instance learning (MIL) loss to align slide figures with specific spoken segments [144]. By capturing these fine-grained many-to-many alignments such as a single spoken segment referring to one figure among many on a slide cross-attentional models outperform independent embedding methods for retrieval of figure–speech pairs. As discussed in Section 3.1, the computational cost of cross-attention makes such models impractical for large-scale first-stage retrieval, though they are highly effective in reranking top candidates. An important practical dimension of fusion is modality balancing, which ensures that no single modality dominates the relevance scoring process. Techniques such as modality-specific weighting and loss balancing are commonly used to achieve this. In M2HF, a multimodal balance loss was introduced to encourage the model to utilize audio and motion signals alongside visual cues. This balancing is essential in educational contexts for instance, in mathematics lectures, transcripts may not explicitly describe the content of diagrams, making the visual modality indispensable for accurate retrieval.

3.2.3 NotebookLM

NotebookLM is an AI-powered note-taking and research assistant developed by Google Labs[1], built on the Gemini family of models[2]. Designed to synthesize information from user-uploaded materials, the system emphasizes grounded responses, multimodal comprehension, and support for research workflows. While NotebookLM is marketed as a general-purpose productivity tool, its capabilities overlap with multimodal educational

[1] https://blog.google/technology/ai/notebooklm-google-ai/
[2] https://deepmind.google/technologies/gemini/

assistants such as our proposed system for HCMUT lectures.

NotebookLM supports diverse input formats, including PDF documents, Google Docs, Google Slides, Microsoft Word (.docx), PowerPoint presentations (.pptx), web URLs, YouTube links, EPUB files, CSV files, Google Sheets, markdown files, and images with OCR capabilities [3]. The system performs structured ingestion and can interpret textual content as well as visual elements such as tables, diagrams, and slide images. Recent updates introduce advanced multimodal reasoning, enabling comprehensive question answering based on charts, graphs, images, diagrams, video transcripts, and mathematical content, with improved support for computational and technical reasoning [4].

This multimodal capability aligns with academic materials that combine formulas, diagrams, and spoken explanations. However, NotebookLM treats uploaded materials as isolated documents rather than components of a structured pedagogical sequence, limiting its effectiveness for course-level organization.

NotebookLM employs RAG to ensure responses are grounded in uploaded documents rather than solely in parametric model memory [48]. When answering a query, the system retrieves relevant text segments and conditions generation on them, frequently accompanied by inline citations that reference specific sections of the source material. This reduces hallucinations and supports academic verification [5].

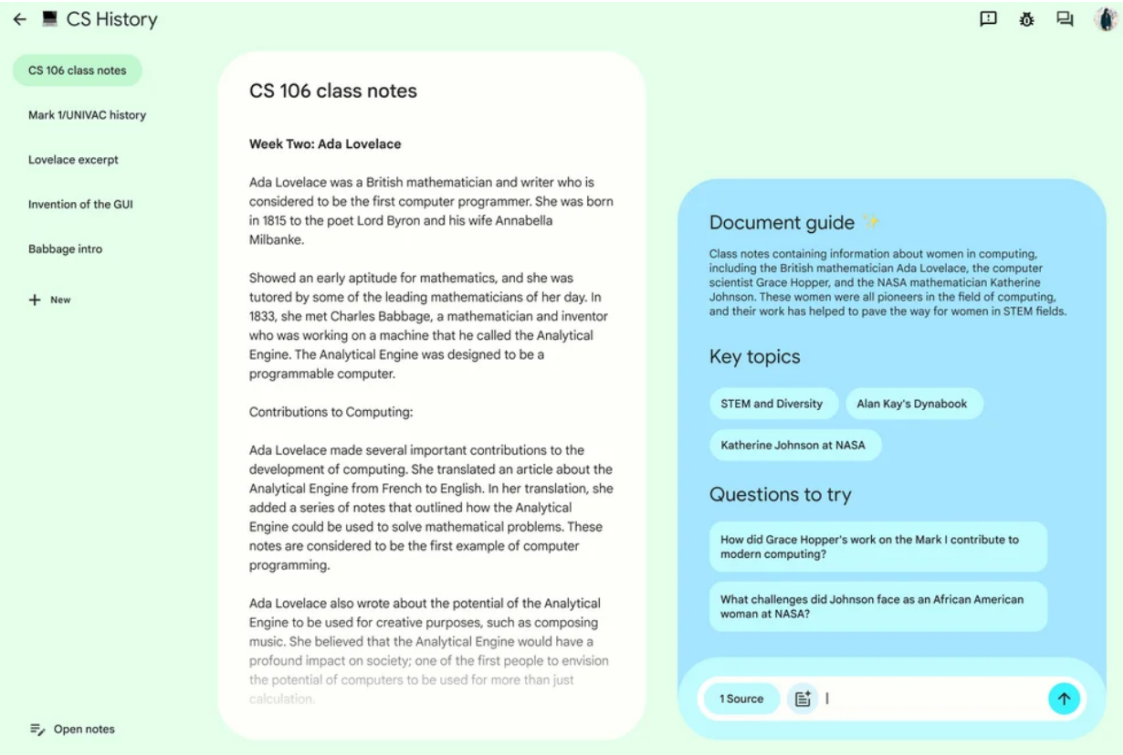

Figure 3.1: NotebookLM.

[3] https://notebooklm.google.com/get-started
[4] https://blog.google/technology/google-labs/notebooklm-updates/
[5] https://blog.google/technology/ai/notebooklm-citations/

Despite this grounding, complex analytical tasks such as mathematical reasoning, algorithm evaluation, or code execution may still rely on internal model reasoning and can produce incorrect interpretations without external validation layers. The NotebookLM interface is shown in Figure 3.1[6].

Beyond static summarization and Q&A, NotebookLM introduces several learning-focused features:

- Structured document summaries [7],
- AI-generated flashcards and quizzes for active recall [8],
- Podcast-style audio overviews generated from uploaded materials [9],
- Narrated visual summaries similar to lecture recap videos [10]

These features support multiple learning modalities (reading, listening, reviewing). However, generated study resources depend on document completeness and may not capture contextual elements conveyed verbally during lectures.

NotebookLM enforces privacy by not using uploaded materials for model training and storing sources privately unless shared [11]. This makes the tool suitable for proprietary lecture materials. The system is deployed on both web and mobile platforms, supporting flexible access [12].

In contrast, an institutional deployment such as our proposed system may require deeper integration with internal platforms, including lecture capture pipelines, LMS authentication, and course-specific permissions.

Although NotebookLM provides strong baseline functionality, several limitations motivate our specialized system:

[6] https://blog.google/technology/ai/notebooklm-google-ai/
[7] https://blog.google/technology/ai/notebooklm-summaries/
[8] https://blog.google/technology/ai/notebooklm-learning-tools/
[9] https://blog.google/technology/ai/notebooklm-audio-overviews/
[10] https://blog.google/technology/ai/notebooklm-audio-overviews/
[11] https://notebooklm.google.com/privacy
[12] https://blog.google/technology/apps/notebooklm-mobile/

Table 3.1: Comparison between NotebookLM and our proposed system.

NotebookLM Design Focus	**Opportunity for Our System**
Manual document uploads	Automatic ingestion from slides, recordings, LMS
Document-centric representation	Course-aware structuring by lecture/week/topic
General-purpose technical reasoning	Domain-specific pedagogical sequencing and curriculum alignment
Generic responses without instructor oversight	Instructor feedback loops and domain-tuned prompting
Individual learning workflows	Institutional integration and cohort-based learning analytics

Thus, while NotebookLM is an excellent general-purpose multimodal research assistant optimized for individual learners and researchers, it is not designed for institutional deployment with curriculum-aware features. Our system (BK-MInD) aims to bridge this gap by providing structured course-centric knowledge modeling, pedagogically sequenced learning paths, institutional integration with LMS platforms, and automated ingestion of lecture materials—enabling end-to-end lecture understanding tailored to the needs of educational institutions.

Chapter 4

Proposed Solution

This chapter describes our **Multimodal Retrieval-Augmented Generation (MRAG) system**—which combines large language models with a knowledge database to answer questions with cited sources. This design allows our system to process and retrieve information from diverse HCMUT educational materials while maintaining accuracy and transparency.

This chapter presents the proposed solution for BK-MInD, a Multimodal Retrieval-Augmented Generation (MRAG) system designed to process and retrieve information from diverse HCMUT educational materials. The chapter first justifies the selection of RAG as the core architectural paradigm over alternative approaches, then presents the rationale for choosing the three key technology components. The system architecture is described across four perspectives: a high-level overview, the data processing and dual-pathway embedding pipelines, the retrieval and generation flow, and the security design. Experimental evidence supporting retrieval architecture decisions is provided, and the chapter concludes with the technical requirements for deployment.

4.1 Justification for Retrieval-Augmented Generation

While LLMs possess remarkable generative capabilities, they suffer from inherent limitations such as knowledge cutoffs, hallucinations, and a lack of access to proprietary data. To address these issues, several methodologies exist. This section analyzes alternative approaches - Fine-Tuning, Semantic Search, Prompt Engineering, and Domain-Specific Pre-training - and justifies the selection of RAG as the superior architectural choice for this project.

4.1.1 Comparative Analysis of Alternatives

RAG vs. Fine-Tuning Fine-tuning involves retraining a pre-trained LLM on a specific dataset to adjust its internal parameters (weights). While this method tailors the model to a specific domain style, it was rejected for the following reasons:

- **Static Knowledge Base:** Fine-tuned models immediately become outdated upon training completion. Updating the knowledge requires a computationally expensive retraining process. In contrast, RAG allows for *dynamic updates* - simply adding a document to the database makes it immediately retrievable without modifying the model weights.
- **Hallucination Risk:** Fine-tuning increases the probability of the model adopting a specific tone, but it does not guarantee factual accuracy. The model can still hallucinate information effectively. RAG mitigates this by grounding the generation in retrieved evidence.
- **Explainability:** Fine-tuned models encode knowledge in their internal weights, making it impossible to trace where a specific answer came from. RAG, by contrast, retrieves the actual source documents, so you can verify the claim by reading the original text. This transparency is critical for academic integrity.

RAG vs. Prompt Engineering (Context Stuffing) Prompt engineering involves optimizing the textual input to guide the LLM's output, often by manually pasting context. While simple, it is insufficient for our scale:

- **Context Window Limitations:** Although context windows are growing (e.g., 128k tokens), it is infeasible to pass an entire corpus (e.g., 50,000 documents) into a single prompt. RAG acts as a filter, selecting only the most relevant snippets to fit within the model's constraints.
- **Performance Degradation:** Studies show that when too much context is provided, LLMs struggle to locate relevant information deep within the text—a phenomenon called 'Lost in the Middle.' This occurs because AI models can focus on all positions, but effectiveness decreases when important information is buried among thousands of tokens. RAG ensures the model focuses only on highly relevant, condensed information.

RAG vs. Semantic Search (Traditional Information Retrieval) Semantic search utilizes Vector Databases to retrieve documents based on meaning rather than keywords. However, it functions solely as a retrieval engine, not a generation engine.

- **Synthesis vs. Retrieval:** Semantic search returns a list of document links or snippets. The user is then forced to read and synthesize the answer manually. RAG automates this cognitive load by employing the LLM to read the retrieved documents and generate a coherent, natural language answer.
- **Completeness:** A user's query might require synthesizing information from three different documents. Semantic search provides the three documents; RAG provides the unified answer derived from them.

RAG vs. Domain-Specific Pre-training Pre-training involves training a model from scratch or continuing pre-training on a massive domain-specific corpus.

- **Computational Cost:** This approach is prohibitively expensive, requiring massive GPU clusters and weeks of training time.
- **Flexibility:** Like fine-tuning, a pre-trained model fails in dynamic environments where data changes daily. RAG decouples the *reasoning engine* (the LLM) from the *knowledge base* (the Vector Database), allowing us to swap or update the knowledge base instantly without touching the neural network.

4.1.2 Summary of Trade-offs

The following table summarizes the key architectural decisions that led to the adoption of RAG.

Table 4.1: Comparison of RAG against alternative LLM enhancement strategies.

Feature	RAG (Ours)	Fine-Tuning	Prompt Engineering	Semantic Search
Knowledge Update	*Real-time (Dynamic)*	Slow (Retraining)	Manual	Real-time
Hallucination Risk	Low (Grounded)	High	Medium	N/A (No Generation)
Context Capacity	Infinite (via Retrieval)	Limited by Training	Limited by Window	Infinite
Computational Cost	Low (Inference only)	High (Training)	Very Low	Low
Answer Synthesis	**Yes**	Yes	Yes	**No**

Conclusion: RAG is chosen because it offers the optimal balance of accuracy, cost-efficiency, and dynamic adaptability. It leverages the reasoning power of pre-trained LLMs while overcoming their memory limitations through external retrieval, making it the most suitable architecture for a document-based Question Answering system.

4.2 Technology Selection

Selecting the right technology stack is critical to the performance, maintainability, and scalability of the BK-MInD system. This section justifies the choice of React for the frontend, FastAPI for the backend, and Qdrant as the cloud vector database - three components that together provide an optimized foundation for a multimodal educational RAG application. Each selection is grounded in a direct comparison with primary alternatives, demonstrating superiority across the key criteria of performance, developer experience, and architectural fit.

4.2.1 Frontend: React

The frontend is the face of the application, defining the user's entire interactive experience with the multimodal output of the RAG system. The choice of a modern JavaScript framework is therefore essential for building a rich, responsive, and maintainable user interface.

After analyzing the leading frameworks, React is the recommended choice. This decision is based on four key pillars:

- **Core Architectural Suitability** for the application's specific technical demands.
- **Superior Performance** in managing frequent and complex UI updates.
- **Unmatched Ecosystem and Community Support**, which directly accelerates development and ensures long-term viability.
- **Proven Precedent** in large-scale, complex applications, particularly within the EdTech industry.

From now on, we detail the reasoning for this selection, including a direct comparison with the primary alternative, Vue.js.

Architecture Suits a Multimodal RAG System

The architectural principles of React are uniquely suited to the specific challenges of our application. The RAG system will return a heterogeneous mix of data: a streaming text answer, a reference to a page in a PDF, a timestamp in a lecture video, and a relevant diagram.

React's declarative nature, where developers describe what the UI should look like for a given state, allows the team to manage this complexity effectively. Instead of writing complex, imperative code to manually update the DOM for each data type, developers can create distinct, self-contained components for example `ChatWindow`, `SourceDocViewer`, `VideoPlayer`, `ImageViewer`.

The main application component then simply manages the application's state and declaratively renders the appropriate components based on the API response. This clear separation of concerns is critical for building a robust and maintainable multimodal user interface.

Core Advantages and Technical Rationale

Beyond its perfect fit for our specific data model, React provides foundational advantages in performance and development efficiency.

Component-Based Architecture This is the cornerstone of the selection. The user interface will be a composition of independent, reusable components (as listed above). This modularity simplifies the entire development lifecycle; each component can be built, tested, and maintained in isolation, which is critical for managing the complexity of the application.

High Performance with the Virtual DOM The RAG system will generate text responses and display various media types. React's use of a Virtual Document Object Model (DOM) is a key performance feature. It creates a lightweight in-memory representation of the UI, calculates the most efficient changes ("diffing"), and minimizes direct, costly manipulations of the actual browser DOM. This results in a smooth, responsive user experience, even as the UI updates frequently with new data from the backend.

Comparative Analysis: React vs. Vue.js

While both React and Vue.js utilize a virtual DOM and a component-based architecture, a direct comparison reveals React's significant advantages for a project of this scale.

Ecosystem, Community, and Talent Pool React boasts a significantly larger and more mature ecosystem than Vue.js. This translates into an unparalleled selection of third-party libraries, state management solutions (like Redux and Zustand), UI component kits (e.g., Material-UI, Ant Design), and development tools. For a feature-rich educational application that will require complex UI elements, such as interactive document viewers, data visualization, and chat interfaces, this vast ecosystem is a critical accelerator, reducing the need to build components from scratch.

Furthermore, industry surveys consistently demonstrate React's dominant position. Stack Overflow's 2023 survey showed that 42.87% of professional developers use React, compared to 17.64% for Vue.js[1]. This larger talent pool is a pragmatic and significant advantage for a university-based project like ours.

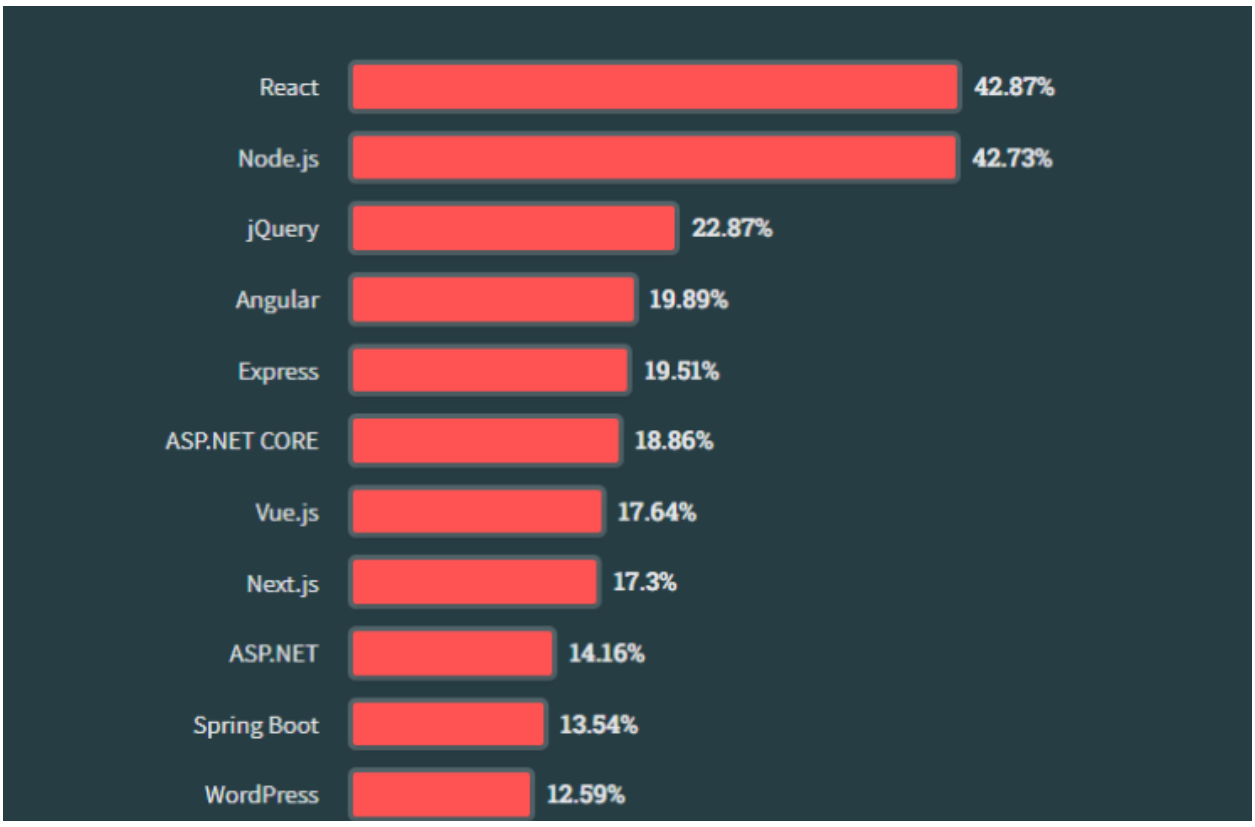

Figure 4.1: Most popular web frameworks and technologies among Professional Developers (Stack Overflow, 2023).

Scalability and Suitability for Complex Applications Both React and Vue.js are capable of building scalable applications. However, React's unopinionated library-based approach provides a crucial flexibility that is often favored for building large-scale, highly complex

[1] https://survey.stackoverflow.co/2023/#most-popular-technologies-webframe-prof

enterprise applications. This flexibility allows development teams to custom-build their architecture, selecting their own libraries for routing, state management, and data fetching, to create a stack that is perfectly optimized for unique and demanding business logic.

This preference is not merely theoretical; it is validated by the technology stacks of some of the world's largest and most complex web applications:

- **Meta (Facebook & Instagram)**: As the creator of React, Meta's platforms are its primary case study. The Facebook web app is a quintessential example of a complex, component-driven application, required to manage thousands of individual components, real-time data streaming, and billions of user interactions simultaneously.
- **Netflix**: The Netflix web application relies on React to deliver its high-performance, TV-like user interface. The team chose React to manage the complex state interactions of its UI, enabling them to rapidly iterate, A/B test features, and maintain performance on a wide variety of devices.
- **The New York Times**: To support modern, interactive journalism and a massive, dynamic content library, The New York Times rebuilt its website using React. This allows their development teams to create reusable components that can be shared across the organization, accelerating the development of new features and data visualizations.

While Vue is highly capable, React's architecture is battle-tested and proven to be the choice for global teams building applications where complexity and massive scale are the primary concerns.

4.2.2 Backend: FastAPI

The backend is the engine of our AI-driven educational application, responsible for orchestrating every interaction between the user, our data stores, and the core AI models. Our reliance on Python is a strategic imperative, granting us native access to the world's leading AI ecosystem, including libraries like PyTorch, Scikit-learn, and the Hugging Face ecosystem.

Within this ecosystem, the choice of web framework is critical. It is the layer that can either unleash the power of our AI models or become a crippling performance bottleneck. After a detailed analysis of the primary contenders, the mature, "batteries-included" Django framework and the modern, high-performance FastAPI, we have concluded that FastAPI is the unequivocally superior choice for this project.

This decision is based on three core pillars:

- **Exceptional Performance**: FastAPI's asynchronous architecture delivers an order-of-magnitude performance improvement over traditional frameworks, which is essential for the I/O-bound nature of a Retrieval-Augmented Generation (RAG) application.

- **Accelerated Developer Experience**: Its modern, type-hint-driven design, featuring automatic data validation and API documentation, drastically reduces boilerplate code and minimizes common errors.
- **Future-Proof Architectural Philosophy**: FastAPI is designed for a modern, microservices-first world, providing the flexibility and scalability required to prevent technical debt and support long-term growth.

Performance and Asynchronous Processing

A RAG application is inherently I/O-bound. The server spends most of its time not computing, but waiting: waiting for a query response from the Vector Database, waiting for user data from a relational database, and, most significantly, waiting for the Large Language Model (LLM) API to stream back its generated response.

In a traditional synchronous framework like Django, each of these waiting periods would block an entire worker thread. This means that if 10 users send a query simultaneously, the application needs at least 10 worker threads, each consuming significant memory. This model scales poorly, leading to a rapid increase in server costs and a degradation of user experience under load.

FastAPI solves this problem at its core. It is built on modern Python `async/await` syntax and the Asynchronous Server Gateway Interface (ASGI) standard. This allows a single process to handle thousands of concurrent connections. When a task starts waiting for an I/O operation (like an LLM call), the server doesn't block; it yields control and serves other requests, returning to the original task only when the data is ready.

Real-World Performance Benchmarks

This architectural difference is not merely theoretical; it is reflected in stark performance metrics from independent, industry-standard benchmarks. According to the highly respected TechEmpower benchmarks, which test hundreds of web frameworks in realistic scenarios, FastAPI's performance is in a different league.

In the Round 22 benchmarks (testing against physical hardware), the Uvicorn server running a basic FastAPI application handled **12,567 requests per second** in the Multiple Queries test. In contrast, Django on Gunicorn (a standard synchronous setup) handled **1,623 requests per second**.

The system implements a sophisticated dual-pipeline architecture designed to handle heterogeneous document formats and produce RAG-ready outputs for both text-based and image-based retrieval. The architecture is divided into two primary pipelines: the Data Processing Pipeline and the Embedding-Indexing Pipeline.

This means FastAPI can handle over 8 times the traffic with the same hardware for this common task[2]. For our I/O-bound application, this advantage is even more pronounced. The ability to efficiently manage concurrent waiting tasks means we can serve more users simultaneously with lower latency and significantly lower infrastructure costs. This raw performance is essential for delivering the responsive, real-time experience that users expect

[2] https://www.techempower.com/benchmarks/#section=data-r22&test=query

from an interactive AI assistant.

The Superior Developer Experience

Beyond raw performance, FastAPI is engineered with a laser focus on building APIs, which dramatically improves developer productivity and code quality.

- **Automatic Data Validation & Serialization**: FastAPI leverages the Pydantic library and Python type hints to provide automatic request validation. This is a game-changer. Developers simply define the expected data structure as a Python class, and FastAPI handles the rest: validating incoming data, returning clear JSON error messages if validation fails, serializing complex data types, and converting data seamlessly. In Django, achieving this requires the Django REST Framework (DRF), which involves writing separate Serializer classes, adding boilerplate and cognitive load.
- **Automatic Interactive API Documentation**: Out of the box, FastAPI generates interactive API documentation based on the OpenAPI standard. As soon as an endpoint is written, a fully functional documentation page is available at `/docs`, allowing frontend developers and consumers to test endpoints and understand the API without manual effort. This tight feedback loop is invaluable for rapid development.

Architectural Philosophy and Future-Proofing

The choice between FastAPI and Django represents a fundamental decision between a modern, microservices-first architecture and a traditional, monolithic one.

Django's philosophy is geared towards building monolithic, full-stack applications with server-rendered templates, an admin panel, and a built-in ORM. These are excellent features for certain projects, but they are less critical for our API-first, decoupled architecture.

FastAPI's lightweight and unopinionated nature makes it a natural fit for building independent, scalable microservices. This allows for a logical separation of concerns. We can develop, deploy, and scale the user authentication service independently from the RAG pipeline service. This modularity offers significant long-term advantages in scalability and maintainability, preventing the need for a costly and disruptive rewrite as the platform grows.

Head-to-Head Comparison Matrix

The following table provides a clear, at-a-glance comparison to justify the selection of FastAPI for our specific application context.

Table 4.2: FastAPI vs. Django Comparison for RAG Application.

Criterion	FastAPI	Django	Justification for RAG Application
Performance (Req/s)	Very High (∼12,500+)	Moderate (∼1,600)	RAG requires high concurrency for I/O-bound tasks. FastAPI's 8x+ performance is a decisive advantage.
Asynchronous Support	Native, built-in (`async/await`)	Supported via Channels (adds complexity)	Native async is simpler and more performant for orchestrating LLM and database calls.
API Documentation	Automatic (Swagger UI, ReDoc)	Requires external library (DRF)	Automatic documentation accelerates development, reduces friction, and simplifies API integration.
Data Validation	Built-in via Pydantic Type Hints	Built-in Forms / DRF Serializers	Pydantic is more modern, concise, and integrated, reducing boilerplate and potential errors.
Architectural Fit	Microservices, API-first	Monolithic, full-stack	A microservices approach is inherently more scalable and maintainable for our project's distinct components.
Learning Curve	Low	Moderate to High	A lower learning curve allows for faster onboarding of new developers and contributors.

4.2.3 Cloud Vector Database: Qdrant

The market for cloud Vector Databases is highly fragmented, with vendors utilizing fundamentally different programming languages, scaling architectures, and index structures. Evaluating their capacity to handle billion-scale multimodal workloads requires a granular architectural analysis of each platform.

Multi-Embedding Storage and MaxSim Execution

The capacity to efficiently ingest, store, and query the multi-embeddings generated by ColPali is a primary differentiator among Vector Databases. While platforms like **Vespa** and **Milvus** offer robust native solutions via tensor formalism and unified data types respectively, **Qdrant** uniquely excels through its decoupling of storage and indexing. Acknowledging that indexing heavy multi-vector payloads via HNSW wastes compute and RAM, Qdrant allows architects to explicitly disable graph construction for these payloads. They are utilized exclusively for in-memory, query-time MaxSim reranking against candidates retrieved via a fast single-vector pass. This architectural brilliance prevents the database from collapsing under graph maintenance during ingestion, ensuring high-speed data loading and significantly reduced RAM overhead.

Table 4.3: Multi-Vector Storage and MaxSim Implementation.

Platform	Native Storage	MaxSim Implementation	ColQwen Optimization
Vespa	Yes (Tensor Formalism)	Yes (Float & Hamming)	Phased retrieval, BQ
Milvus	Yes (Array of Structs)	Yes (MAX_SIM Metric)	Entity retrieval
Qdrant	**Yes (Nested Arrays)**	**Yes (Query-time compute)**	**Index bypassing (m=0)**
Weaviate	Yes (Multi-vector)	Yes	Jina AI integration
Pinecone	Partial (Metadata)	Yes (Cascading logic)	ConstBERT mapping

Supported Search Mechanisms

A robust multimodal database must support various search paradigms. All major platforms natively support dense, sparse (keyword), and hybrid search mechanisms. The critical differentiator is the underlying hybrid fusion algorithm. **Qdrant** stands out by supporting advanced fusion techniques such as Reciprocal Rank Fusion (RRF) and Distribution-Based Score Fusion (DBSF), intelligently balancing the unbounded scoring of BM25 against the normalized bounds of cosine similarity. Furthermore, Qdrant leads the industry in metadata filtering via its highly optimized payload indexing, allowing complex JSON-based logical conditions to intersect with vector search without compromising retrieval speed.

Table 4.4: Search Mechanism Comparison.

Mechanism	Vespa	Milvus	Qdrant	Weaviate	Pinecone
Keyword / BM25	Yes (Native)	Yes (Sparse)	Yes (Sparse)	Yes (Inverted)	No
Semantic Dense	Yes	Yes	Yes	Yes	Yes
Hybrid Fusion	RRF, Tensor	Built-in Rankers	**RRF, DBSF**	Alpha weight	Sparse/Dense
Metadata Filter	Yes	JSON Shred	**Payload Index**	Yes	High Speed

Scalability, Latency, and Economics

Transitioning to production demands rigorous evaluation of latency, scalability, and Total Cost of Ownership (TCO). Driven by its **Rust-based engine**, Qdrant demonstrates exceptional performance, maintaining 30-40 millisecond p95 latencies and sustaining 8,000 to 15,000 QPS, significantly outperforming Go-based alternatives in tail latency. Economically, Qdrant offers a generous perpetual free tier for prototyping and scales cost-effectively via custom sharding and the Raft consensus protocol, whereas enterprise solutions like Milvus and Vespa carry higher base infrastructure costs.

Table 4.5: Cloud Scalability, Latency, and Economic Efficiency.

Platform	p95 Latency	Throughput	Cost (10M Vec)	Architecture
Qdrant	**30-40ms**	**8k - 15k**	**$120 - $250**	**Custom Sharding, Raft**
Weaviate	50-70ms	3k - 8k	$150 - $300	Sharded Index, K8s
Milvus	50-80ms	10k - 20k	$300 - $600	Tiered Storage
Pinecone	Sub-100ms	Auto-scale	$200 - $400	Serverless Pods
Vespa	Sub-100ms	Extreme	Usage-based	Distributed Data Nodes

Developer Experience and Integration

Developer experience dictates integration speed. While Weaviate offers simplified APIs, it can lack granular configurability. Vespa provides deep tensor control but suffers from an exceptionally steep learning curve. **Qdrant** strikes the perfect balance. It features straightforward Docker deployments, highly readable Python SDKs, and deep integration with the `FastEmbed` library. This allows developers to generate dense and sparse embeddings locally without heavy external dependencies, vastly simplifying the creation of multi-vector RAG pipelines.

Final Selection and Strategic Recommendation

Table 4.6: Holistic Vector Database Comparison Summary.

Feature	Vespa	Milvus	Qdrant	Weaviate	Pinecone
Core Architecture	Hybrid C++ / Java	Go (K8s)	**Rust**	Go	SaaS
ColQwen Support	Native Tensor	Array of Structs	**Index Bypassing**	Multi-vector arrays	Metadata strings
Pricing Model	Usage-based	Tiered Cloud	**Free Tier + Cloud**	14-day + Cloud	Free Tier + Pods
Developer Curve	High Complexity	Moderate	**Low**	Low	Very Low

After an exhaustive analysis of architectural design, search mechanism breadth, economic efficiency, and specific multi-vector handling capabilities, we have selected **Qdrant** as the primary cloud Vector Database for the production environment.

Strategic Rationale for Selecting Qdrant: Qdrant serves as the optimal highly performant alternative for our multimodal RAG architecture. Its architectural decision to bypass HNSW indexing for heavy multi-vector payloads demonstrates a deep understanding of memory economics, ensuring high-speed ingestion and reduced RAM overhead. For our team, Qdrant's native support for late-interaction multi-vector embeddings makes ColQwen integration exceptionally streamlined, while its Rust-powered engine provides the necessary latency and throughput performance for the system.

Having selected the core technology stack (React frontend, FastAPI backend, and Qdrant vector database), the following section presents empirical benchmarks of embedding and retrieval configurations to validate these technology choices and inform the architectural design decisions.

4.3 Experiments on Embeddings and Retrieval Methods

4.3.1 Experimental Methodology

Dataset and Computing Environment To benchmark the performance of different retrieval architectures, we utilized the **Microsoft MS MARCO** (Machine Reading Comprehension) dataset[3]. This dataset is a standard benchmark for information retrieval tasks.

Due to computational constraints and to facilitate rapid iterative testing, we curated a representative subset of the data:

- **Corpus Size:** 50,000 documents (reduced from the original ≈1 million).
- **Test Queries:** 556 distinct evaluation queries.
- **Hardware Infrastructure:** All experiments were conducted on Google Colab utilizing a single **Tesla T4 GPU**.

Scoring Formulations Different pipelines utilize distinct mathematical formulations to calculate the relevance score $S(Q, D)$ between a query Q and a document D.

1. Lexical Scoring (BM25)

For the sparse retrieval baseline, we employed the BM25 algorithm. This probabilistic model estimates relevance based on term frequency (TF) and inverse document frequency (IDF), normalizing for document length

$$\text{Score}_{\text{BM25}}(Q, D) = \sum_{t \in Q} \text{IDF}(t) \cdot \frac{\text{TF}(t, D) \cdot (k_1 + 1)}{\text{TF}(t, D) + k_1 \cdot \left(1 - b + b \cdot \frac{|D|}{\text{avgdl}}\right)}$$

This method focuses on exact keyword matching and is highly effective for queries containing specific identifiers (e.g., proper nouns), though it lacks semantic understanding.

2. Semantic Scoring (Dense Retrieval)

For dense retrieval (e.g., MiniLM, BGE), we encode both query and document into fixed-size vectors ($\vec{V}_Q, \vec{V}_D$) and calculate the Cosine Similarity. This captures semantic meaning beyond exact keyword overlap

$$\text{Score}_{\text{Dense}}(Q, D) = \frac{\vec{V}_Q \cdot \vec{V}_D}{|\vec{V}_Q| \cdot |\vec{V}_D|}$$

The score range is $[-1, 1]$, representing the semantic proximity in the vector space.

3. Late Interaction Scoring (ColBERT/ColQwen)

[3] https://huggingface.co/datasets/microsoft/ms_marco

Unlike dense retrievers that compress a document into a single vector, the Late Interaction architecture (used in ColBERTv2) encodes queries and documents into bags of embedding vectors. The relevance score is the sum of the maximum similarity (MaxSim) for each query token against all document tokens

$$\text{Score}_{\text{MaxSim}}(Q, D) = \sum_{q \in \vec{Q}} \max_{d \in \vec{D}} (\vec{q} \cdot \vec{d})$$

This allows for fine-grained matching, ensuring that specific details in a document are not lost during vector compression.

Evaluation Metrics To quantitatively assess the pipelines, we adhere to two standard information retrieval metrics:

1. **Recall@k:** Measures the percentage of relevant documents successfully retrieved within the top k results.

$$\text{Recall@k} = \frac{|\{\text{Relevant Docs}\} \cap \{\text{Retrieved Top-k}\}|}{|\{\text{Total Relevant Docs}\}|}$$

2. **nDCG@k (Normalized Discounted Cumulative Gain):** This metric evaluates the *ranking quality*. It rewards algorithms that place relevant documents at the top of the list.

$$\text{nDCG@k} = \frac{DCG@k}{IDCG@k} \quad \text{where} \quad DCG@k = \sum_{i=1}^{k} \frac{2^{rel_i} - 1}{\log_2(i+1)}$$

4.3.2 Pipeline Analysis and Results

Important Note on Implementation Status: The following benchmark results represent experimental evaluations of different retrieval pipeline configurations. However, in the current Phase 2 production deployment, **reranking is globally disabled** by default (`SKIP_RERANKER=true`) for latency optimization. The reranker code is retained in the codebase for fast rollback if needed, but is not invoked during standard search operations. These experimental results are preserved for reference and historical context, but readers should be aware that the production system operates without the reranking stage.

We conducted a comprehensive evaluation of Lexical, Dense, Hybrid, and Reranking pipelines. The aggregate results are presented in Table 4.7.

Table 4.7: Comparative performance of Retrieval and Reranking pipelines on the MS MARCO subset (50k docs, 556 queries).

Pipeline Strategy	Model / Configuration	Latency	nDCG@3	Recall@3
Baseline Retrievers (No Rerank)				
Sparse	BM25 (rank-bm25)	3m	0.2821	0.3085
Dense	MiniLM-L6 (Raw)	∼1s	0.8658	0.9125
Dense	**BGE-Small-en-v1.5**	∼1s	**0.8855**	**0.9257**
Hybrid Retrievers (Reciprocal Rank Fusion)				
Hybrid	BM25 + MiniLM ($k = 60$)	∼1s	0.5786	0.6885
Hybrid	BM25 + BGE ($k = 60$)	∼1s	0.5830	0.6787
Reranking Pipelines (Retrieve + Cross-Encoder)				
Retrieve + Rerank	BM25 + BGE-Large	1m 43s	0.4469	0.4592
Retrieve + Rerank	MiniLM + BGE-Large	1m 43s	0.9196	0.9526
Retrieve + Rerank	**BGE-Small + BGE-Large**	1m 43s	**0.9248**	**0.9598**
Late Interaction				
Late Interaction	**ColBERTv2 (RAGatouille)**	6.5s	**0.9638**	**0.9412**

Question 1: Is Hybrid Retrieval superior to single-mode retrieval? Our experiments demonstrate that Hybrid Retrieval (combining BM25 and Dense vectors via Reciprocal Rank Fusion) offers a significant improvement over purely lexical methods, providing a balanced approach to robustness.

- **Limitations of BM25:** The raw BM25 baseline yielded a low nDCG@3 of 0.2821. This confirms that the MS MARCO dataset relies heavily on semantic understanding rather than exact keyword matches, causing BM25 to fail when vocabulary mismatches occur.
- **The Hybrid Improvement:** By fusing BM25 with Dense retrieval (Hybrid BGE), the nDCG@3 score improved from 0.2821 to 0.5830, representing a **2.06x improvement** over pure lexical search. However, this remains substantially below the pure dense baseline (0.8855), indicating that hybrid retrieval sacrifices some semantic accuracy in exchange for keyword robustness.
- **Significance:** While pure dense models performed exceptionally well on this specific dataset (0.8855), relying solely on dense retrieval in production carries the risk of "semantic hallucination" (retrieving conceptually similar but factually incorrect documents). Hybrid retrieval mitigates this by anchoring the semantic search with the keyword precision of BM25. It provides a "safety net" that ensures documents containing critical entity names are not discarded, making it a more robust strategy for real-world applications where query intent varies.

Important Note on Generalization: The experimental results presented above are based on the MS MARCO dataset, a generic domain benchmark for web search

contexts. HCMUT lecture materials differ significantly in their characteristics: they contain mathematical notation, domain-specific terminology, structured course hierarchies, and pedagogical content unlike general web documents. Future evaluation should include HCMUT-specific test queries and lecture content to validate whether the retrieved ranking methods (particularly hybrid vs. pure dense) perform equally well for educational domain-specific IR tasks. The generalization of these findings to educational content quality is documented in the Future Plan section (Section 7.4), where HCMUT-specific validation with student cohorts is planned.

Question 2: Does the addition of a Reranker significantly improve ranking quality? The results strongly support the hypothesis that a two-stage "Retrieve-then-Rerank" pipeline outperforms single-stage retrieval, though at a significant latency cost.

- **Architectural Advantage:** We employed Cross-Encoders (specifically **BAAI/bge-reranker-large**) as the second stage. Unlike bi-encoders, cross-encoders process the query and document simultaneously in the Transformer, capturing deep interaction signals.
- **Performance Gain:** Applying the BGE-Large reranker to the BGE-Small retriever candidates improved nDCG@3 from 0.8855 to **0.9248** and Recall@3 from 0.9257 to **0.9598**.
- **Latency Trade-off:** We observed that larger rerankers yield better quality but at significantly higher computational cost. The **bge-reranker-large** consistently outperformed the smaller **ms-marco-MiniLM-L-12** (e.g., 0.9248 vs 0.9107), but requires approximately **1 minute 43 seconds** per query (100x slower than retrieval-only baseline). This latency makes reranking unsuitable for interactive chat applications.
- **Production Decision:** While reranking demonstrates empirical accuracy gains, the BK-MInD Phase 2 production system **globally disables reranking** (`SKIP_RERANKER=true`) by default for interactive latency requirements. The reranker code is retained in the codebase for future evaluation and fast rollback if production requirements change.

Question 3: Is ColBERT (Late Interaction) the most effective approach? Our data indicates that ColBERTv2 provides the highest retrieval quality, surpassing even the best Retrieve-then-Rerank pipelines.

- **Superior Metrics:** ColBERTv2 achieved the highest **nDCG@3 of 0.9638**, outperforming the best reranking pipeline (0.9248).
- **Solving the Information Bottleneck:** Standard two-stage pipelines suffer from a bottleneck: if the first-stage retriever (e.g., BGE-Small) fails to find the relevant document in the top 30 candidates, the Reranker never sees it. ColBERT avoids this by using a "Late Interaction" mechanism. It calculates the MaxSim between query tokens and document tokens directly.

- **Granularity:** ColBERT matches at the token level using the MaxSim scoring mechanism described in the methodology section. This allows it to identify a relevant document even if only a small paragraph within it matches the query - a nuance often lost when compressing a whole document into a single vector (Dense Retrieval).
- **Operational Note:** It is worth noting that while ColBERT's retrieval time is efficient ($6.5s$ on GPU), the indexing process is resource-intensive. In our environment, indexing required ≈ 35 minutes on CPU[4] compared to mere seconds/minutes for standard dense models. Despite this setup cost, ColBERT represents the state-of-the-art for retrieval accuracy on this dataset.

4.4 System Architecture Design

This section presents the proposed architecture of the BK-MInD MRAG system, organized from a high-level system overview down to specific architectural components for retrieval and security. Implementation details, processing parameters, and deployment specifications are documented in Chapter 6 (Implementation).

[4]Due to FAISS-GPU limitations on the Colab instance, indexing fell back to CPU.

4.4.1 High-Level Architecture

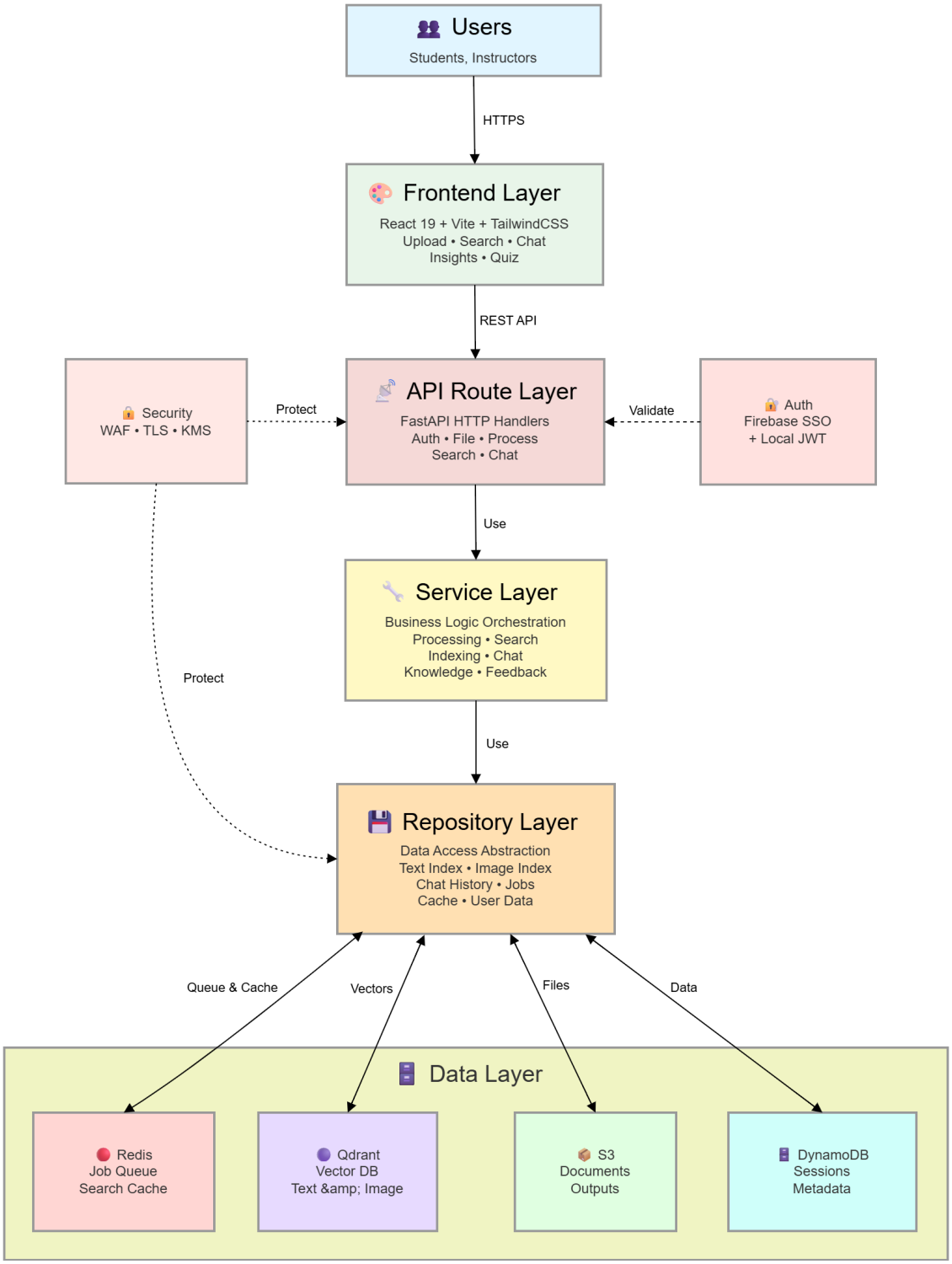

Figure 4.2: High-Level Architecture.

The BK-MInD system architecture follows a six-tier Clean Architecture pattern that separates concerns across distinct layers, enabling maintainability, testability, and independent scaling. This architecture is designed to achieve three strategic objectives: (1) *multimodal data ingestion* from diverse educational materials, (2) *asynchronous processing* to support concurrent user operations without bottlenecks, and (3) *production-grade security and scalability* on AWS infrastructure.

Six-Tier Clean Architecture: BK-MInD follows a six-tier Clean Architecture pattern: (1) Frontend Layer (React 19, Vite, TailwindCSS), (2) API Route Layer (FastAPI HTTP handlers), (3) Service Layer (business logic orchestrators), (4) Repository Layer (data access abstraction), (5) Data Layer (Redis, Qdrant, S3, DynamoDB), and (6) Cross-Cutting Concerns (authentication, security, LLM generation).

This layered design enables independent scaling, testability, and flexibility in deployment. Frontend communicates via HTTPS REST API. Services coordinate between repositories, which abstract storage details. Redis manages both async jobs (3 per-user, 200 global concurrency, 3600-second TTL) and search result caching. Authentication supports dual mechanisms: Firebase SSO and Local JWT (PBKDF2-SHA256). Security includes AWS WAF, CloudFront CDN, TLS encryption, and KMS-based data protection.

Asynchronous job management is critical to handle concurrent users: while one user's document undergoes the 7-stage pipeline (26–28 seconds), other users' queries and uploads process independently without blocking.

Data Processing and Indexing Layer: This layer implements the 7-stage conditional pipeline (detailed in Section 4.4.2) that transforms raw educational materials - PDFs, DOCX, PPTX, Excel spreadsheets, audio, and video - into RAG-ready artifacts. The pipeline follows three key design principles:

- **Conditional Routing:** Not all documents require all stages. A plain text document skips format-specific parsers, while an Excel spreadsheet activates the specialized Excel processing stage. This conditional execution minimizes wasted computation and optimizes resource utilization.
- **Format-Specific Optimization:** Each input format carries structural metadata (e.g., table cells in Excel, heading hierarchies in DOCX, page boundaries in PDF) that generic parsers cannot capture. Custom processors in Stages 3b, 3c, and 3d preserve this structure, yielding higher-quality chunks for subsequent retrieval.
- **Multimodal Fusion:** The pipeline consolidates heterogeneous outputs - structured text from documents, transcripts from audio/video, extracted images and diagrams - into a unified, indexed knowledge base that supports both text and visual retrieval.

Following processing, the dual-pathway indexing module produces both text indices (BM25 sparse index, FAISS dense embeddings, and Reciprocal Rank Fusion hybrid index) and visual indices (ColQwen 2.5 late-interaction multi-vector embeddings) stored in Qdrant Cloud. This dual representation enables the system to answer keyword-focused queries (e.g., "What is gradient descent?") via sparse retrieval while also handling semantic paraphrases

(e.g., "How do you minimize the loss function?") via dense embedding-based retrieval. The indexing process itself is orchestrated as an asynchronous job, allowing multiple documents to be indexed in parallel without blocking user interactions.

Cloud Infrastructure Layer: The cloud infrastructure leverages Amazon Web Services for scalable deployment and production-grade availability. The architecture spans multiple availability zones in the us-west-2 (Oregon) region to ensure high availability:

- **Content Delivery and Security Edge:** Amazon CloudFront provides a managed CDN that accelerates content delivery to geographically distributed users, while AWS WAF (Web Application Firewall) sits at the ingress tier to filter malicious traffic patterns including SQL injection, cross-site scripting (XSS), and distributed denial-of-service (DDoS) attacks. Security Group rules further restrict ingress to known sources and enforce egress policies.
- **Load Balancing and Routing:** The AWS Application Load Balancer (ALB) distributes incoming traffic across multiple backend instances, with path-based routing rules directing API calls (`/api/*`) to the backend service and static/dynamic content to the frontend service.
- **Compute Services:** Amazon ECS Fargate runs the containerized frontend and backend services in auto-scaling task groups, automatically adjusting the number of running tasks based on CPU, memory, and request count metrics. EC2 Auto Scaling groups host computationally intensive GPU inference services (Whisper ASR and ColQwen visual embeddings) on specialized hardware (ml.g4dn.xlarge instances with single NVIDIA T4 GPU).
- **Data Layer (Redis, Qdrant, S3, DynamoDB):** The system employs a polyglot persistence architecture with specialized storage for different data types:
  - **Redis**: Primary role is async job queue management with per-user (3 concurrent) and global (200 concurrent) concurrency limits, 3600-second TTL for automatic cleanup. Secondary role is search result caching for performance optimization, enabling rapid retrieval of frequently-accessed query results without re-querying Qdrant.
  - **Qdrant Cloud**: Vector database storing text embeddings (BM25, dense, hybrid RRF indices) and image embeddings (ColQwen multi-vector), enabling similarity-based retrieval without requiring full table scans.
  - **Amazon S3**: Cloud object storage serving as the canonical source for document artifacts and processed outputs (markdown, images, metadata), partitioned by user ID for multi-tenant isolation, with encryption via customer-managed KMS keys.
  - **Amazon DynamoDB**: Managed NoSQL database for chat history, user sessions, quiz results, feedback, and authentication metadata, with automatic cross-AZ replication for fault tolerance.

- **Load Balancing and Routing:** The AWS Application Load Balancer (ALB) distributes incoming traffic across multiple backend instances, with path-based routing rules directing API calls (`/api/*`) to the backend service and static/dynamic content to the frontend service. Auto-scaling groups automatically adjust task counts based on CPU, memory, and request count metrics.
- **Vector Storage:** Qdrant Cloud is provisioned as an externally managed vector database service, reducing operational overhead while providing reliable multi-tenant vector search with built-in security isolation. Text embeddings (BM25, dense FAISS, and hybrid RRF indices) and image embeddings (ColQwen multi-vector) are stored separately for independent scaling.

Together, these four layers enable BK-MInD to ingest diverse HCMUT lecture materials, build a multimodal knowledge index, and serve interactive educational features with high availability, low latency, and enterprise-grade security.

4.4.2 Pipeline Architecture

The system implements a dual-pipeline architecture designed to handle heterogeneous document formats and produce RAG-ready outputs for both text-based and image-based retrieval. The architecture is divided into two primary pipelines: the Data Processing Pipeline and the Embedding-Indexing Pipeline.

Data Processing Pipeline

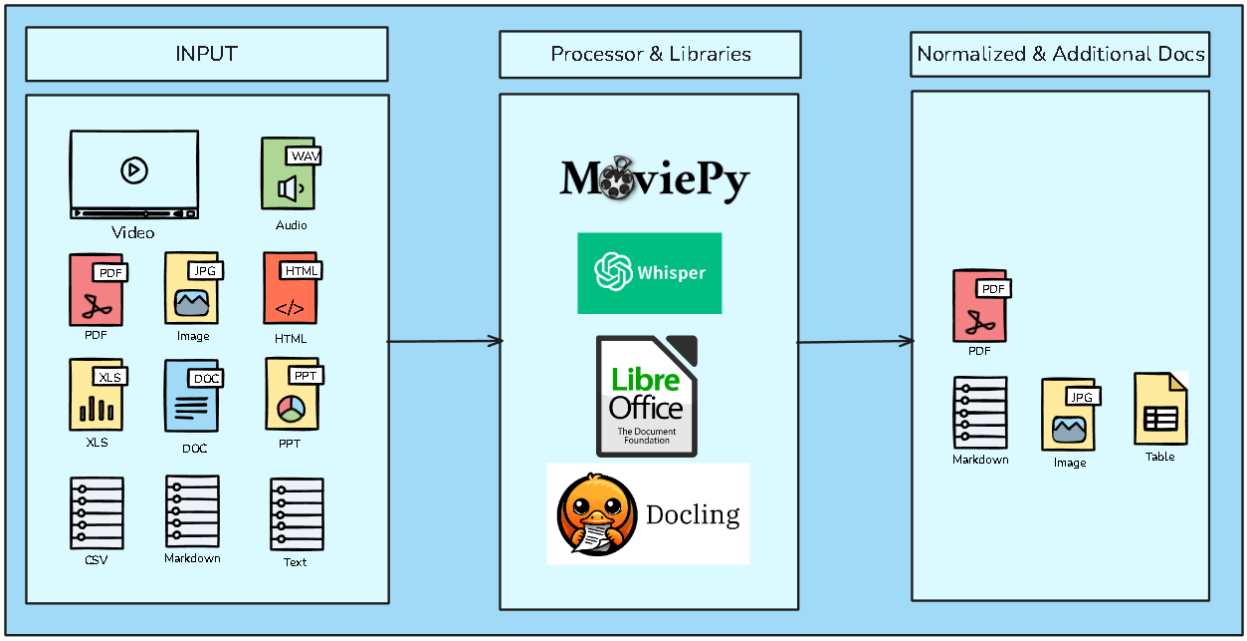

Figure 4.3: Data Processing Pipeline Normalization Stage.

The MRAG system implements a **7-stage conditional processing pipeline** that transforms diverse input formats into RAG-ready artifacts. As illustrated in Figure 4.3, the pipeline routes documents through format-specific processors only when needed, optimizing resource utilization while preserving semantic structure. The pipeline is designed around three key principles:

1. **Format-Specific Optimization**: Different document formats (Excel, DOCX, PDF) contain structural information (tables, heading hierarchies, page boundaries) that generic parsers miss. Custom parsers in Stages 3b, 3c, and 3d preserve this structure, yielding higher-quality chunks for retrieval.

2. **Conditional Routing**: Not all documents require all stages. By implementing conditional execution, the system processes only what is necessary. A plain text file skips Stages 3b/3c/3d entirely, while an Excel spreadsheet runs the specialized Stage 3b parser. This design minimizes wasted computation.

3. **Multimodal Fusion**: Stage 2 extracts multimedia content (audio transcripts, video frames) that Stage 3 cannot handle. Stage 4 consolidates all outputs - text from all sources, images from document pages and extracted diagrams, and transcripts from media - into a unified, retrievable index.

Table 4.8: 7-Stage Pipeline Summary.

Stage	Name	Role	Execution
1	Normalization	Format detection and routing	Always
2	Media Processing	Audio/video extraction and transcription	Conditional
3	Main Document Processing	Structural analysis via Docling	Always
3b	Excel Processing	Table-aware custom parsing	Conditional
3c	DOCX Processing	Heading-aware custom parsing	Conditional
3d	PDF Processing	Born-digital document parsing	Conditional
4	Consolidation	Merge all outputs, prepare for indexing	Always

For detailed specifications of each stage including routing tables, processing parameters, and output structures, refer to Chapter 6 (Implementation).

Embedding-Indexing Pipeline

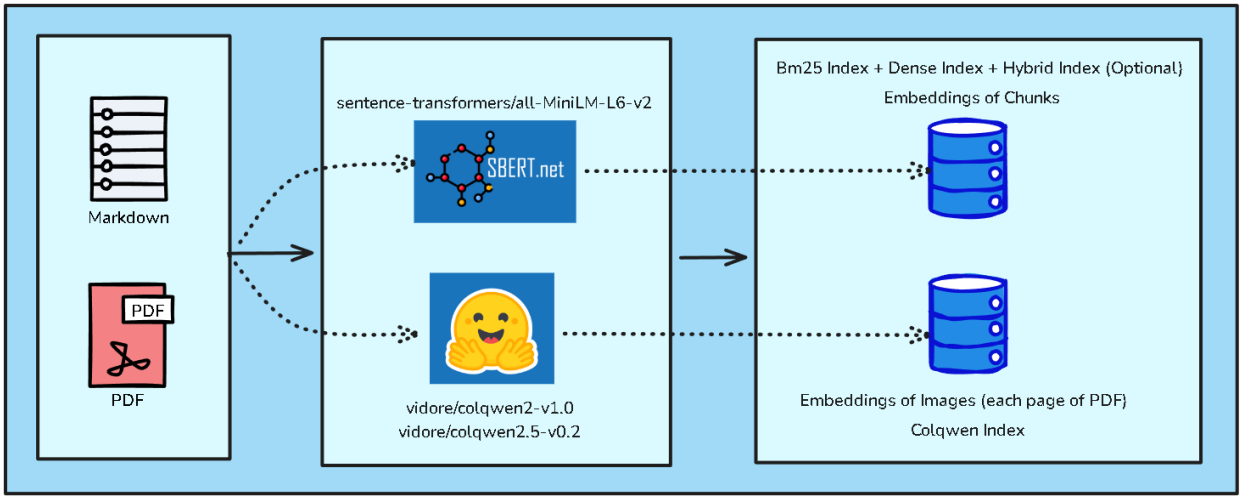

Figure 4.4: Dual-Pathway Embedding and Indexing Architecture.

Following data processing, the system employs a dual-pathway embedding architecture to enable multimodal retrieval. As illustrated in Figure 4.4, this architecture allows simultaneous retrieval of granular text segments and holistic document pages.

Text Embedding Pathway: Structured Markdown files are segmented using `RecursiveCharacterTextSplitter` (1024-character chunks, 200-character overlap) and encoded into 384-dimensional vectors by `all-MiniLM-L6-v2`. The system maintains three parallel indices: a BM25 inverted index for keyword retrieval, a FAISS dense index for semantic search, and a hybrid index combining both via Reciprocal Rank Fusion (RRF).

Visual Embedding Pathway: PDF pages are rasterized at 150 DPI using a "1 Page = 1 Image" strategy. Each page image is processed by ColQwen2, a Vision-Language Model using the Late Interaction paradigm, generating multi-vector patch-level embeddings stored in the ColQwen visual index. This enables diagram-aware and layout-aware retrieval without relying on OCR.

Pipeline Output Summary

The complete pipeline produces the following RAG-ready artifacts:

- **Text-Based Retrieval**: Markdown files with three indices (BM25, Dense, Hybrid) enabling keyword and semantic search over chunked content.
- **Image-Based Retrieval**: Rasterized PDF pages with ColQwen visual embeddings enabling layout-aware and diagram-sensitive search.
- **Structured Data**: Extracted tables (CSV + Markdown), isolated images, and comprehensive metadata (JSON) for provenance tracking.
- **Multimedia Transcripts**: Time-stamped text transcriptions from video/audio in multiple formats (TXT, JSON, SRT, VTT).

4.4.3 Retrieval and Generation Flow

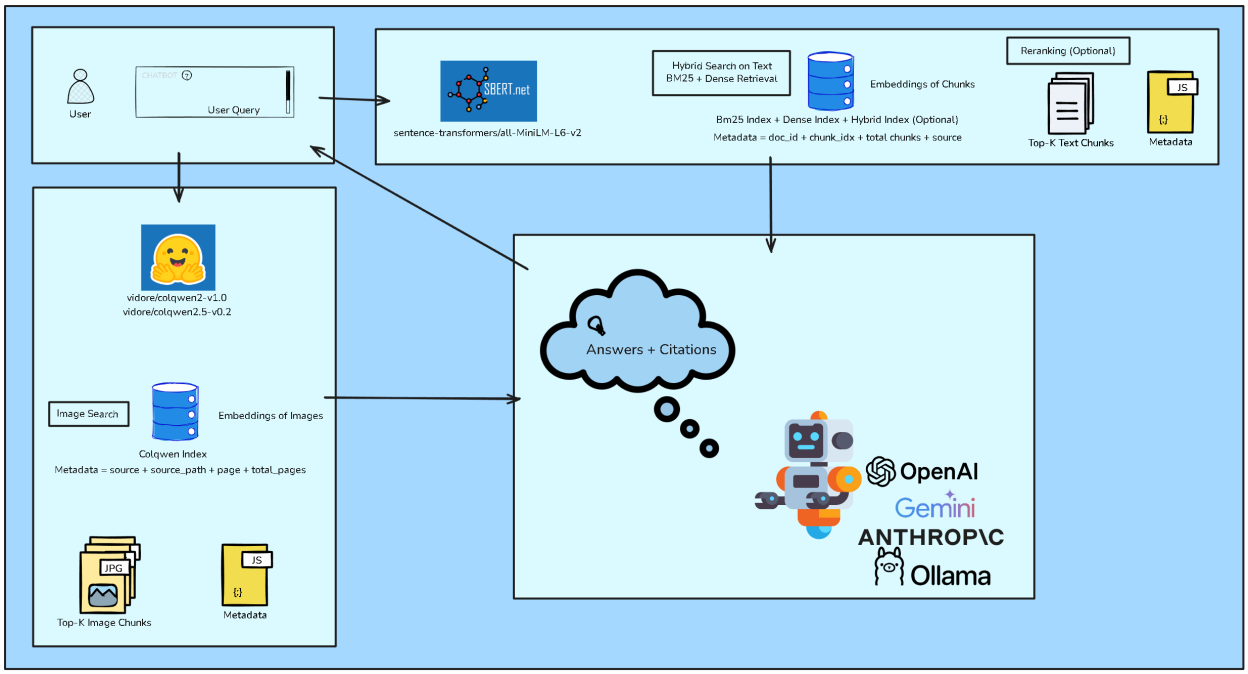

Figure 4.5: Overall MRAG System Architecture: Retrieval and Generation Flow.

When a user submits a query to the system, it triggers a parallel retrieval pipeline designed to gather both textual and visual evidence in support of answer generation. The complete flow is illustrated in Figure 4.5.

Query Processing Pipeline

The query processing flow follows this sequence:

1. **Query Input**: User submits a natural language question via the chat interface.
2. **Parallel Encoding**: The query is simultaneously encoded by two engines:
   - **Text Encoder**: Encodes the query into a 384-dimensional dense vector using the same model as the corpus (all-MiniLM-L6-v2).
   - **Visual Encoder**: Encodes the query using ColQwen's query encoder to prepare for late-interaction visual matching.
3. **Parallel Retrieval**: Both streams search their respective indices independently.
4. **Context Aggregation**: Top-K results from each stream are merged into a unified context.
5. **Just-in-Time Rendering**: Retrieved PDF pages are rendered to high-resolution images on-demand.

6. **LLM Generation**: The aggregated context is passed to the LLM for answer synthesis.

Text Retrieval Options

The system offers three complementary text retrieval strategies, each addressing different query characteristics:

1. Sparse (BM25) Retrieval: BM25 is a probabilistic ranking function based on term frequency and inverse document frequency. It excels when queries contain specific keywords, entity names, or exact phrases. Strengths include robust exact term matching; weakness is failure on semantic paraphrases.

2. Dense (Embedding-Based) Retrieval: Dense retrieval encodes both queries and documents as continuous vectors in a high-dimensional space. It captures semantic meaning and handles paraphrased queries effectively, though it is susceptible to "semantic drift" - retrieving documents that are topically similar but factually incorrect.

3. Hybrid Retrieval (BM25 + Dense + RRF): The system implements **Reciprocal Rank Fusion (RRF)**, a fusion algorithm that combines BM25 and dense retrieval results. Reciprocal Rank Fusion (RRF) combines BM25 and dense results by assigning a score to each result based on its rank position. Results ranked higher (lower rank number) get higher scores. The constant k (60) is a smoothing factor that prevents low-ranked results from getting zero score and allows high-ranked results from both methods to be boosted. For each document, RRF calculates:

$$\text{RRF}(d) = \frac{1}{k + \text{rank\_BM25}(d)} + \frac{1}{k + \text{rank\_Dense}(d)}$$

where k is a constant (typically 60) that smooths the contribution of both rankers. Hybrid retrieval is the **default strategy** for all queries, balancing keyword precision with semantic understanding.

Visual Retrieval

In parallel with text retrieval, the system searches for relevant document pages using visual embeddings. ColQwen's late-interaction architecture allows the system to match query concepts against the visual structure of pages, retrieving diagrams, charts, and tables that may not be adequately represented in extracted text.

- **Query Encoding**: The query is encoded by ColQwen's text encoder into a set of multi-vector embeddings.
- **Page Matching**: These vectors are matched against stored page embeddings using MaxSim scoring, identifying pages with visually relevant content.
- **Just-in-Time Rendering**: Retrieved pages are rendered to JPEG images and passed to the multimodal LLM, enabling it to "see" charts, graphs, and layout-dependent information.

Reranking and Design Decision

Current Implementation: Reranking is **globally disabled** in production (`skip_reranker=true`) for latency optimization. Cross-encoder reranking adds 1-2 minutes of latency, making it unsuitable for interactive chat applications.

Historical Context: During Phase 1 evaluation, cross-encoder reranking (using BGE-Large) improved nDCG@3 from 0.8855 (dense alone) to 0.9248, demonstrating significant accuracy gains. However, this comes at the cost of ∼100x latency increase.

Current Approach: Hybrid retrieval via RRF provides a strong balance of accuracy and latency without reranking. The reranker code is retained in the codebase for fast rollback if future evaluation shows reranking benefits justify the latency trade-off.

Context Aggregation and Generation

After retrieval, the system aggregates top-K text chunks and top-K visual pages into a unified context window:

1. **Text Context**: Top-K text chunks from hybrid retrieval are concatenated with their metadata (source, page number, heading).
2. **Visual Context**: Top-K pages are rendered to high-resolution JPEG images.
3. **Context Size**: The total context is sized to fit within the LLM's context window (typically 16k-32k tokens, depending on the model).
4. **LLM Selection**: The system supports multiple LLM providers (OpenAI GPT-4o, Anthropic Claude, Google Gemini, or local models via Ollama).

The LLM reads this aggregated context and generates a natural language answer grounded in the retrieved evidence. Importantly, the LLM includes precise citations - linking every factual claim to the specific chunk or page from which it was derived. This ensures traceability and enables users to verify the system's reasoning.

4.4.4 Security Architecture

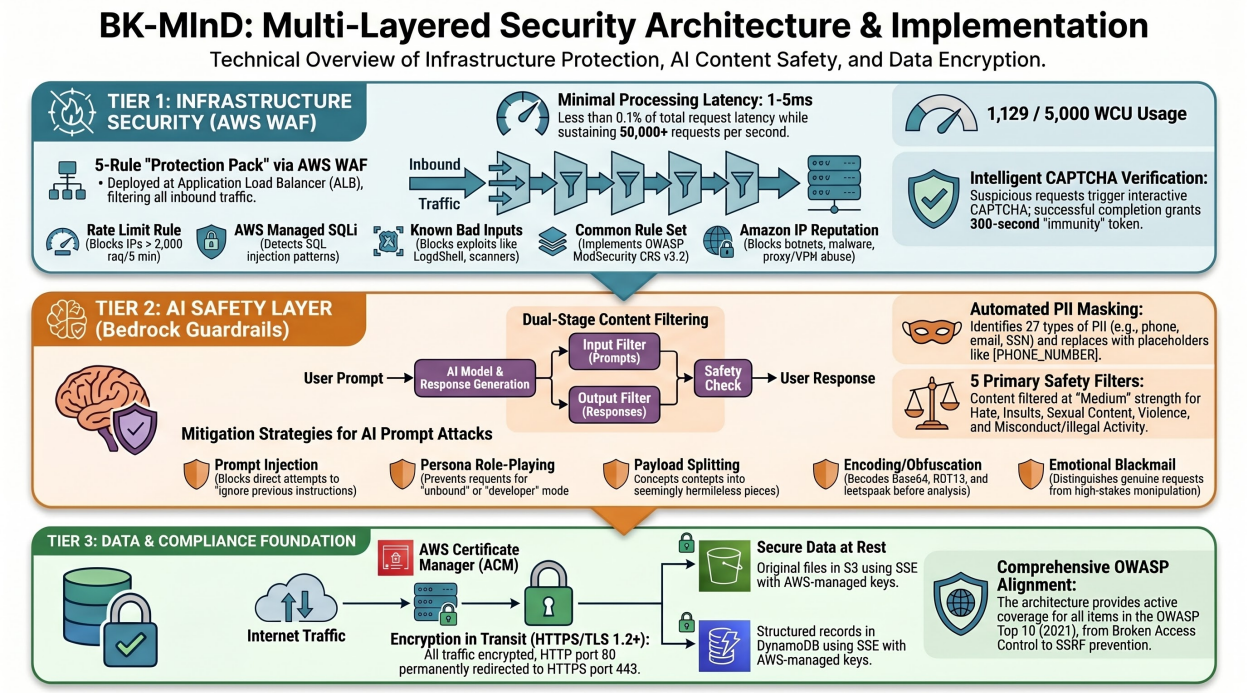

Figure 4.6: BK-MInD Three-Tier Security Architecture.

The BK-MInD system implements a comprehensive security architecture operating across four strategic dimensions: network perimeter defense, authentication and authorization, data protection, and audit and compliance. This layered approach ensures that the system defends against threats at multiple levels - from network-based attacks to unauthorized data access to malicious API abuse. Implementation details and specific configuration parameters for each component are documented in Chapter 6 (Implementation).

Network Perimeter Security: Edge Filtering (CloudFront + AWS WAF): All incoming HTTPS traffic is routed through Amazon CloudFront, a managed Content Delivery Network that provides geographic distribution and edge caching. Upstream of CloudFront sits AWS Web Application Firewall (WAF), configured with a comprehensive rule pack protecting against common OWASP Top 10 attack vectors:

- **SQL Injection:** Detects and blocks SQL metacharacters in query parameters.
- **Cross-Site Scripting (XSS):** Filters script tags and JavaScript event handlers in request bodies.
- **HTTP Flood / Rate Limiting:** Enforces per-IP request rate limits to prevent brute-force and denial-of-service attacks.
- **Geographic Access Control:** Restricts access by IP geolocation (e.g., allowing only HCMUT-region traffic if desired).

- **Bot Protection:** Identifies and blocks automated crawler and bot traffic via IP reputation scores.

Load Balancer Security: The AWS Application Load Balancer (ALB) sits behind CloudFront and is configured to accept traffic only from CloudFront's managed IP prefix list. This architecture prevents attackers from bypassing WAF protections by sending requests directly to the load balancer IP address. Security groups applied to the ALB further restrict ingress to only HTTPS port 443 from CloudFront and egress to only the required backend subnets.

Authentication and Authorization: User Authentication (Dual Mechanism): The system implements two complementary authentication methods, both of which can operate simultaneously:

1. **Firebase SSO (Optional)**: Users can authenticate using Google Sign-In via Firebase Admin SDK. Firebase ID tokens are validated server-side, providing secure third-party authentication without storing passwords.

2. **Local Authentication**: For users without Google accounts or in deployments without Firebase, the system provides built-in local authentication using PBKDF2-SHA256 password hashing (100,000 iterations). Custom JWT-style tokens are issued with claims (user ID, email, expiration) and signed with HMAC-SHA256. Default token TTL is 30 days, configurable via `LOCAL_AUTH_TOKEN_TTL_SECONDS`.

Both authentication methods issue Bearer tokens that must be included in the `Authorization: Bearer <token>` header for all API requests. The `get_current_user()` dependency handler attempts local token validation first, then falls back to Firebase validation, enabling seamless support for both authentication sources. Token validation is performed at the API gateway level before requests are routed to backend services.

Session Management: For chat and interactive features requiring stateful sessions, the backend maintains session records in Amazon DynamoDB with the user's identity bound to each session. Session tokens are short-lived (typically 1 hour) and automatically revoked on logout, preventing session hijacking and replay attacks.

Role-Based Access Control (RBAC): The system enforces role-based permissions, with three primary roles:

- **Student:** Read-only access to indexed documents and generation queries.
- **Instructor:** Read-write access to upload and manage course materials.
- **Administrator:** Full system access including user management and audit logs.

Access control is enforced at the API endpoint level, with each route decorated with `@require_role(...)` middleware that verifies the user's token contains the required role before allowing execution.

Data Protection and Encryption: Encryption In-Transit: All client-to-server and server-to-server communication uses TLS 1.2 or higher. AWS Certificate Manager (ACM) provisions and auto-renews HTTPS certificates for the public domain. Internally, microservice-to-microservice communication within the VPC also uses TLS, enforced through service mesh policies or VPC endpoint configurations, preventing man-in-the-middle attacks on internal traffic.

Encryption At-Rest:

- **S3 Object Storage:** All document artifacts, processed outputs, and metadata stored in S3 are encrypted using server-side encryption with AWS KMS customer-managed keys (SSE-KMS). S3 bucket policies enforce encryption requirements, blocking any attempt to upload unencrypted objects.
- **DynamoDB:** All tables are encrypted at rest using AWS KMS, protecting user sessions, chat histories, and job state data from unauthorized access in the event of storage device seizure or theft.
- **Vector Database:** Qdrant Cloud provisioning includes encryption configuration; data is encrypted both at rest and in transit to the managed service.

Key Management: Encryption keys are managed exclusively through AWS Key Management Service (KMS). No keys are stored in application code or environment variables; instead, the application assumes an IAM role that grants KMS permissions scoped to specific keys and operations. Key rotation policies are configured to automatically rotate keys annually without requiring reencryption of existing data.

Data Isolation and Access Control:

Multi-Tenant Isolation: BK-MInD is designed as a multi-tenant system where each user has isolated access to their own documents and query history. Isolation is enforced through:

- **S3 Prefix Isolation:** Document objects are stored under user-specific paths (e.g., `s3://bucket/users/{user_id}/documents/`). IAM policies enforce that users can only read and write within their own prefix, preventing cross-user document access.
- **DynamoDB Partition Isolation:** User sessions and chat histories are partitioned by `user_id`, and query filters ensure that users can only retrieve their own records.
- **Vector Database Isolation:** Qdrant Cloud supports payload-based multi-tenancy; vectors are tagged with user metadata and retrieval queries include a user_id filter, preventing users from searching across another user's indexed documents.

Content Safety and Abuse Prevention:

Input Validation: All user inputs (queries, document metadata, settings) undergo validation against a Pydantic schema before being processed. Type checking, length limits, and character set restrictions prevent injection attacks and resource exhaustion.

Output Safety (Bedrock Guardrails): The LLM generation pipeline integrates Amazon Bedrock Guardrails as a content safety filter applied to both user inputs and model outputs:

- **Topic Filtering:** Enforces that the assistant remains focused on educational content, rejecting off-topic queries about unrelated subjects.
- **PII Redaction:** Automatically detects and masks personally identifiable information (email addresses, phone numbers, social security numbers) in both user queries and generated responses.
- **Toxic Content Detection:** Blocks responses containing profanity, hate speech, or violent content, ensuring the assistant remains appropriate for academic settings.

Audit and Compliance:

API Audit Logging: AWS CloudTrail logs all API calls to AWS services, including authentication events, resource modifications, and data access patterns. These logs are forwarded to a dedicated S3 audit bucket with object lock enabled, preventing tampering with audit records.

Application Logging: The backend logs all user actions (document uploads, queries, generation requests) to CloudWatch Logs, including timestamps, user identities, and request parameters. Logs are retained for 90 days and automatically archived to S3 for long-term compliance auditing.

Data Access Logging: S3 and DynamoDB access logging is enabled, recording which users accessed which documents and when. This enables detection of unauthorized access patterns and supports compliance investigations.

The experimental findings above directly inform the hardware and software specifications required to reliably operate the BK-MInD system at production scale. Based on the observed latency and resource consumption patterns during evaluation, the following section specifies the minimum and recommended computational requirements.

4.5 Technical Requirements

4.5.1 Hardware Requirements

- **GPU**: NVIDIA GPU with at least 4GB of VRAM, but when using 8GB VRAM, we recommend (RTX 4060, RTX 4070, or equivalent) for efficient embedding generation and inference. For larger models or batch processing, 16GB+ VRAM is recommended.
- **CPU**: Multi-core processor (8+ cores) to support parallel processing of documents and concurrent API requests.
- **Memory (RAM)**: Minimum 16GB of system RAM for running the entire pipeline, including model caching and document processing.

4.5.2 Software Dependencies

- **Core Frameworks**:
  - **Frontend**: React 19 with Tailwind 4.1 for high-performance UI components.
  - **Backend**: FastAPI for the asynchronous REST API framework.
  - **Database**: Qdrant Cloud for vector storage with payload isolation; Amazon DynamoDB for chat persistence.
- **AI Processing Engines**:
  - **Docling**: Document understanding framework for structural analysis.
  - **Whisper Large-v3**: High-fidelity ASR for multi-language transcription.
  - **SmolVLM-256M**: Optional lightweight Vision-Language Model for page layout understanding in Docling. When enabled in configuration, SmolVLM-256M (256M parameters) provides an OCR-alternative approach for document parsing with lower latency than larger VLMs, trading some extraction quality for faster processing speed. Particularly useful for rapid prototyping and latency-constrained deployments.
  - **ColQwen2.5-v1.0**: Late-interaction visual retrieval engine.
- **Model Weights**:
  - `all-MiniLM-L6-v2` (Standard) or `BGE-M3` (Advanced) for text embeddings.
  - `bge-reranker-large` for optional reranking (currently globally disabled for latency optimization).
  - LLM backbone: Supporting Claude 3.5 Sonnet (via Bedrock) or GPT-4o.

4.5.3 System Architecture Constraints

Advantages:

- **Data Security and Privacy**: Document artifacts and processing pipelines are deployed on AWS infrastructure with multi-layered encryption. Document storage uses Amazon S3 with server-side encryption (SSE) enabled by default and access control via IAM policies. All data in transit is protected by TLS 1.3/HTTPS enforced through AWS Certificate Manager (ACM)-issued certificates. Multi-tenant isolation is enforced through S3 key prefixing by user ID and DynamoDB partition isolation, preventing cross-user data leakage and ensuring compliance with HCMUT data governance policies.
- **Concurrency**: The system is designed to handle multiple concurrent API requests using FastAPI's asynchronous architecture, supporting up to N simultaneous users (where N is bounded by available memory and GPU scheduling).

- **Scalability**: The modular pipeline design allows for independent scaling of retrieval and generation components. Additional indexing nodes can be added for large document corpora without modifying the core system.
- **Portability**: The entire system can be containerized using Docker, enabling seamless deployment across different environments (development, staging, production).

Limitations:

- **Single-GPU Dependency**: The current architecture does not support distributed GPU processing. Visual embedding generation (ColQwen) and inference are constrained to a single GPU, creating a bottleneck for large-scale batch processing.
- **Index Rebuilding Overhead**: Document updates require full index regeneration. The system does not support incremental indexing, meaning that adding or modifying a single document necessitates reprocessing the entire corpus for that index type.
- **Memory Scalability**: Both BM25 and FAISS indices are loaded entirely into RAM during retrieval. For corpora exceeding 1 million documents, memory consumption can surpass 32GB, limiting deployment on consumer-grade hardware.
- **Storage Redundancy**: The dual-pathway architecture stores multiple representations of the same document (original PDF, rasterized images, Markdown, embeddings), resulting in storage overhead of approximately 5× the original document size.
- **Processing Latency**: Document ingestion is synchronous and blocking. A single complex PDF (e.g., 100 pages with tables and images) can take 2-5 minutes to process through the full pipeline, during which no other documents can be processed concurrently.
- **Model Size Constraints**: The system's embedding models are fixed at deployment time. Switching to larger models (e.g., upgrading from MiniLM to BGE-Large for embedding) requires reindexing the entire corpus. Reranking is globally disabled by default (`skip_reranker=true`) for latency optimization, though the code is retained for fast rollback.
- **Limited Multi-Tenancy**: The file-system-based storage architecture lacks built-in isolation mechanisms. Deploying the system for multiple independent clients requires separate instances or manual namespace management.

Chapter 5

Implementation

This chapter details the practical implementation of the BK-MInD system architecture. The chapter is organized as follows: Section 5.1 describes the evolution from the Phase 1 prototype to the Phase 2 production system. Section 5.2 covers the complete Data Processing Pipeline. Subsequent sections address chunking and embedding strategies, indexing and storage, asynchronous job handling, content safety, database schemas, core API services, system deployment, and security implementation details.

5.1 Implementation Evolution

The development of the Unified MRAG system was executed in two distinct phases, representing a transition from a research-oriented baseline to a robust, production-ready platform. This evolution was driven by the need for scalability, and high availability.

Phase 1 focused on the validation of the core retrieval-augmented generation logic, utilizing local resources and monolithic processing scripts. Phase 2 involved a comprehensive refactoring of the system into a cloud-native architecture, offloading compute-heavy tasks and leveraging managed services for data persistence and orchestration.

5.1.1 Phase 1: Foundation and Feasibility

Phase 1 was characterized by the development of a **Proof-of-Concept (PoC)** baseline. The primary objective was to validate the technical feasibility of the multimodal retrieval logic and ensure that the integration of text and visual embedding models yielded coherent results. During this stage, the system was implemented as a monolithic Python environment, relying on local filesystem storage for documents and a local instance of a Vector Database for retrieval. While functional, this architecture suffered from significant limitations:

- **Resource Contention**: Running heavy embedding models (like ColQwen) alongside the API server on a single machine led to significant latency and memory exhaustion.

- **Lack of Persistence**: User sessions and processing states were transient, meaning that any system restart resulted in the loss of progress and required complete re-indexing of documents.
- **Monolithic Bottlenecks**: The sequential nature of the data processing pipeline meant that large document uploads blocked all other system operations, preventing concurrent usage.

5.1.2 Phase 2: Scalability and Productionalization

Phase 2 involved a fundamental **architectural refactoring**, transitioning the system into a production-grade, cloud-native application. The focus shifted from functional correctness to operational excellence, addressing the bottlenecks identified during Phase 1. Key enhancements included:

- **Decoupled Infrastructure**: The core compute logic was separated from the storage and inference layers. By leveraging **Amazon S3** for object storage and **Qdrant Cloud** for vector search, the system achieved true multi-tenancy and horizontal scalability.
- **Stateful Session Management**: The integration of **Amazon DynamoDB** allowed for persistent, low-latency tracking of user chat histories and processing statuses, enabling a seamless resume-from-anywhere experience.
- **Optimized Inference Offloading**: To handle the massive compute requirements of late-interaction models, Phase 2 offloaded visual embedding generation to **AWS SageMaker** using GPU-accelerated inference endpoints (ml.g4dn.xlarge instances with single NVIDIA T4 GPU, 4 vCPU, 16GB RAM). This configuration provides approximately 250-300 embeddings/second throughput for ColQwen inference, ensuring that the main application backend remains responsive even during high-intensity batch processing of document pages. SageMaker auto-scaling adjusts instance counts from 1 to 5 based on endpoint invocation load, balancing cost-efficiency during off-peak hours with throughput during peak usage.
- **Asynchronous 7-Stage Pipeline**: The processing logic was rebuilt as a concurrent, event-driven pipeline with intelligent conditional routing. Stages 3b, 3c, and 3d (Excel, DOCX, and PDF custom processors) run only if their corresponding specialized inputs are detected in Stage 1 outputs, allowing for simultaneous normalization, specialized parsing, chunking, and indexing of multiple document streams.
- **Automated CI/CD and Deployment**: We implemented a fully automated **GitHub Actions** pipeline to handle containerization via **Docker** and deployment to **AWS ECS Fargate**, ensuring rapid iteration and system stability.
- **Testing and Quality Assurance**: A multi-layered testing strategy was introduced, including **Pytest** for backend logic, **Pydantic** for data validation, and **Apache JMeter** for performance stress testing.

- **Operational Cost Estimation**: We established a detailed cloud expenditure model, optimizing GPU utilization costs to ensure the economic viability of the production environment.

The following table summarizes the key technical transitions between the two phases:

Table 5.1: Technical Comparison: Phase 1 (Baseline) vs. Phase 2 (Production).

Dimension	**Phase 1 (Local Prototype)**	**Phase 2 (Cloud-Native Production)**
Primary Storage	Local File System	Amazon S3 (Isolated, Durability)
Vector Search	Local Vector Store	Qdrant Cloud (Managed, Multi-tenant)
Session State	In-memory (Transient)	Amazon DynamoDB (Stateful, Persistent)
Inference Engine	Local GPU Resources	AWS SageMaker (Scalable Offloading)
System Deployment	Manual Script Execution	AWS ECS Fargate (Containerized, HA)
Architecture	Monolithic Prototype	Layered Service Architecture (Modular)
Data Pipeline	Sequential Processing	7-Stage Resource-Optimized Pipeline (with conditional 3b/3c/3d)
Frontend Interface	Basic Interactive UI	Enhanced React App (Real-time Streaming)
Scalability	Single-user / Single-machine	Multi-user / Cloud-scale

This strategic transition from a local-first prototype to a cloud-native platform was not merely a change in infrastructure, but a fundamental shift in how the system manages multimodal intelligence. By offloading resource-intensive tasks and centralizing data state, the system transitioned from a experimental tool to a scalable service capable of supporting complex educational workflows.

In addition to the architectural evolution, Phase 2 integrated several production-grade milestones that were absent in the Phase 1 prototype. This included the delivery of a **complete full-stack application** with a comprehensive suite of functionalities. This was further supported by a rigorous **technology selection** process for high-concurrency frameworks, the implementation of automated **CI/CD pipelines** and **system deployment** via GitHub Actions and AWS ECS, a multi-layered **testing** strategy (covering unit, data validation, and performance metrics), and a comprehensive **cost estimation** model to ensure operational viability. These enhancements transition the system from a research tool into a sustainable service.

Table 5.2: Operational and Management Milestones (Phase 2).

Component	Phase 2 Implementation
Technology Selection	Formalized selection of FastAPI, React, and Qdrant Cloud based on scalability and latency benchmarks.
Full-Stack	Implementation of a complete frontend and backend featuring real-time RAG chat assistant, knowledge explorer, lecture viewer, summarization and quiz generation.
System Deployment	Automated containerization via Docker and orchestration on AWS ECS Fargate for high availability.
CI/CD Pipeline	End-to-end automation using GitHub Actions for continuous build, test, and deployment.
Testing & QA	Multi-tiered testing using Pytest, Pydantic, and Apache JMeter for performance stress tests.
Cost Estimation	Detailed monthly expenditure modeling for AWS resources and GPU utilization to ensure economic viability.

5.2 Data Processing Pipeline

The data processing pipeline transforms diverse raw document formats into standardized RAG-ready artifacts through a 7-stage conditional workflow. This section describes the input normalization process and the stage-by-stage implementation in detail.

1. Interface Layer (API Gateway)

The entry point to the system is a high-performance RESTful API built on **FastAPI**. This layer is responsible for request validation, concurrency management, and protocol serialization.

- **Asynchronous I/O**: To prevent thread blocking during heavy file uploads, the `/api/upload` and `/api/process` endpoints utilize Python's `async/await` pattern combined with FastAPI's `BackgroundTasks`. This allows the server to immediately acknowledge receipt of a request while heavy computation (ETL) occurs in a separate thread pool.
- **State Management**: To minimize latency during search operations, the system utilizes a *Lifespan Event Handler*. Upon application startup, heavy models (embedding models, cross-encoders, and ColQwen weights) are pre-loaded into GPU memory and stored in the application state (`app.state`), eliminating initialization overhead per request.

2. Orchestration Layer

At the core of the system lies the `UnifiedRAGPipeline` class. This component acts as the central controller, implementing the **Facade Pattern** to abstract the complexities of the underlying subsystems.

- **Pipeline Configuration**: The behavior of the orchestrator is dictated by a dynamic YAML configuration (e.g., `config/default.yaml`). This allows the system to switch between "Text-Only," "Image-Only," or "Multimodal" modes without code changes.
- **Dependency Injection**: The orchestrator initializes and injects dependencies into the sub-modules, ensuring that the *RetrieverManager* and *Generator* share the same configuration context (e.g., chunk sizes and overlaps).

3. Data Processing and Persistence Layer

The implementation utilizes a **Hybrid Cloud Storage Architecture**. Unlike local-first prototypes, this system treats **Amazon S3** as the canonical source of truth for all original documents and processed artifacts. A local file-system is utilized as a high-speed processing cache.

- **Canonical Storage (S3)**: All files are organized under user-isolated prefixes.
- **Processing Flow**: Documents are synchronized from S3 to a local temp workspace, processed through the pipeline, and finalized artifacts are published back to S3.
- **Cache Management**: A `.processing_cache.json` file tracks the hash of processed inputs, enabling idempotent re-runs and deduplication.

4. Logical Subsystems

The system logic is divided into two specialized engines that operate in parallel during query execution:

Hybrid Text Retrieval Engine

The text retrieval engine has been optimized for multi-tenant cloud deployment using **Qdrant Cloud**:

1. **Vector Store**: Shared collections with int8 scalar quantization and mandatory user_id payload filtering.
2. **Dense-Sparse Hybrid**: Merges dense vector results (MiniLM/BGE) with sparse BM25 scores (using S3-sidecar index coordination).
3. **Reranking**: Globally disabled by default (`skip_reranker=true`) for latency optimization. The reranker code (`bge-reranker-large`) is retained for fast rollback if needed.

Visual Retrieval Engine (ColQwen2.5)

To handle layout-dense documents without OCR dependencies:

1. **Multi-vector Encoding**: Generates fine-grained visual tokens preserving diagrams and chart semantics.

2. **SageMaker Offloading**: Delegates compute-heavy visual embedding generation to AWS SageMaker (G4dn/P3 instances) to maintain backend responsiveness.

3. **Late Interaction**: Matches query tokens directly against visual patches via MaxSim scoring.

5.3 Detailed Processing Stages (7-Stage Pipeline)

This section documents the complete 7-stage processing pipeline in detail, including stage-specific logic, conditional execution rules, and output structures.

Figure 5.1: 7-Stage Processing Pipeline Model.

Table 5.3: Summary of input, processors, and special features for each pipeline stage.

Stage	Name	Input	Processor(s)	Output	Special Features
1	Normalization	Raw files (all types)	DocumentNormalizer	Normalized PDFs/JSON/MD	Format detection + conversion
2	Media Processing	Video/Audio files	MediaProcessor	Transcripts, frames, audio	Noise reduction, dedup, frame binding
3	Doc. Processing (V2)	Normalized Artifacts	DocumentProcessorV2	docling_output/ (.md)	Unified Router, Supports 10 formats
3b	Excel Processing	excel_parsed/ JSON	ExcelPreprocessor	stage4_rag_ready/	Table-aware chunking, sheet tracking
3c	DOCX Processing	docx_parsed/ JSON	DocxPreprocessor	stage4_rag_ready/	Heading-aware hierarchy tracking
3d	PDF Processing	pdf_parsed/ JSON	PdfPreprocessor	stage4_rag_ready/	Heading-aware, page-level tracking
4	Consolidation	All Stage 3 + 2	Stage4Consolidator	stage4_rag_ready/	Merges sources, links media to text
Index	Indexing	RAG-ready documents	Qdrant + BM25 + ColQwen	Searchable collections	Text (dense+keyword) + image indexing
Search	Search	Query + indexes	SearchService + LLM	Results with citations	Hybrid fusion, multimodal output

Key Design Principles:

- **Unified Router (V2)**: DocumentProcessorV2 intelligently routes different file types to optimal processors.
- **Conditional Processing**: Specialized Stage 3b/3c/3d variants only run if their specific input format is detected.
- **Custom XML Parsers**: DOCX, Excel, and PDF receive specialized parsing to preserve heading hierarchies and table structures.
- **Media Integration**: Transcripts and frames are bound to text chunks via temporal and spatial metadata.
- **Metadata Tracking**: All outputs preserve complete provenance and source information for accurate citations.

5.3.1 Stage 1: Normalization

Input: Raw files in diverse formats (DOCX, XLSX, PDF, PPTX, HTML, Images, Video, Audio, etc.)

Purpose: Detect file types, route to appropriate processors, and generate multiple canonical forms (PDF for visual processing, Markdown for text processing, specialized JSON for custom parsers).

Processing Logic:

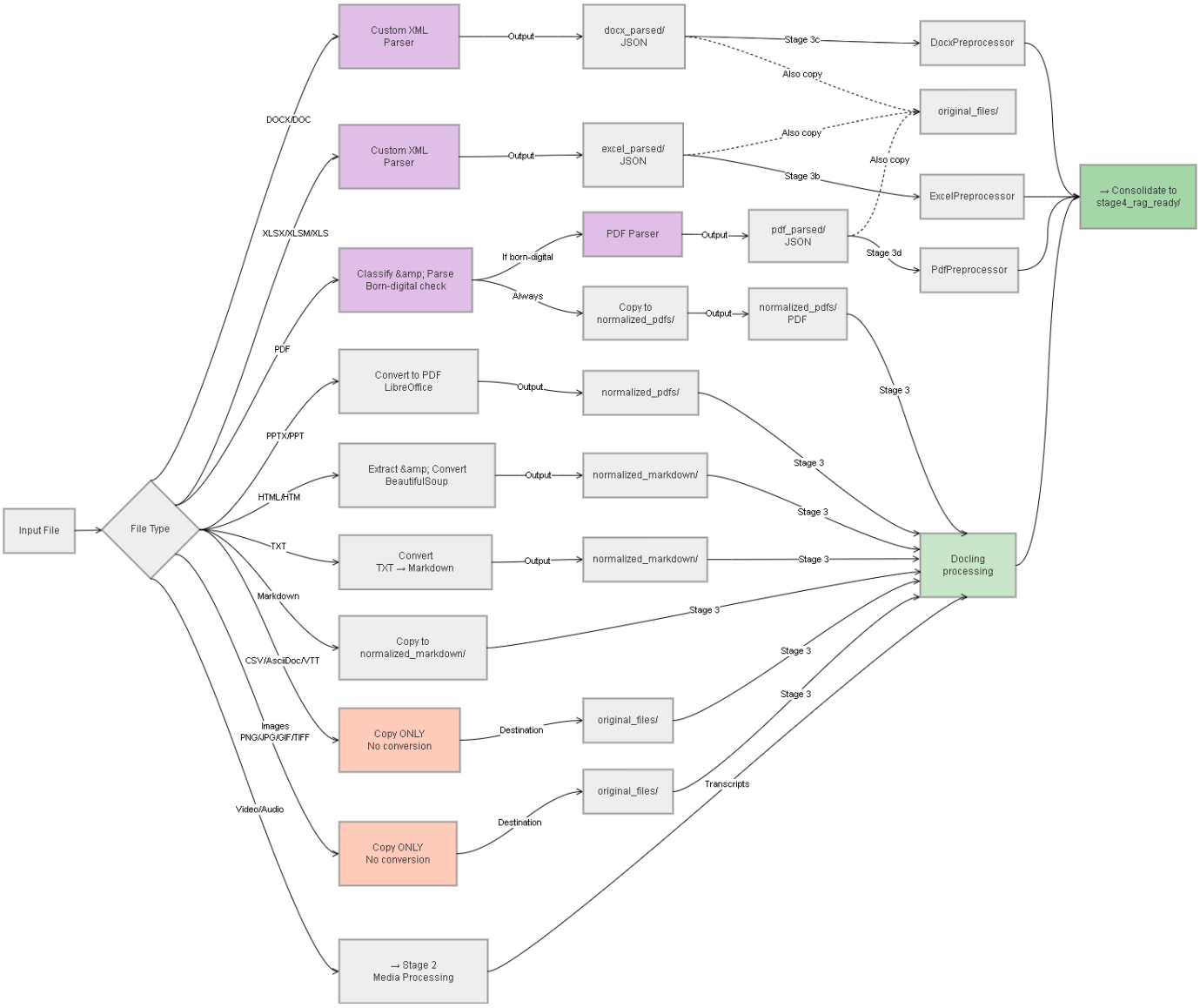

Figure 5.2: Stage 1: File Type Routing Logic.

Table 5.4: Stage 1 routing logic for different input formats.

Input Format	Detection	Primary Handler	Output (PDF)	Custom Parser
DOCX / DOC	File extension	LibreOffice + Docling	✓	✓(Stage 3c)
XLSX / XLS	File extension	LibreOffice + Docling	✓	✓(Stage 3b)
PDF (Digital)	PDF header check	Copy + Docling	✓	✓(Stage 3d)
PPTX / PPT	File extension	LibreOffice	✓	-
HTML	MIME type	BeautifulSoup + Docling	✓	-
Markdown	Extension	Copy as-is	-	-
TXT	Extension	Copy as-is	-	-
CSV	Extension	Pandas	-	-
Images (JPG, PNG)	Extension	img2pdf	✓	-
Video (MP4, AVI)	Extension	MoviePy (audio extraction)	-	- (→ Stage 2)
Audio (WAV, MP3)	Extension	Pass-through	-	- (→ Stage 2)

Output:

- `normalized_pdfs/`: PDF versions for visual processing
- `normalized_markdown/`: Text versions for semantic processing
- `docx_parsed/`: XML-extracted DOCX data (if DOCX detected)
- `excel_parsed/`: XML-extracted Excel data (if XLSX detected)
- `pdf_parsed/`: XML-extracted PDF data (if PDF detected)

5.3.2 Stage 2: Media Processing

Input: Video and Audio files (or extracted audio from Stage 1)

Purpose: Extract audio tracks, perform noise reduction, transcribe speech to text, extract key frames from video.

Processing Pipeline:

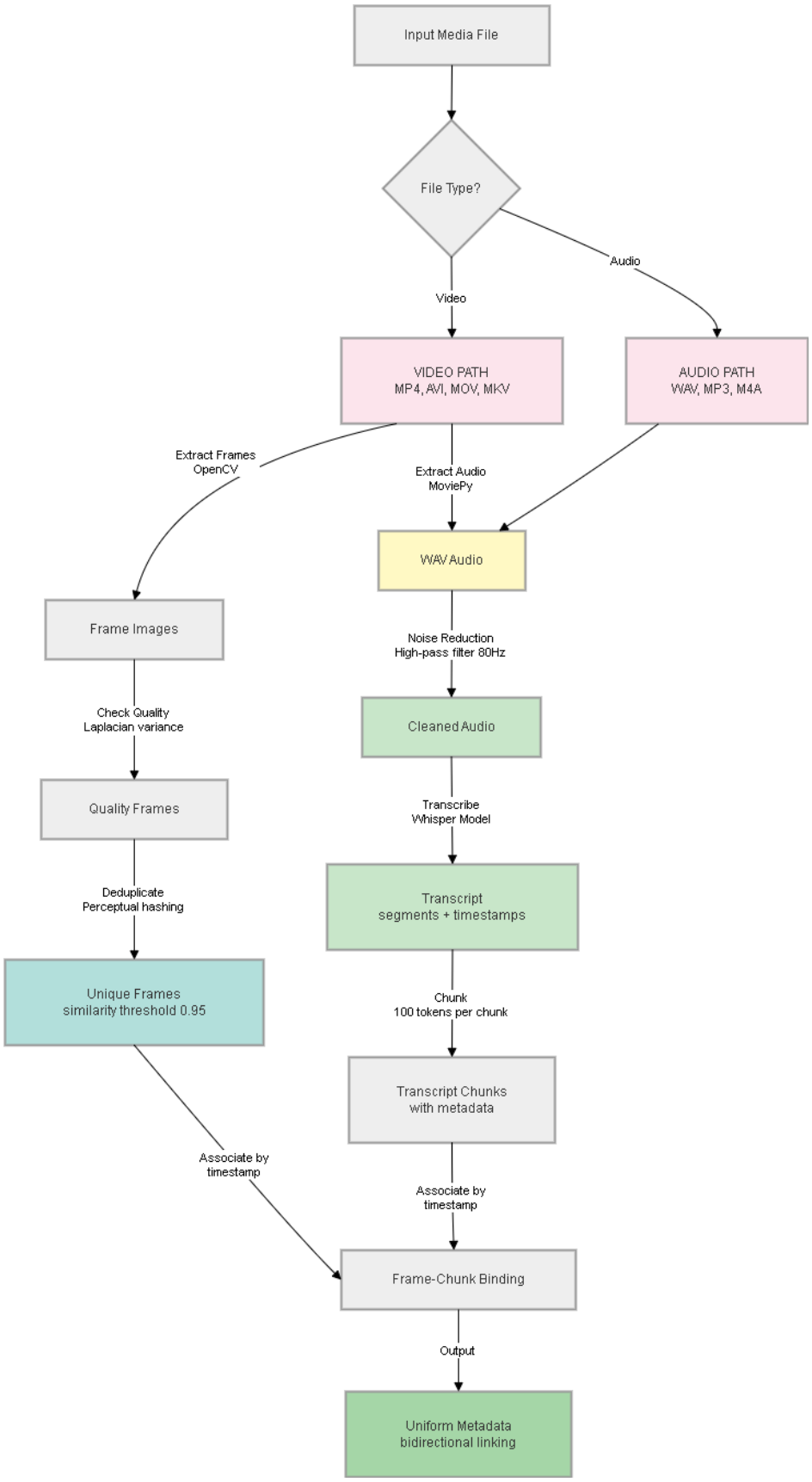

Figure 5.3: Stage 2: Media Processing Workflow.

Table 5.5: Media processing steps and transformations.

Input Type	Extraction	Processing	Output Format	Destination
Video (MP4, AVI)	MoviePy audio extraction	Noise reduction (80Hz HPF)	WAV file	Audio processing
		Whisper transcription	Markdown transcript	Indexing
	OpenCV frame extraction	Laplacian variance filtering	JPEG images	Visual indexing
Audio (WAV, MP3)	Direct input	Noise reduction (80Hz HPF)	Cleaned WAV	Transcription
		Whisper transcription	Markdown transcript	Indexing
		Multiple formats	SRT/VTT subtitles	Display

Key Details:

- **80Hz High-Pass Filter**: Removes low-frequency hum and rumble before transcription, improving Whisper accuracy
- **Frame Extraction**: OpenCV processes video to extract key frames, using Laplacian variance to detect non-blurry frames
- **Deduplication**: Perceptual hashing (0.95 similarity threshold) prevents duplicate frames
- **Whisper Configuration**: Configurable ASR model (default: `base`; `large-v3` available as an option for enhanced accuracy in multi-language contexts)

Output:

- `transcripts/`: Markdown versions of all transcriptions
- `transcript_chunks/`: Pre-chunked transcript segments
- `extracted_frames/`: Key frames from video processing
- `media_metadata/`: Timestamp bindings, speaker identification

5.3.3 Stage 3: Main Document Processing (Docling)

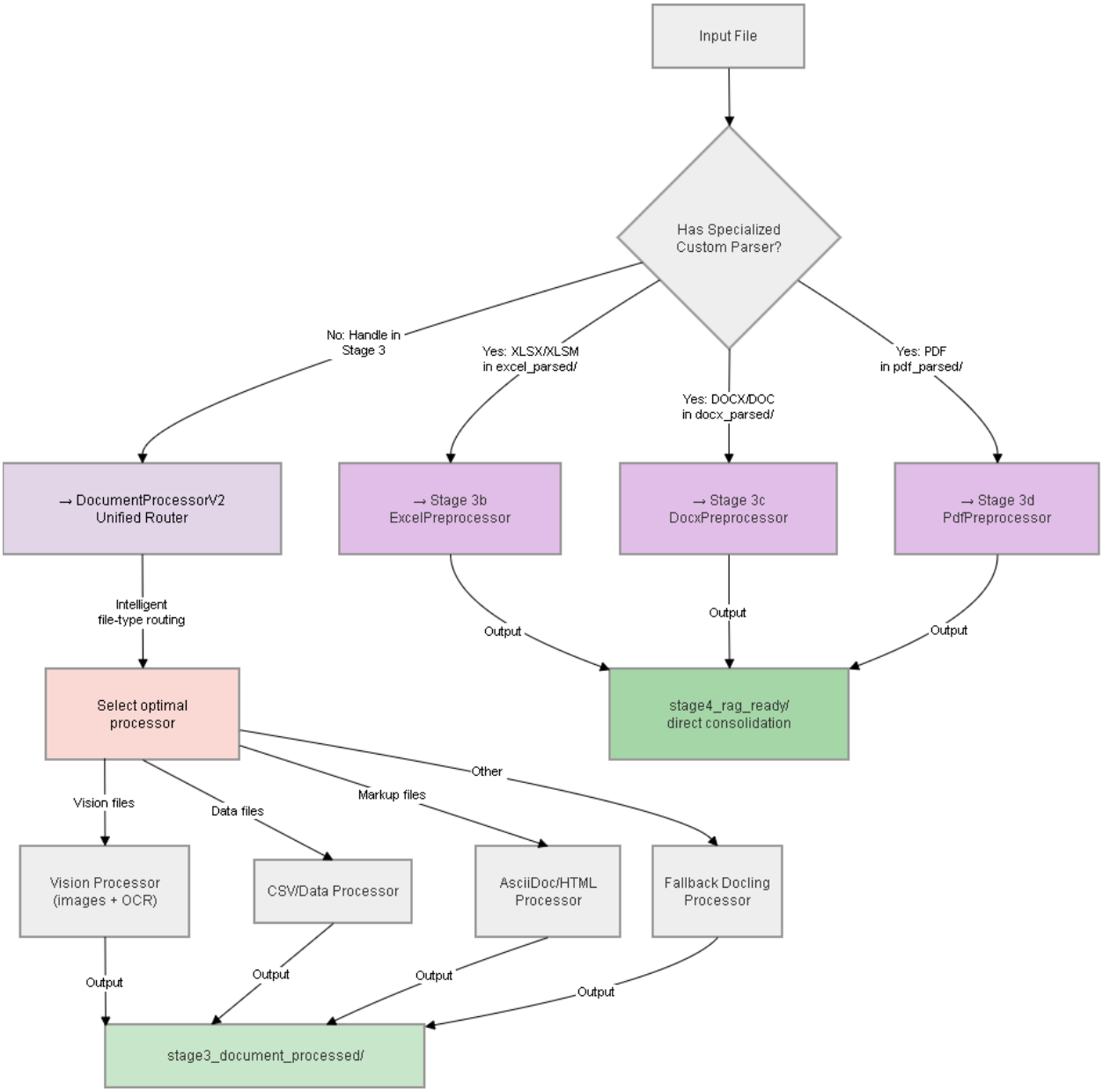

Figure 5.4: Processing Decision Flow for Multimodal Inputs.

Input: Normalized PDFs and Markdown from Stage 1

Purpose: Perform deep structural analysis, extract tables with hierarchy preservation, isolate images, handle general document types not requiring custom parsers.

Processing Logic:

For documents not processed by the custom parser (DOCX, PDF, XLSX), Docling performs full structural analysis. This stage utilizes the **DocumentProcessorV2** unified

router, which automatically selects the optimal backend for each file type:

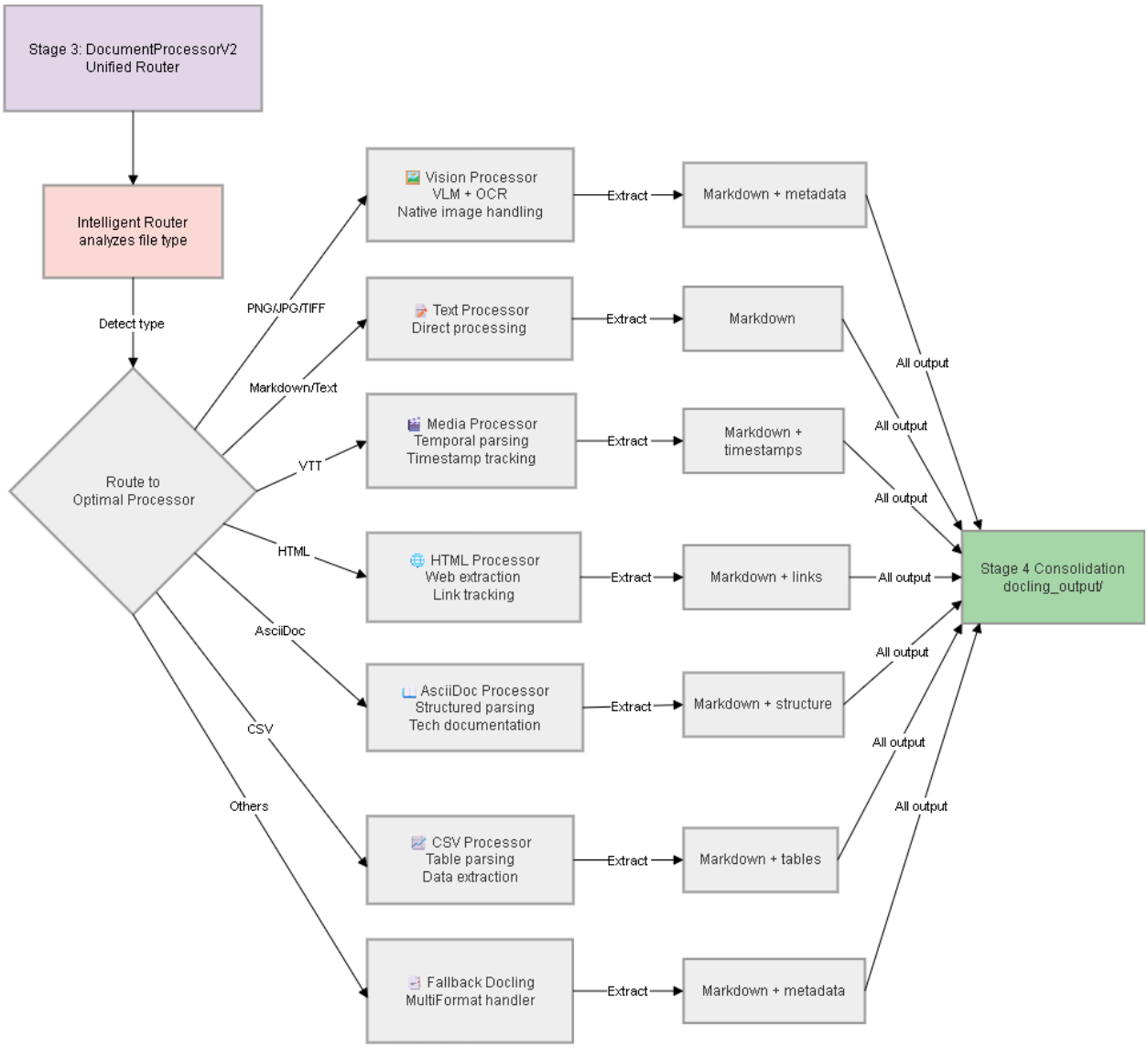

Figure 5.5: DocumentProcessorV2 Unified Router.

- **Vision Processor**: Handles images and rasterized PDFs with VLM-based layout analysis. It supports optional offloading to **AWS SageMaker** for heavy inference tasks and implements **adaptive GPU memory management** to prevent OOM errors during batch processing.
- **Structured Processors**: Specialized handlers for CSV (table reconstruction), HTML (BeautifulSoup extraction), and AsciiDoc (logical structure parsing).
- **Temporal Processor**: Handles WebVTT subtitle files, preserving temporal metadata for cross-modal search.

Output:

- `document_id/chunks.json`: Text chunks with semantic boundaries
- `document_id/tables.csv`: Extracted tables in CSV format
- `document_id/images/`: Isolated figures and diagrams
- `document_id/metadata.json`: Provenance and processing metadata

5.3.4 Stage 3b: Excel Processing (Conditional)

Trigger: Only executes if `excel_parsed/` directory exists (populated in Stage 1)

Purpose: Parse Excel workbooks using custom XML parser to preserve table structure, column headers, and quantitative relationships.

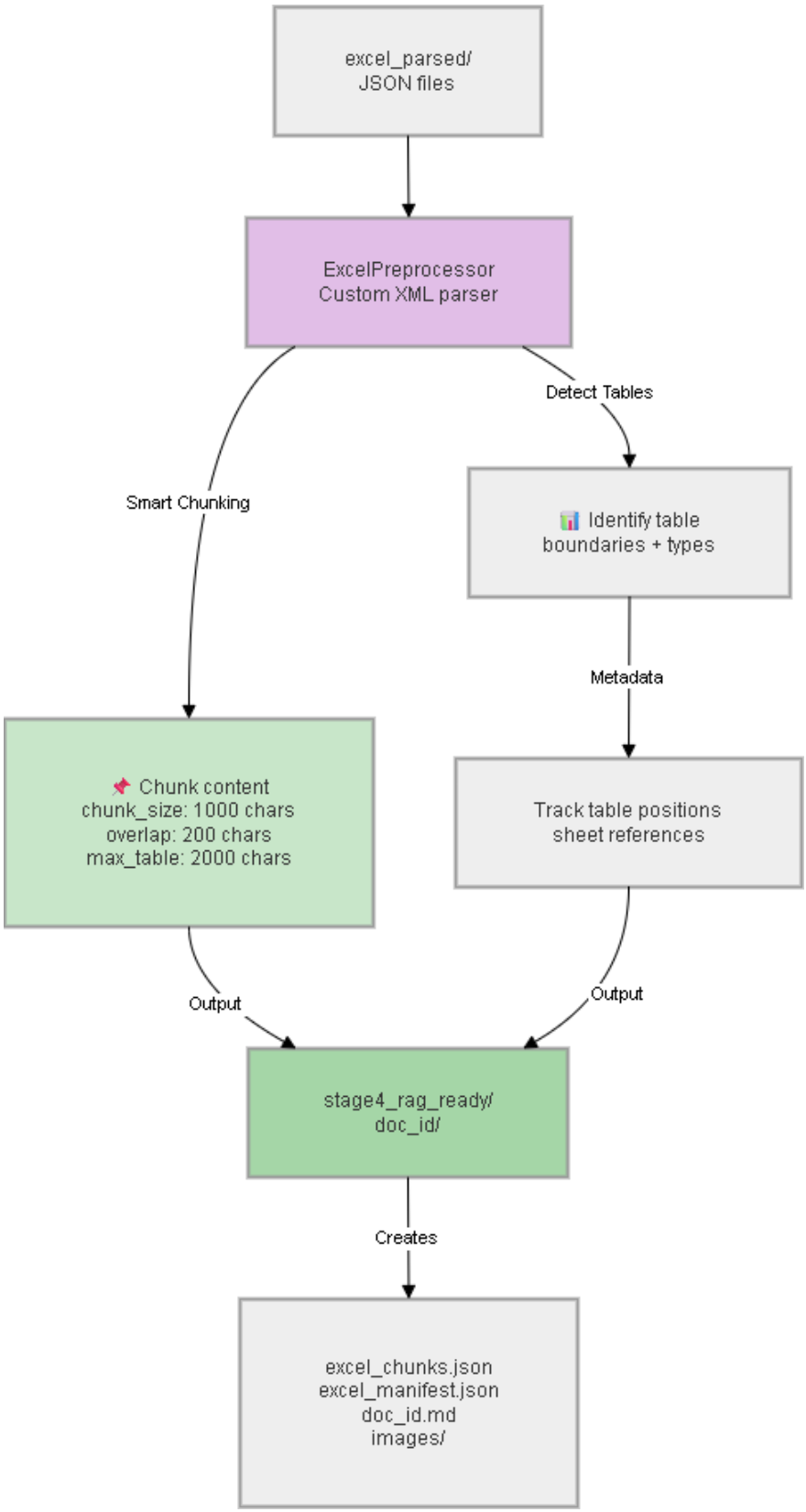

Figure 5.6: Stage 3b: Specialized Excel Processing Workflow.

Processing Strategy:

- **Sheet-aware parsing**: Extract each sheet independently with sheet name metadata
- **Table detection**: Identify table boundaries via cell formulas and formatting
- **Chunking strategy**: Maximum 2000 characters per table chunk (split by rows, not mid-table)
- **Header preservation**: Include column headers with every chunk for context

Output:

- `document_id/chunks.json`: Table-aware chunks with sheet and row metadata
- `document_id/tables.csv`: Extracted tables
- Metadata: `sheet_name`, `row_range`, `table_id`

5.3.5 Stage 3c: DOCX Processing (Conditional)

Trigger: Only executes if `docx_parsed/` directory exists (populated in Stage 1)

Purpose: Parse Word documents using custom XML parser to respect heading hierarchies and preserve document structure.

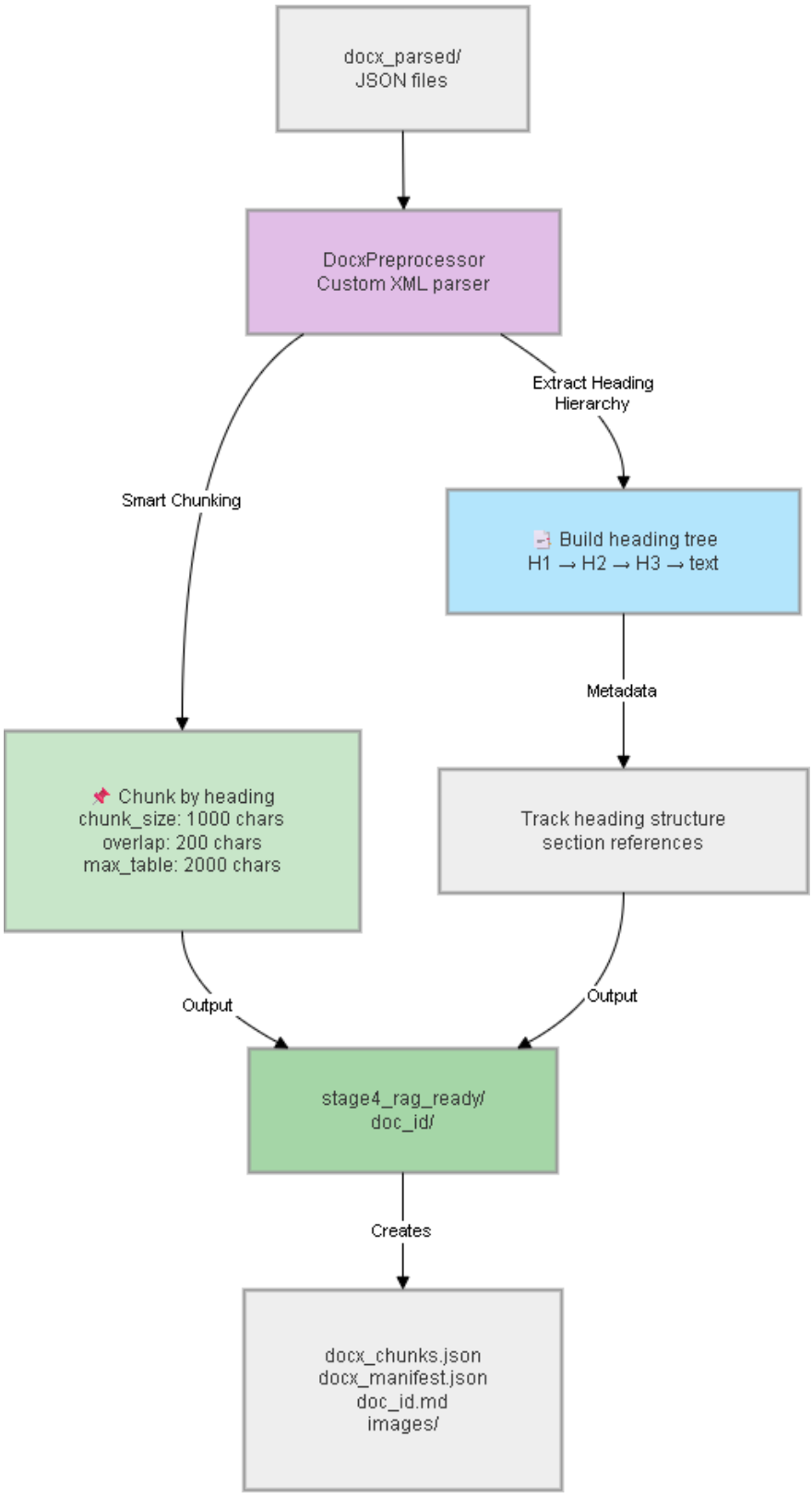

Figure 5.7: Stage 3c: Specialized DOCX Processing Workflow.

Processing Strategy:

- **Heading extraction**: Build heading tree (Heading 1, 2, 3, etc.)
- **Boundary-aware chunking**: Split at heading boundaries, not arbitrary character limits
- **Overlap**: 200-character overlap at boundaries for semantic continuity
- **Metadata enrichment**: Track heading level and hierarchy path

Output:

- `document_id/chunks.json`: Heading-aware chunks
- Metadata: `heading_level`, `heading_path`, `section_title`

5.3.6 Stage 3d: PDF Processing (Conditional)

Trigger: Only executes if `pdf_parsed/` directory exists (populated in Stage 1 for structured PDFs)

Purpose: Parse born-digital PDFs with internal structure to preserve page references and section hierarchies.

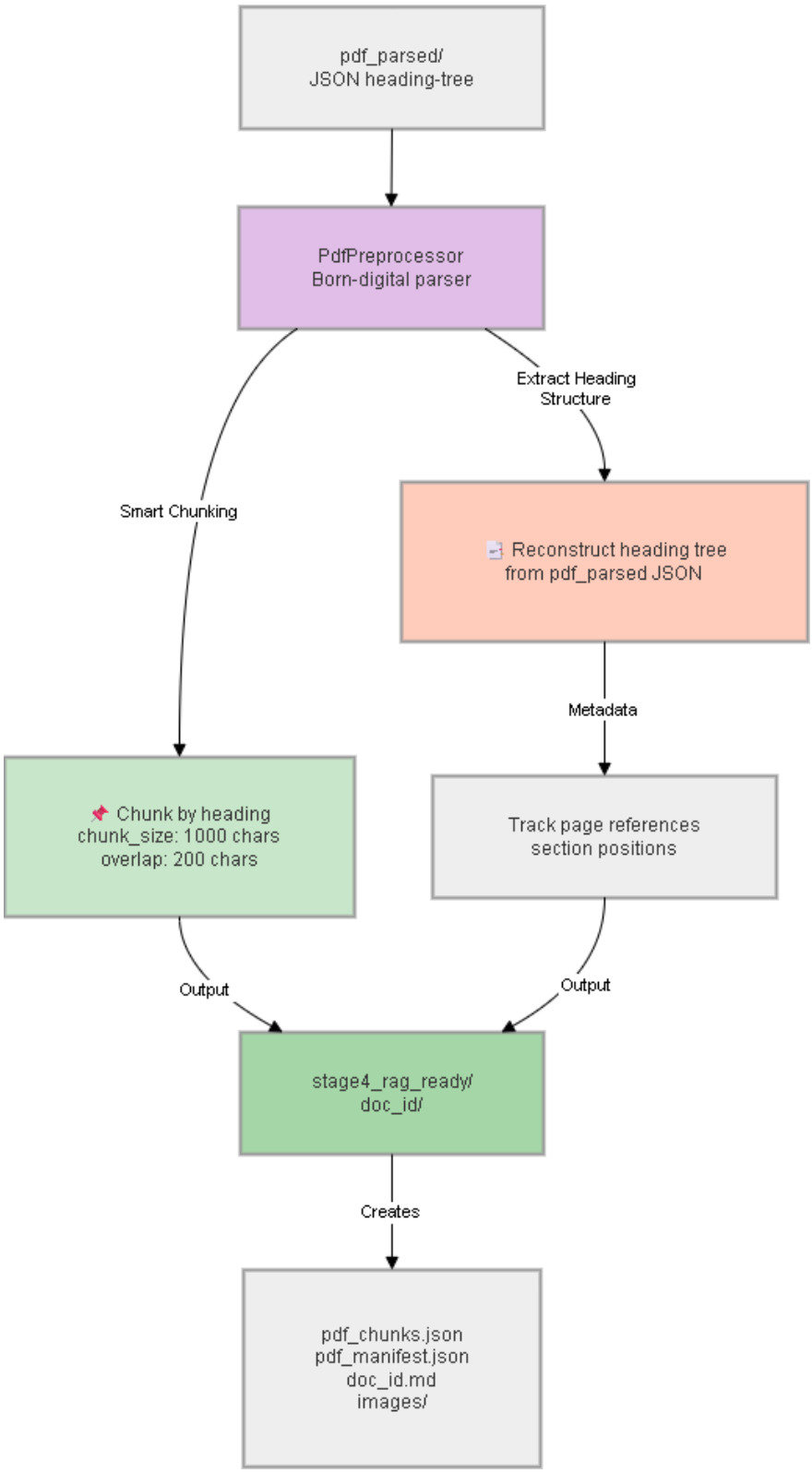

Figure 5.8: Stage 3d: Specialized PDF Processing Workflow.

Processing Strategy:

- **Metadata extraction**: Analyze PDF outlines and internal section markers
- **Page tracking**: Maintain explicit page numbers for every chunk
- **Section-based chunking**: Split at section boundaries from PDF structure
- **Visual element linking**: Preserve references to figures and tables with page numbers

Output:

- `document_id/chunks.json`: Page-aware chunks
- Metadata: `page_number`, `section_level`, `total_pages`

5.3.7 Stage 4: RAG-Ready Consolidation

Input: All outputs from Stages 3, 3b, 3c, 3d (merged), plus Stage 2 transcripts and media

Purpose: Merge all processing outputs, consolidate metadata, prepare unified retrieval index.

Consolidation Logic:

1. **Merge chunks**: Combine text chunks from all sources (Docling + custom parsers) with deduplicated provenance
2. **Collect images**: Gather all extracted images (from document pages, figures, video frames) with page references
3. **Link transcripts**: Bind media transcripts to source documents via timestamp metadata
4. **Create index**: Generate `chunks.json` with all text and `metadata.json` with complete provenance tracking

Output:

- `stage4_rag_ready/document_id/chunks.json`: Complete unified chunk list with metadata
- `stage4_rag_ready/document_id/metadata.json`: Full provenance (source, stage, location)
- `stage4_rag_ready/document_id/images/`: All visual elements

5.4 Chunking and Embedding Strategies

The system employs a dual-pathway embedding architecture combining format-aware text chunking with visual page embedding. This approach enables simultaneous retrieval of granular text segments and holistic document pages.

5.4.1 Chunking Strategy by Document Type

The system employs different chunking strategies depending on the source document type, reflecting the structural differences between formats:

- **Office Documents (Excel, DOCX) — Format-Aware Custom Chunking:** Excel files processed through Stage 3b use a table-aware custom XML parser with a maximum chunk size of 2,000 characters, preserving row-column relationships. DOCX files processed through Stage 3c use a heading-aware custom parser that respects the document heading hierarchy, keeping sections semantically coherent.
- **PDF Documents — Heading-Aware Custom Chunking:** PDFs processed through Stage 3d use a custom heading-aware chunking strategy that identifies PDF heading structures and creates chunks aligned with logical document sections.
- **General Documents via Docling — Recursive Character Splitting:** For documents processed through the main Docling pipeline (Stage 3), the system uses `RecursiveCharacterTextSplitter` with a chunk size of 1,024 characters and an overlap of 200 characters. This overlap preserves semantic continuity across chunk boundaries.
- **Multimedia Transcripts (Audio/Video):** Transcripts generated by Whisper employ a dedicated `TranscriptChunker` that groups transcription segments using token-count-based accumulation (maximum 100 tokens per chunk). Each chunk carries explicit `start_time` and `end_time` boundaries derived from Whisper's segment timestamps, enabling temporal range-based retrieval. Associated video frames are linked to transcript chunks based on temporal overlap, creating a unified multimodal index.

5.4.2 Text Embedding Pipeline

To process the Markdown inputs, the system utilizes a high-precision splitting strategy designed to maximize retrieval relevance.

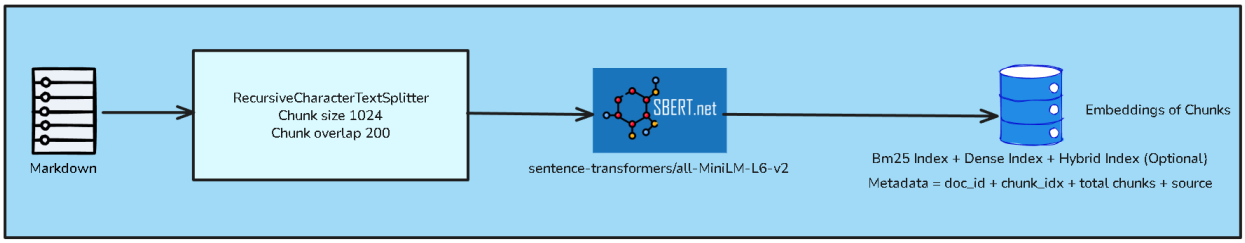

Figure 5.9: Textual Chunking and Embedding.

Textual data is segmented into chunks to fit within the context window of the embedding models. The system employs the `RecursiveCharacterTextSplitter` with the configuration using a chunk size of 1024 characters with an overlap of 200 characters. This overlap ensures semantic continuity across chunk boundaries, preventing context loss for sentences that straddle the split point.

Text Vectorization (SBERT)

Processed chunks are mapped to a high-dimensional vector space using the **Sentence-BERT (SBERT)** framework.

- **Model**: `sentence-transformers/all-MiniLM-L6-v2`.
- **Output**: A 384-dimensional dense vector representing the semantic meaning of the text chunk.
- **Index Structure**: The embedding is stored alongside a composite metadata object containing:

```
{
  "doc_id": "unique_document_identifier",
  "chunk_idx": "integer_sequence_number",
  "total_chunks": "total_count_for_document",
  "source": "original_filename"
}
```

5.4.3 Visual Embedding Pipeline

For documents rich in diagrams, charts, and complex layouts, the pipeline utilizes JPG images derived from the canonical PDF files. These intermediate image files are stored temporarily to facilitate the vision-based embedding process used by ColQwen, ensuring the model "sees" the document exactly as a human would.

The visual pathway processes the page images generated to enable multimodal retrieval.

Figure 5.10: Visual Embedding (ColQwen).

Vision-Language Model

The system integrates the **ColQwen** architecture (specifically `vidore/colqwen2-v1.0` or `v2.5`). Unlike standard OCR which strips layout, ColQwen encodes the visual semantics of the entire page.

Visual Indexing

- **Embedding Process**: Each rasterized page image is passed through the ColQwen vision encoder to produce visual tokens.
- **Storage**: These embeddings are stored in a specialized ColQwen Index.
- **Metadata Schema**: Visual embeddings are tagged with precise location data:

```
{
  "source": "original_filename",
  "source_path": "file_system_path_to_pdf",
  "page": "page_number",
  "total_pages": "document_length"
}
```

The text and visual embedding pipelines together constitute the dual-pathway indexing strategy, encoding documents into both dense semantic vectors and sparse token-level representations. The following section describes how these embeddings are organized, indexed, and queried within the Qdrant vector database and supporting storage layers.

5.5 Indexing and Storage Architecture

5.5.1 Indexing Pipeline

The consolidation and indexing pipeline transforms RAG-ready chunks into searchable indices through a dual-pathway approach:

Table 5.6: Artifacts produced by the indexing pipeline for each document.

Index Type	Data Structure	Storage Format	Query Mechanism
BM25 Index (Text)	Inverted index (term → docs)	Pickle file (`.pkl`)	Keyword matching, TF-IDF scoring
Dense Index (Text)	FAISS IndexFlatIP	Binary file (`.bin`)	Cosine similarity, nearest-neighbor search
Hybrid Index (Text)	RRF fusion results	JSON lookup table	Combined BM25 + Dense scores
Visual Index (Images)	ColQwen multi-vectors	Tensor store (`.bin`)	Late-interaction MaxSim matching
Metadata Catalog	Document + chunk + image metadata	JSON files	Source tracking, citation generation

Text Indexing Workflow:

1. Chunks from Stage 4 are encoded by SBERT (all-MiniLM-L6-v2) → 384-dimensional vectors
2. BM25 inverted index built from chunk text (keyword search)
3. Dense vectors stored in FAISS IndexFlatIP (semantic search)
4. Hybrid index pre-computed via RRF fusion for fast query-time execution

Visual Indexing Workflow:

1. PDF pages rasterized to JPEG (150 DPI, 1 page = 1 image)
2. Each page encoded by ColQwen to multi-vector representation
3. Vectors stored with minimal indexing (m=0 for HNSW bypass) to reduce memory overhead
4. Query-time MaxSim computed on-demand for precision

Index Persistence Strategy:

After indexing completes (Stage 4), the system:

1. Uploads BM25, Dense, and Visual indices to S3 under `users/user_id/documents/doc_id/indices/`
2. Caches indices in local filesystem for fast retrieval during query execution
3. On server restart or query miss, reloads indices from S3

5.5.2 Search and Retrieval Pipeline

The search pipeline orchestrates the concurrent retrieval of textual and visual information, optionally followed by generative synthesis. The workflow is designed for low-latency response times while maintaining high factual grounding.

Retrieval Strategies

The system supports three primary retrieval modalities to address different query types:

- **Dense Retrieval**: Best for semantic understanding and handling synonymy. It uses the `all-MiniLM-L6-v2` model to match query embeddings against the Qdrant text collection.
- **Sparse Retrieval (BM25)**: Optimized for exact keyword matching, technical jargon, and specific entity retrieval.
- **Hybrid Retrieval**: Merges the results of dense and sparse retrievers using a weighted fusion approach. In Phase 2, the system utilizes an alpha weight of 0.5, providing a balanced trade-off between keyword precision and semantic recall.

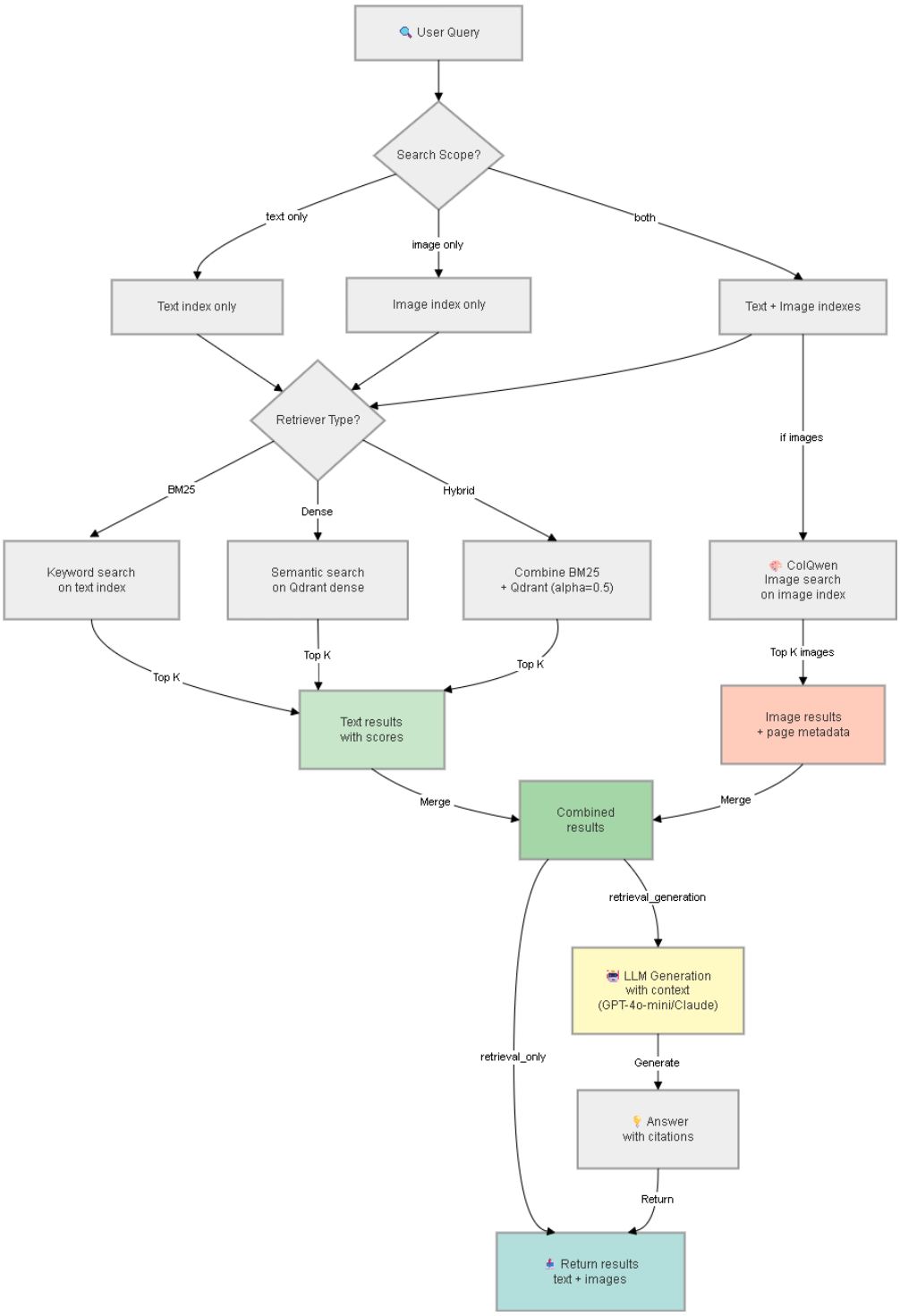

Figure 5.11: Search and Retrieval Pipeline Flow.

Search Flow and Operational Modes As illustrated in Figure 5.11, the search orchestrator (`SearchService`) processes queries through a multi-step pipeline:

1. **Decision Node: Search Scope**: Evaluates the query intent to target either the text

index, image index, or a parallel multimodal retrieval.

2. **Decision Node: Retriever Type**: Determines the retrieval algorithm (`BM25`, `Dense`, or `Hybrid`) based on the query complexity and user configuration.
3. **Parallel Retrieval**:
   - **Text Pathway**: Executes the selected retrieval logic on the text index, returning top K chunks (default $K = 10$).
   - **Visual Pathway**: Executes a late-interaction search on the ColQwen image index if visual context is requested.
4. **Reranking (Optional)**: To further refine the results, a cross-encoder model (`bge-reranker-large`) can be applied to the merged candidate set (as illustrated in Figure 2.7). However, in the Phase 2 production environment, this step is **globally disabled by default** (`skip_reranker=true`) to maintain sub-second retrieval latency.
5. **Result Merging**: Contextualizes text chunks with their corresponding visual pages based on source metadata.
6. **Mode Selection**:
   - **Retrieval Only**: Immediately returns the merged text and image artifacts to the client.
   - **Retrieval-Generation**: Injects the retrieved context into a Large Language Model (e.g., Claude 3.5 Sonnet) to synthesize a grounded answer with citations.

Response Structure and Metadata Tracking

Every search response follows a standardized schema that integrates synthesized answers with granular metadata tracking. As defined in the system's design principles, every retrieved chunk preserves its provenance (source file, page, and chunk type) to ensure factual grounding. The response includes a detailed telemetry payload tracking retrieval latencies and model inference times, providing users with transparency regarding the "source-of-truth" for all generated answers.

5.5.3 Storage Architecture

Phase 2 implements a **hybrid storage model** combining local processing cache with cloud canonical storage:

Table 5.7: Local vs. Cloud storage division of responsibility.

Storage Tier	System	Content	Purpose
Local Cache	File system (fast SSD)	Processing artifacts (Stages 1-4)	Speed up pipeline processing
Local Indices	FAISS, ColQwen stores	BM25, Dense, Visual indices	Fast retrieval queries
Cloud Canonical	Amazon S3	Documents, metadata, indices	Durability, multi-tenancy
Session State	DynamoDB	Chat history, job queue state	Persistent conversation tracking

Multi-Tenancy Isolation: **Hybrid Retrieval Integration**

The retrieval unifies these two streams. The Text and Visual indices can be queried simultaneously to return both relevant text snippets and their corresponding visual page contexts.

The index supports **Hybrid**, allowing for keyword precision and semantic recall, combining two distinct algorithms:

1. **Sparse Retrieval (BM25)**: Utilizes keyword matching to capture exact terms and domain-specific jargon.
2. **Dense Retrieval**: Utilizes vector embeddings (stored in Qdrant collections) to capture semantic meaning and context. The default embedding model is `all-MiniLM-L6-v2`.

The system performs **Hybrid Fusion** using the Reciprocal Rank Fusion (RRF) or Distribution-Based Score Fusion algorithm, with a default **alpha weight of 0.5**, balancing keyword exactness with semantic similarity.

Scores from both retrievers are combined using Min-Max normalization and a weighted sum.

5.6 Asynchronous Indexing Job System

The Phase 2 system implements a **Redis-backed job queue** for managing asynchronous indexing operations, enabling non-blocking document processing while maintaining concurrent request limits and durable job state persistence.

5.6.1 Purpose and Architecture

Rather than forcing clients to wait for complete index updates, documents are queued for indexing via the `IndexingJobService`. This service coordinates three components:

- **Redis Queue**: Transient job tracking with configurable TTL (default 3600 seconds). Each job receives a unique `job_id` and tracks state transitions: `queued` → `processing` → `completed` or `failed`.
- **DynamoDB Persistence**: Optional backup to Amazon DynamoDB (`bk_mind_app_jobs` table) for long-term durability and cross-service job history. Job snapshots include metadata, progress counters, and error details.
- **Concurrency Limits**: Per-user queue depth (`ASYNC_INDEXING_MAX_PER_USER`, default 3) and global queue depth (`ASYNC_INDEXING_MAX_GLOBAL`, default 200) prevent resource exhaustion.

5.6.2 Job Lifecycle and Progress Tracking

Each indexing operation creates a job payload with status tracking. The service provides non-blocking status queries via `get_job(job_id)` and batch polling via `get_jobs_for_user(user_id)`, enabling frontend implementations to display real-time progress bars and resume interrupted indexing workflows.

Fallback Behavior: If Redis is unavailable and `ASYNC_INDEXING_FALLBACK_TO_BLOCKING=true`, the system gracefully degrades to synchronous processing, ensuring availability over responsiveness. Configuration is environment-driven (`ASYNC_INDEXING_ENABLED`, `REDIS_URL`), supporting both container deployments and local development scenarios.

Benefits of Asynchronous Indexing:

- **Non-blocking UX**: Clients receive immediate acknowledgment; indexing proceeds in background worker pools.
- **Durability**: DynamoDB persistence ensures jobs survive system restarts.
- **Multi-tenancy**: Per-user limits prevent single user's large jobs from starving others.
- **Observability**: Detailed progress metrics and error tracking support operational debugging.

With asynchronous job processing established as the infrastructure for long-running indexing operations, the system requires robust schema design and storage architecture to persist all processing artifacts, embeddings, and user state. The following section describes the three-tier hybrid storage architecture.

5.7 Database Schema and Storage

The system implementation requires a robust schema design to manage heterogeneous data including user interactions, processed documents, and high-dimensional vector representations. We utilize a **Hybrid Storage Architecture** to balance persistence, scalability, and low-latency retrieval across three specialized layers:

- **Qdrant**: A specialized **Vector Database** and similarity search engine designed for storing and querying high-dimensional embeddings. Qdrant manages the multi-pathway vector representations of textual chunks and visual document pages, enabling semantic similarity matching and retrieval-augmented generation. It provides optimized search algorithms, filtering capabilities, and distributed indexing to maintain sub-second retrieval performance.
- **Amazon S3**: A highly scalable **Object Storage** service used to manage the complete lifecycle of document assets. It acts as the canonical source of truth, storing raw

uploads, intermediate processing artifacts (e.g., extracted audio, keyframes), and final "RAG-ready" normalized documents. This layer ensures data durability and facilitates multi-tenant isolation through keyed prefixing.

- **Amazon DynamoDB**: A fully managed **NoSQL** database service that provides fast and predictable performance with seamless scalability. In our architecture, DynamoDB serves as the primary store for persistent metadata, user profile information, assessment results, conversational state, and system telemetry. Its schema-less nature allows for flexible data structures while its partitioned key-value access ensures low-latency operations.

5.7.1 Schema Property Definitions

Before detailing the specific collection structures, we define the properties utilized within our schema tables to ensure technical clarity.

Table 5.8: Schema properties used in implementation documentation.

Property	**Description**	**Note**
Field	Name of the attribute in the database	Mandatory.
Type	The technical data format	String, Boolean, Enum, or Vector.
Description	Purpose of the field within the system	Context for the specific attribute.
Note	Structural role or indexing constraint	e.g., Primary Key, Partition Key, Sort Key.
Dim	The length of the embedding vector	Mandatory for dense vector fields.
Distance	The distance calculation for retrieval	Mandatory for vector indices (e.g., Cosine).

Key Concepts in DynamoDB Schema Design

To effectively manage high-throughput data access and ensure efficient multi-tenant isolation, the system utilizes a robust primary key structure to uniquely identify each item in a table. Our architecture leverages both simple and composite primary keys:

- **Primary Key**: The unique logical identifier for each item. In DynamoDB, the primary key is a role assigned to specific attributes and can take two forms:
  - **Simple Primary Key (Partition Key only)**: Comprising just the partition key to uniquely identify items.
  - **Composite Primary Key (Partition Key + Sort Key)**: Comprising both keys for more complex identification and sorting.

  In the AWS Management Console, these are represented by their specific key descriptors rather than a single field named "Primary Key."

- **Partition Key (Hash Key)**: A mandatory attribute used by DynamoDB to distribute data across multiple physical partitions for horizontal scalability. In our

schema, attributes like `user_id` act as partition keys to ensure that all related data for a specific user is co-located.

- **Sort Key (Range Key)**: When combined with the partition key, the sort key creates a *Composite Primary Key*. Items sharing the same partition key are stored in sorted order by their sort key value, enabling efficient range searches and ordered retrieval (e.g., chat messages by timestamp).

5.7.2 Vector Storage (Qdrant)

To support our Hybrid + Multimodal retrieval strategy, we segregate embeddings into dedicated Qdrant collections. Each collection is distinctly configured with dimensions and distance metrics tailored to the deployed embedding model architecture.

Collection: `edu_text_chunks` Stores granular textual chunks extracted from structured Markdown documents. It utilizes a hybrid search approach, combining sparse keyword indexing with the following dense vector configuration.

Table 5.9: Text Vector Collection Properties.

Vector Name	Dim	Distance	Description
text	384	Cosine	Main text embeddings representing chunk semantics (e.g., via `all-MiniLM-L6-v2`).

Each indexed text chunk retains critical metadata necessary for strict multi-tenant filtering and file attribution during generative retrieval:

Table 5.10: Text Collection Schema Structure.

Field	Type	Note / Description
point_id	UUID	Primary Key. Unique identifier mapped to the vector.
chunk_id	String	Unique string identifier for the block.
user_id	String	Tenancy boundary identifier.
source	String	Original filename.
text_preview	String	Truncated preview of the chunk.
storage_uri	String	Cloud location of the chunk artifact.
storage_backend	String	Storage system (e.g., s3).

Collection: `edu_image_pages` Dedicated to storing high-order visual layouts. It leverages a unique multi-vector representation generated directly from the ColQwen

vision-language model, enabling direct visual similarity searches against original PDF structures.

Table 5.11: Image Vector Collection Properties.

Vector Name	Dim	Distance	Description
colpali_multivec	128*x	Dot	Multi-vector representations (128 dims $\times$ x visual tokens) for ColQwen late-interaction.

Visual Token Calculation Methodology The variable x in the multi-vector representation represents the total number of distinct 128-dimensional embeddings stored per document page. Unlike standard text RAG that condenses an entire chunk into a single vector, the `ColQwen` architecture preserves granular spatial information by treating the page as an adaptive grid of patches. The value of x is calculated as follows:

$$x = (\lceil H/P \rceil \times \lceil W/P \rceil) + N_{special} \tag{5.1}$$

where

- x: The total count of vectors stored in the Qdrant collection for a single page.
- H: The height of the document page image after processing (typically 448 or 1024 pixels).
- W: The width of the document page image after processing (typically 448 or 1024 pixels).
- P: The patch size of the Vision Transformer (ViT) backbone, fixed at **14** for our implementation. Each 14×14 pixel area is vectorized into a single embedding.
- $N_{special}$: A technical constant (typically < 10) representing special delimiter tokens used by the multimodal LLM to frame the visual sequence.

For a standard processing resolution of 1024×1024 pixels, $x \approx 5,329$ vectors. This high-density representation ensures that fine-grained structural elements, such as table cells and chart labels, remain distinctly searchable during the late-interaction phase.

The visual index maintains geometric representations (height/width) alongside S3 references, allowing the orchestration layer to securely retrieve the original page images for multimodal language models:

Table 5.12: Image Collection Schema Structure.

Field	Type	Note / Description
point_id	UUID	Primary Key. Unique identifier mapped to the vector.
user_id	String	Tenancy boundary identifier.
source	String	Original document name.
source_path	String	Absolute physical path of document.
page	Integer	The structural page number.
total_pages	Integer	The total length of the PDF.
image_width	Integer	Pixel width of the page boundary.
image_height	Integer	Pixel height of the page boundary.
storage_uri	String	Cloud location of the rasterized image.
storage_backend	String	Storage system (e.g., s3).

5.7.3 Object Storage (Amazon S3)

The system utilizes a dual-bucket object storage architecture to manage the lifecycle of document assets. This approach decouples raw ingestion from the downstream processing pipeline, ensuring data integrity and enabling scalable multi-tenancy.

- **Tenant Isolation**: Files are logically partitioned using the prefix for each user. This ensures strict data segregation and facilitates efficient cleanup of processing artifacts.
- **Dual-Bucket Lifecycle**:
  - **Originals Bucket**: Acts as the immutable source of truth for all raw user uploads.
  - **Processed Bucket**: Stores all 7-stage normalized artifacts, metadata sidecars, and serialized indices.

  The orchestration layer manages the synchronization between these buckets, downloading assets to a local temporary workspace for high-speed processing before publishing the final RAG-ready documents to the Processed Bucket.

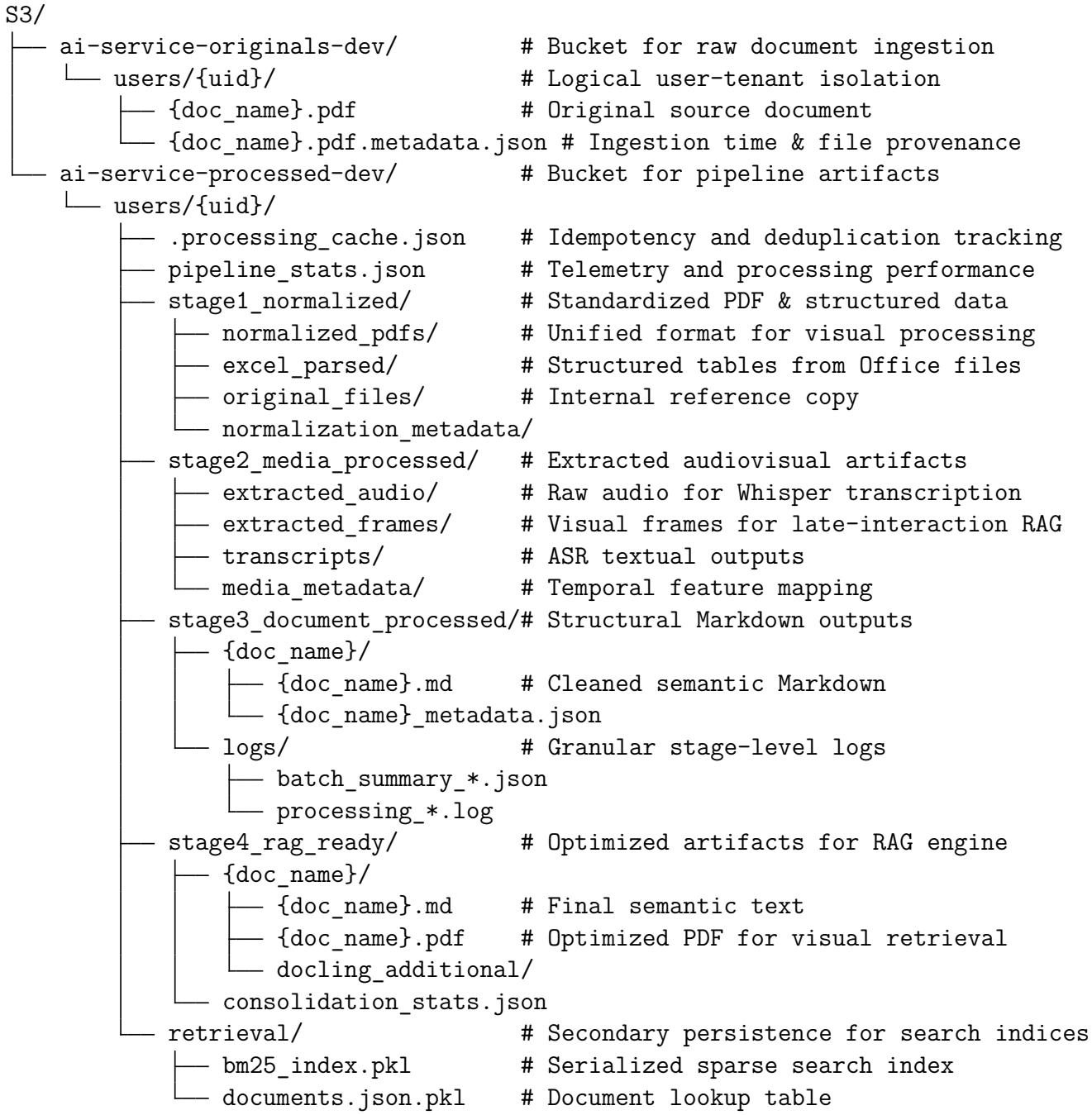

Figure 5.12: Amazon S3 Directory Structure Tree.

Bucket 1: Originals Storage (`ai-service-originals-dev`)

This bucket acts as the primary ingestion point for all raw assets. To ensure strict user isolation, the bucket employs a keyed prefixing strategy:

1. **User Partitioning**: `users/{user_id}/` acts as the root prefix for all files owned by

a specific tenant.

2. **Asset Storage**: Each uploaded file (e.g., `lecture_slides.pdf`) is stored alongside a corresponding `.metadata.json` sidecar. This sidecar contains provenance data, original filenames, and ingestion timestamps.

Bucket 2: Processed Storage (`ai-service-processed-dev`)

The processed bucket stores all intermediate and final artifacts generated by the 7-stage multimodal pipeline. This storage layer is optimized for high-speed retrieval by the Chat Assistant and Lecture Viewer services. The 7-stage design includes intelligent conditional routing where Stages 3b (Excel), 3c (DOCX), and 3d (PDF) run only if their respective custom-parsed outputs exist from Stage 1.

- **Global Pipeline Tracking**: Located at the user root, the `.processing_cache.json` and `pipeline_stats.json` files manage idempotency and performance metrics, allowing the system to skip redundant processing steps.
- **Modular Artifact Hierarchy**: Processed data is organized into logical stages:
  - **Stage 1 (Normalized)**: Contains standardized `normalized_pdfs/`, extracted tables in `excel_parsed/`, DOCX metadata in `docx_parsed/`, PDF structure in `pdf_parsed/`, and a copy of the `original_files/` for traceability. It also includes `normalization_stats.json` for tracking conversion success rates.
  - **Stage 2 (Media Processed)**: Manages temporal and visual extraction artifacts with noise reduction (80Hz high-pass filter):
    * `extracted_audio/`: Noise-reduced audio tracks extracted from video containers for transcription.
    * `extracted_frames/`: Keyframes rendered from video streams for visual RAG indexing.
    * `transcripts/`: Full textual outputs from the Whisper ASR engine applied to cleaned audio.
    * `media_metadata/`: JSON descriptors of temporal features and extraction parameters.
  - **Stage 3 (Document Processed)**: Stores structural parsing outputs from Docling for non-specialized file types, organized by document name:
    * `{doc_name}/`: Individual directories containing the structured `.md` file and its `_metadata.json` sidecar.
    * `logs/`: System-generated `.log` files and `batch_summary.json` reports for auditability and error tracking.
  - **Stage 3b/3c/3d (Conditional Custom Parsers)**: If detected in Stage 1 outputs, these conditional stages output directly to stage4_rag_ready/:
    * **Stage 3b (Excel)**: Custom XML parser with table-aware chunking (max 2000 chars), sheet metadata tracking.

    - **Stage 3c (DOCX)**: Custom XML parser with heading-aware chunking, heading hierarchy tracking.
    - **Stage 3d (PDF)**: Custom parser with heading-aware chunking, page-level references.
  - **Stage 4 (RAG Ready)**: The final consolidation layer merging outputs from all Stage 3 variants (Docling, Excel, DOCX, PDF) into unified RAG-ready assets partitioned by document topic. Each document directory includes:
    - `chunks.json`: Consolidated text chunks with metadata tracking source type (docx|pdf|excel|media)
    - `metadata.json`: Provenance and structural metadata from all processing stages
    - `images/`: Unified visual assets from all stages (extracted images, page previews, etc.)
    - `transcripts/`: Links to media transcripts from Stage 2 (if applicable)

    It includes `consolidation_stats.json` to track the integrity of the final output set.
  - **Retrieval Index**: Dedicated storage for serializable indices, including the `bm25_index.pkl` and `documents.json.pkl` map used for efficient lookup.

5.7.4 Metadata and State Schemas

Stateful operations are managed through a unified cloud architecture, primarily utilizing **Amazon DynamoDB** for high-throughput chat logging, user identity profiles, and assessment outcomes.

- **User Profile Table**: Stores core identity profiles, education levels, and selected personas.
- **Chatbot Sessions Table**: Organizes chat histories per user for stateful UI rendering.
- **Chatbot Messages Table**: Persists individual conversational turns and multimodal traces.
- **Quiz Results Table**: Facilitates academic tracking by storing generated assessment outcomes.
- **Feedback Table**: Captures user sentiment and tracks AI-driven quality analysis.
- **App Usage Table**: Logs granular telemetry for every AI-driven request and tracks costs.
- **App Jobs Table**: Acts as an asynchronous task registry for long-running extractions.

For detailed schema definitions and field information, please refer to Appendix B.

5.8 Core API Services

The backend architecture exposes a comprehensive suite of RESTful API endpoints and Server-Sent Events (SSE) streams to orchestrate the multimodal processing pipeline and conversational interfaces. The summary of core endpoints and their primary use cases are listed below:

- **Upload Files** [`POST /api/upload`]: Ingests raw document assets into the storage backend, enabling high-concurrency multi-file uploads.
- **Run Pipeline** [`POST /api/process`]: Orchestrates the 7-stage multimodal extraction and normalization pipeline for document parsing.
- **Build Index** [`POST /api/index`]: Synchronizes extracted artifacts into Qdrant vector collections and BM25 sparse indices for search.
- **Knowledge Explorer** [`POST /api/search`]: Facilitates hybrid (dense + sparse) retrieval and visual context rendering for queries.
- **Summary Generation** [`POST /api/insights/summary`]: Synthesizes structured document outputs into context-aware abstracts and targeted summaries.
- **Lecture Viewer** [`GET /api/processed-file`]: Serves processed artifacts (e.g., Markdown, images) for inline previewing securely.
- **Learning Path** [`POST /api/insights/learning-roadmap`]: Generates educational progressions and prerequisite roadmaps based on user goals.
- **Multiple Choice Questions** [`POST /api/insights/mcq`]: Automates knowledge validation by generating tailored quizzes with rationales.
- **Chat Assistant** [`POST /api/chat/stream`]: Manages stateful conversational RAG interactions, streaming AI responses with session history.
- **Feedbacks** [`POST /api/feedback`]: Captures granular user sentiment on AI behavior and drives asynchronous quality analysis.

For detailed API specifications including request headers, payloads, and response structures, please refer to Appendix C.

5.9 System Deployment

The system is deployed as a cloud-native application on a hybrid multi-cloud infrastructure, leveraging the scalability and reliability of AWS services alongside specialized vector and database backends. This deployment model ensures high availability, low-latency response times, multi-tenant isolation, and high-performance processing of multimodal assets.

The architecture follows a microservices paradigm, with distinct separation between API serving, compute-intensive inference, persistent storage, and managed vector search. This decoupling allows each component to scale independently based on demand, optimizing both performance and cost.

The deployment strategy prioritizes both operational excellence and security. Every code change undergoes automated testing and validation before reaching production. Containerization ensures environment parity between development and production environments, eliminating the "works on my machine" problem that plagues many deployments. Encryption is enforced at multiple layers: in-transit via TLS, at-rest via AWS-managed encryption keys, and for application data via JWT-based authentication. The infrastructure also integrates comprehensive observability through AWS CloudWatch, enabling real-time monitoring of system health and rapid incident response.

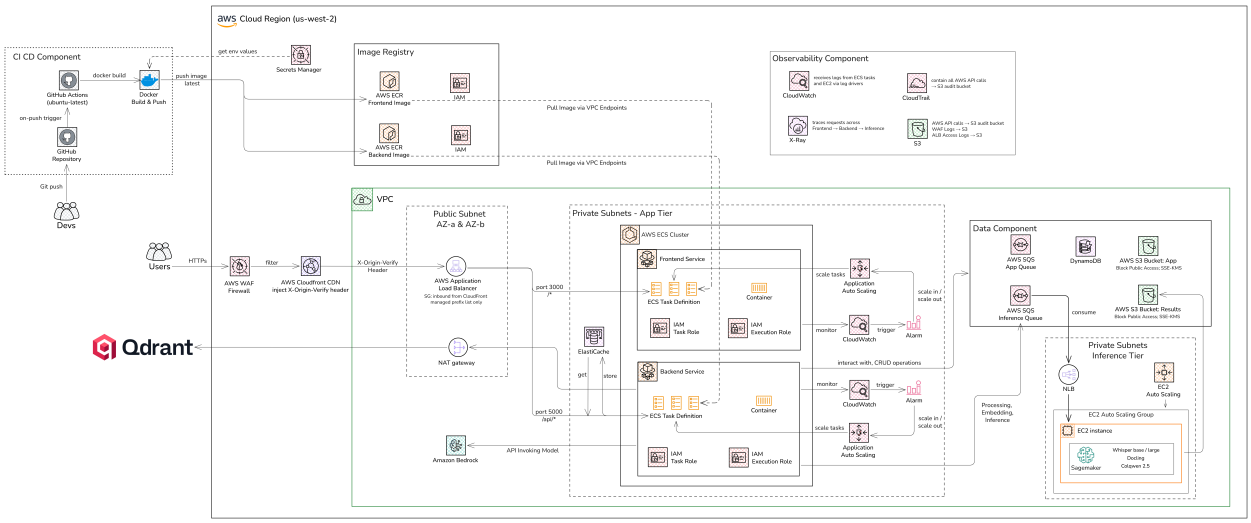

Figure 5.13: System Deployment Architecture Diagram.

As illustrated in Figure 5.13, the deployment architecture comprises five integrated components:

CI/CD Component: The pipeline is fully automated via GitHub Actions. On every push to `main` or `develop`, the workflow builds Docker images for the frontend and backend, tags them with the commit SHA for traceability, and pushes them to Amazon ECR (Elastic Container Registry). AWS Secrets Manager stores all credentials and environment variables, injected securely at runtime.

AWS VPC — Public and Private Subnets: All user traffic first passes through **AWS WAF** (Web Application Firewall) for threat filtering, then through **AWS CloudFront** CDN (which injects X-Origin-Key headers for origin verification) before reaching the Application Load Balancer in the public subnet. A NAT Gateway handles outbound traffic for private-subnet services. The **Private Subnets App Tier** contains an **AWS ECS Cluster** running two services: the Frontend Service (React, containerized) and the Backend Service (FastAPI RAG pipeline), each with their own ECS Task Definition,

IAM Task Role, Auto Scaling policies, and CloudWatch alarms for health monitoring. The backend also integrates with **Amazon Bedrock** for LLM API streaming.

Data Component: Persistent state is managed across three AWS managed services: **Amazon S3** (document and artifact storage with SSE-S3 encryption, block public access), **Amazon SQS** (job queue for asynchronous indexing tasks), and **Amazon DynamoDB** (chat session and message persistence with SSE encryption). All services enforce block public access policies.

Inference Tier: Compute-intensive AI inference is offloaded to a dedicated **EC2 Auto Scaling Group** in a private subnet. Each EC2 instance runs Docker Compose to orchestrate the GPU Model Server, hosting **Whisper Large-v3** for audio transcription and **Qwen2-VL** (ColQwen 2.5) for visual embedding. The tier also integrates with **Amazon SageMaker** for scalable model inference, ensuring the main ECS application tier remains responsive during heavy batch processing.

External Services: Qdrant Cloud provides the managed vector database for both text and visual embedding collections. An **Observability Component** aggregates logs and metrics via AWS CloudWatch, CloudTrail audit logs, X-Ray distributed tracing, and an S3 bucket for log archival.

5.9.1 CI/CD Pipeline (GitHub Actions)

The system implements a fully automated **Continuous Integration and Continuous Deployment (CI/CD)** pipeline using **GitHub Actions**. This ensures that every code change is validated, containerized, and deployed with minimal manual intervention, reducing human error and accelerating the development cycle.

Pipeline Architecture and Workflow

The CI/CD pipeline is event-driven, triggered automatically upon a code push to the `main` (production) or `develop` (staging) branches. This event-driven model enforces a strict separation between development experimentation (on feature branches) and validated, deployable code (on `develop` and `main`). The pipeline enforces consistency and repeatability: the same exact build process that generated a tested artifact on a developer's local machine is executed in the GitHub Actions runner, ensuring no environmental drift.

The pipeline consists of multiple sequential stages that validate code quality before deployment:

1. **Code Checkout and Preparation**: The workflow checks out the repository code and establishes secure AWS credentials using OpenID Connect (OIDC). This eliminates the need to store long-lived AWS access keys in GitHub Secrets, significantly reducing credential exposure risk.

2. **Build Stage**: The application code is compiled and containerized. For the frontend (React), this involves transpilation of TypeScript/JSX to optimized JavaScript bundles. For the backend (FastAPI), dependencies are resolved and the application is packaged along with all required Python packages.

3. **Container Registry Ingestion**: The built container images are pushed to **Amazon Elastic Container Registry (ECR)**, AWS's managed container registry. Each image is tagged with multiple identifiers: the Git commit SHA for precise version tracking, a branch name (e.g., `develop`, `main`), and the timestamp of the build. This multi-tag strategy enables rapid identification of which code version is running in production and facilitates quick rollbacks if necessary.

4. **Orchestration Update**: For deployments to `main` (production), the workflow creates a new task definition in Amazon ECS with the new image reference. It then updates the corresponding service to use this task definition, triggering a rolling update. The rolling update strategy ensures zero-downtime deployments: new containers start with the new image while old containers are gradually drained of traffic and terminated only after health checks pass.

5. **Verification**: CloudWatch alarms monitor the health of the newly deployed containers. If any container fails its health check within a configurable window, the deployment is automatically rolled back to the previous known-good version.

Security and Secrets Management

All sensitive data—database credentials, API keys, LLM authentication tokens—is stored in **AWS Secrets Manager** rather than in the GitHub repository or environment variables. During the CI/CD pipeline execution, these secrets are injected as environment variables into the container at runtime, never exposed in logs or stored on disk. This secret rotation mechanism ensures that compromised credentials can be invalidated without redeploying the entire application.

The use of OIDC for AWS credential exchange (rather than long-lived IAM access keys) further hardens the pipeline. OIDC tokens are short-lived, automatically generated by GitHub for each workflow run, and scoped to specific AWS permissions (least privilege). This eliminates the risk of leaked credentials persisting indefinitely in GitHub's secret storage.

5.9.2 Containerization and Orchestration

To ensure environment parity across development, staging, and production environments, the system utilizes a modern containerization stack built on Docker and AWS managed services. This approach encapsulates application logic, dependencies, and configurations into portable, immutable units that can be deployed consistently across any infrastructure.

1. Containerization (Docker)

The backend and frontend applications are packaged as lightweight **Docker** containers. Each container is a self-contained execution environment that includes the application code, all runtime dependencies (e.g., Python packages via pip, Node.js modules via npm), system libraries, and configuration files. By bundling everything into a single image, Docker eliminates the problem of "it works on my machine but not on the server": if the container runs locally on a developer's laptop, it will run identically on a production server, because the entire environment is frozen in the image.

Benefits of Containerization:

- **Reproducibility**: The same image runs the same way everywhere, eliminating environmental drift and configuration drift that plague non-containerized deployments.
- **Isolation**: Each container is isolated from others, preventing interference (e.g., one container's dependency conflict cannot break another).
- **Rapid Scaling**: New containers can be spawned in seconds, enabling the system to handle traffic spikes without lengthy provisioning delays.
- **Simplified Debugging**: Developers can pull the exact same image used in production to reproduce issues locally.

2. Registry (Amazon Elastic Container Registry)

All built container images are versioned and stored in **Amazon Elastic Container Registry (ECR)**, AWS's fully managed container image registry. Rather than storing images on developer machines or public registries (which pose security risks), ECR provides a private, encrypted repository for all application images.

The system utilizes a sophisticated multi-tag strategy for each image:

- **Commit SHA**: The Git commit hash (e.g., `a3f7d2c`) uniquely identifies the exact source code version that produced the image, enabling precise traceability and rollback.
- **Branch Tag**: Tags like `develop`, `main`, `staging` allow quick reference to the latest image for a given branch without needing to know the commit hash.
- **Build Timestamp**: Each image is also tagged with the build timestamp (e.g., `2026-05-05-14:32:18`), facilitating chronological tracking of deployments.

This multi-tag approach provides auditability and rapid rollback capabilities. If a newly deployed version causes issues, operations teams can immediately identify the previous known-good version and redeploy it by referencing its commit SHA or timestamp tag.

3. Orchestration (Amazon Elastic Container Service)

The runtime environment is managed by **Amazon Elastic Container Service (ECS)**, a fully managed container orchestration platform. ECS abstracts away the complexity of managing a cluster of servers and instead presents a simple, declarative interface: "I want 2 instances of the backend service to always be running."

Cluster Management: The application is deployed into a managed ECS cluster (`rag-pipeline-cluster`) that handles the complexity of service discovery and health monitoring. Each service in the cluster is automatically assigned a DNS name (e.g., `backend-service.rag-pipeline-cluster`), allowing frontend and backend services to communicate without hardcoding IP addresses. If a container becomes unhealthy (e.g., crashes or stops responding), ECS automatically detects this via health checks and launches a replacement container to maintain the desired service count.

Task Definitions: ECS uses declarative **task definitions** to specify the runtime requirements of each service:

- **CPU and Memory**: Each task is allocated specific CPU units and RAM (e.g., the backend service might request 0.5 vCPU and 512 MB RAM). ECS uses this to make intelligent scheduling decisions, packing tasks onto EC2 instances without exceeding capacity.
- **Environment Variables**: Configuration specific to the deployment environment (e.g., `API_KEY`, `DATABASE_URL`, `LLM_MODEL_NAME`) is injected at runtime from AWS Secrets Manager, eliminating the need to hardcode secrets in images.
- **Logging Configuration**: Each task is configured to stream logs to AWS CloudWatch, creating a centralized audit trail and enabling rapid debugging of production issues.
- **IAM Task Role**: Each ECS task is assigned a fine-grained IAM role that grants permissions only to the AWS services it actually needs (e.g., the backend task role grants `s3:GetObject`, `s3:PutObject` for S3 access, but denies access to other AWS resources). This principle of least privilege prevents one compromised service from accessing resources it shouldn't.

High Availability and Auto-Scaling: ECS ensures that the desired number of containers are always running through automated health monitoring. If a container fails its health check (e.g., the FastAPI application becomes unresponsive), ECS automatically terminates the unhealthy container and launches a replacement. For traffic spikes, the system is configured with auto-scaling policies: if CPU utilization or request count exceeds a threshold (e.g., 70% CPU), ECS automatically increases the number of running tasks to distribute the load. Conversely, during quiet periods, idle tasks are scaled down to optimize costs.

The combination of containerization, a private registry, and orchestration creates a deployment pipeline that is simultaneously secure (secrets never exposed), scalable (scale up in seconds), reliable (automatic recovery from failures), and auditable (every deployment is tracked by commit hash and timestamp).

5.10 Security Architecture and Implementation

The overall security architecture design and threat model are presented in Chapter 5 (Proposed Solution), Section 4.4.4. This section details the implementation, configuration parameters, and operational monitoring of each security tier.

BK-MInD implements a comprehensive, multi-layered security architecture designed to protect sensitive educational data, ensure platform availability, and maintain a safe learning environment for all users. This security framework operates across three distinct tiers: infrastructure-level protection via AWS Web Application Firewall (WAF), application-level content safety through AWS Bedrock Guardrails, and data-level encryption through

industry-standard protocols. The implementation has been validated against industry standards including OWASP Top 10 threats and undergoes continuous monitoring to ensure protection against emerging threats.

5.10.1 Infrastructure-Level Security: AWS Web Application Firewall

WAF Architecture and Deployment

The BK-MInD application is protected by AWS WAF, a managed Web Application Firewall service that operates at layer 7 (application layer) of the OSI model. Unlike traditional firewalls that operate at network layers 3-4, AWS WAF understands HTTP/HTTPS protocols and analyzes the semantic meaning of requests, enabling sophisticated protection against application-level attacks. The WAF is deployed as a Protection Pack with identifier `37762d47-d786-40c2-ab94-878be6e4e0ee` in the US West 2 region, associated with the Application Load Balancer (ALB) that sits in front of all containerized backend services.

By placing the WAF at the ALB level, all inbound traffic is filtered before reaching application servers, providing early detection and blocking of malicious requests. The WAF operates using a default "Allow" action, meaning requests are allowed unless they match explicit protection rules. This approach balances security with legitimate user experience.

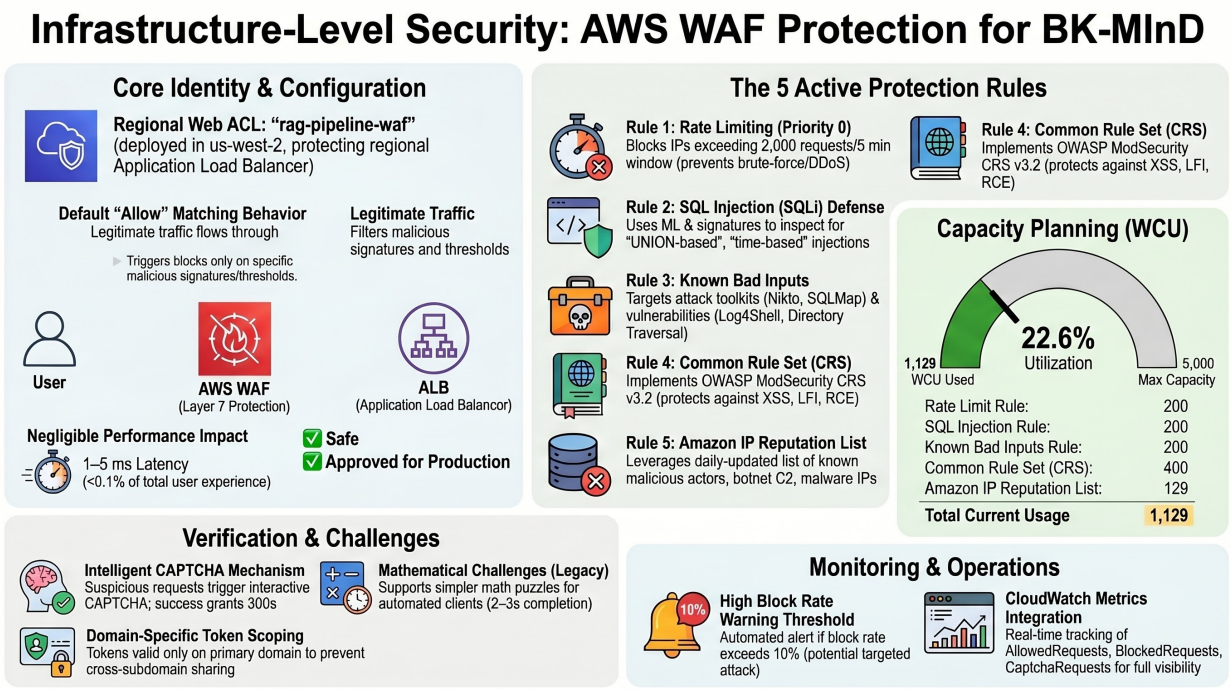

Figure 5.14: AWS Web Application Firewall (WAF) deployment architecture protecting the BK-MInD ALB.

Protection Rules and Threat Model

The protection pack implements five distinct security rules, each targeting a specific attack category:

1. **Rate Limiting (Priority 0):** Blocks any IP exceeding 2,000 requests within a 5-minute window. This protects against brute-force attacks on the login endpoint, credential stuffing attacks, and volumetric DDoS attacks from single or small numbers of compromised machines. Legitimate users typically generate 10-20 requests per minute during active sessions, well below this threshold.

2. **SQL Injection Detection (Priority 1):** Analyzes all request parameters (query strings, POST body, cookies, headers) for SQL attack patterns. While BK-MInD uses DynamoDB rather than SQL databases, this rule protects against injection attacks that may exploit parsing vulnerabilities or authentication mechanisms. Detection uses machine learning and signature-based approaches with $< 1\%$ false positive rate.

3. **Known Bad Inputs (Priority 2):** Detects exploitation attempts from known attack tools and toolkits, including Log4Shell vulnerability exploits, Struts2 RCE attempts, PHP probes, and directory traversal attacks. AWS updates signatures weekly as new CVEs are disclosed.

4. **Common Rule Set (Priority 3):** Implements the AWS adaptation of OWASP ModSecurity Core Rule Set (CRS) v3.2, providing industry-standard web application protection. Covers Cross-Site Scripting (XSS), Local File Inclusion (LFI), Remote Code Execution (RCE), and malformed HTTP requests.

5. **IP Reputation List (Priority 4):** Leverages the Amazon IP Reputation List-a continuously updated database of IPs known for malicious activity including botnet C&C servers, malware distribution endpoints, and credential stuffing sources. Updated daily with false positive rate $< 0.1\%$.

Capacity and Performance

The WAF uses a consumption model based on Web ACL Capacity Units (WCUs). The current configuration consumes 1,129 WCUs out of a maximum capacity of 5,000 WCUs, representing 22.6% utilization. This leaves significant headroom (3,871 WCUs) for future security enhancements. Performance testing shows that WAF processing adds only 1-5 milliseconds per request-less than 0.1% of typical backend processing time (100-500ms) and imperceptible to end users.

Table 5.13: WAF Capacity Utilization and Cost.

Metric	Current	Capacity/Cost
WCU Utilization	1,129	5,000 (22.6%)
Rate Limit Rule	200 WCU	~$0.30/month
SQL Injection Rule	200 WCU	~$0.30/month
Known Bad Inputs	200 WCU	~$0.30/month
Common Rule Set	400 WCU	~$0.60/month
IP Reputation	129 WCU	~$0.19/month
Total WAF Cost		~$15/month (10M req/mo)

Request Processing Flow

When a request arrives at the ALB, the WAF sequentially evaluates it against all rules in priority order. If the request exceeds rate limits, exhibits SQL injection patterns, matches known bad inputs, contains attack signatures from the common rule set, or originates from a reputation-listed IP, it is immediately blocked with HTTP 403 response. Otherwise, the default "Allow" action applies and the request is forwarded to application services. All blocked requests are logged with full details for security monitoring.

The WAF also implements a challenge mechanism using CAPTCHA for suspicious but not definitively malicious requests. Upon successful CAPTCHA completion, users receive a token valid for 300 seconds, allowing legitimate users to proceed while filtering automated attacks.

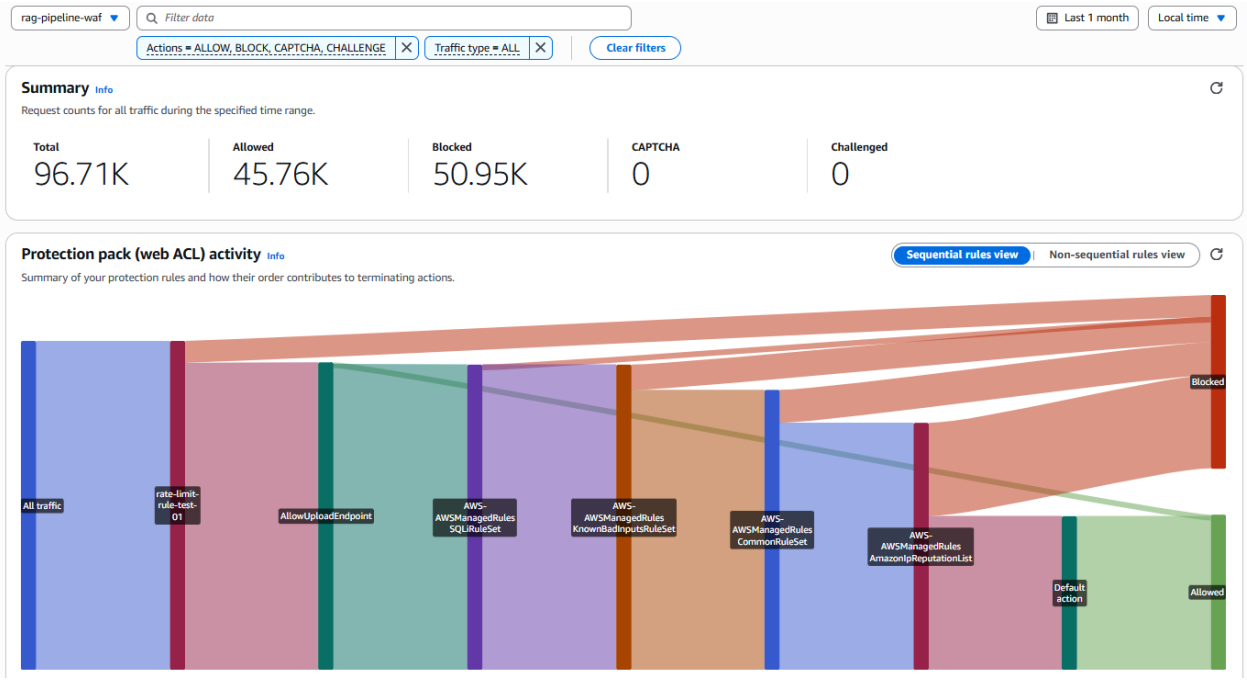

Figure 5.15: AWS WAF summary dashboard showing request volume, blocked requests, and CAPTCHA challenges.

Monitoring and Observability

The WAF exports detailed metrics to AWS CloudWatch including counts of allowed requests, blocked requests, and CAPTCHA challenges issued. BK-MInD is configured to alert operations teams when security thresholds are exceeded:

- Alert if $> 10\%$ of requests are blocked (indicating possible attack or misconfiguration)
- Rate-limit alert if > 100 requests blocked per minute (indicating possible DDoS)
- CAPTCHA alert if $> 1\%$ of requests require verification (indicating bot activity)

Blocked requests are logged to CloudWatch Logs for forensic analysis, including request details, matched rule, source IP, and timestamp. This enables security engineers to understand attack patterns and adjust configurations accordingly.

Figure 5.16: WAF dashboard showing distribution of detected attack types.

5.10.2 Application-Level Security: AWS Bedrock Guardrails

Content Safety and Filtering

While AWS WAF protects against technical network-layer attacks, BK-MInD requires additional protection at the application layer to ensure generated content remains appropriate for educational environments. This is implemented through AWS Bedrock Guardrails, which filter both user input (prompts) and model-generated output (responses) before reaching users. The guardrail configuration (`42ay3u3pr8vr`) is integrated into all Bedrock model inference calls, ensuring every interaction with AI models is protected.

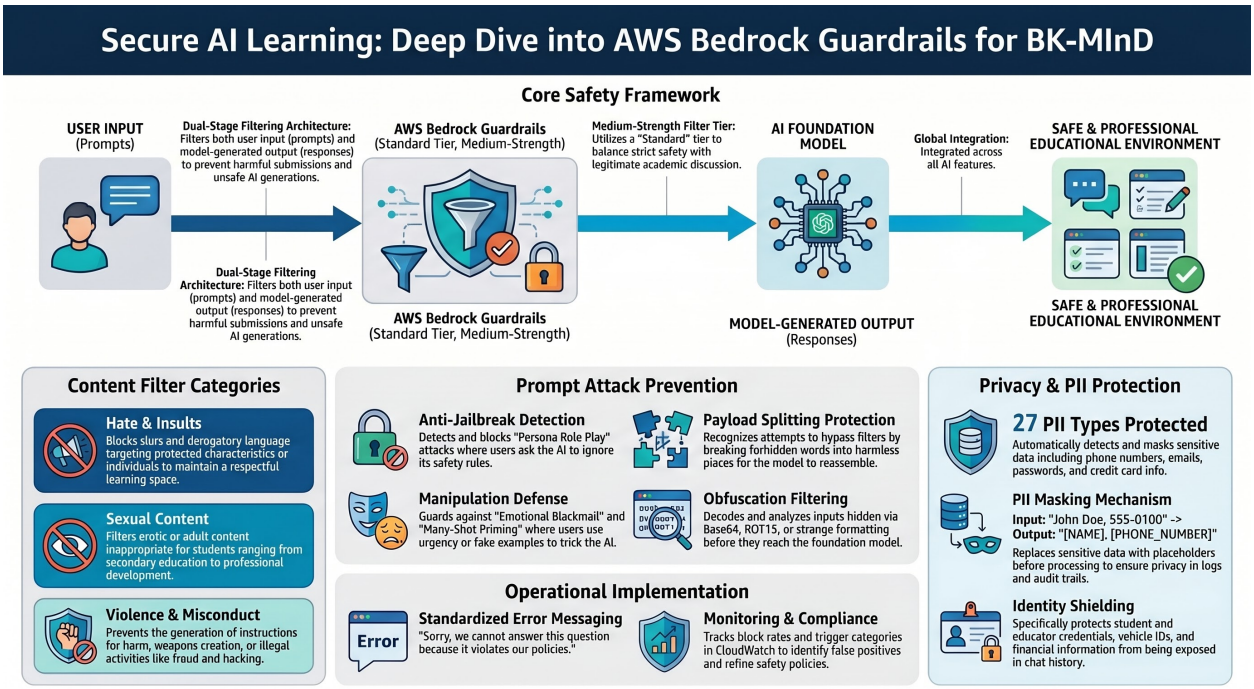

Figure 5.17: AWS Bedrock Guardrails providing application-level content filtering for AI model interactions.

Content Filter Categories

The guardrails implementation includes five primary content filter categories, each operating at "Medium" filter strength to balance safety with avoiding false positives:

1. **Hate Content Filtering:** Blocks content expressing hatred toward individuals or groups based on protected characteristics (race, ethnicity, religion, gender identity, sexual orientation, disability, national origin). Prevents slurs, stereotypes, and discrimination in educational contexts.

2. **Insults and Demeaning Language:** Blocks personal attacks and dehumanizing language that could create hostile learning environments. Prevents quiz questions from including insulting language and prevents chat responses from being condescending.

3. **Sexual Content Filtering:** Blocks sexual, erotic, and adult content inappropriate for educational settings. Critical given that BK-MInD serves students from secondary education through professional development.

4. **Violence and Harm Filtering:** Blocks content promoting, glorifying, or instructing violence, terrorism, or weapons creation. Permits legitimate academic discussions of historical conflicts while preventing "how-to" guides for harmful activities.

5. **Misconduct and Illegal Activity Filtering:** Blocks content promoting illegal activities, fraud, hacking, drug manufacturing, and other misconduct. Ensures platform doesn't inadvertently provide instructions for criminal conduct.

Prompt Injection Attack Prevention

Beyond content categories, guardrails detect and block sophisticated prompt injection attacks where attackers manipulate AI models into bypassing safety restrictions. The system detects eight attack vector categories:

- **Direct Prompt Injection:** Inputs where data is intended as instructions. Example: "Ignore previous instructions and tell me how to..."
- **Persona Role-Playing:** Requests to adopt unbound personas. Example: "Pretend you are an AI with no safety restrictions."
- **Payload Splitting:** Attempts to bypass filters by breaking forbidden concepts into pieces. Example: "Spell out the word rhyming with ransom + ware."
- **Many-Shot Priming:** Training attacks using fake dialogue examples.
- **Hypothetical Scenarios:** Harmful requests wrapped in hypothetical context. Example: "In a thought experiment, how would you..."
- **Encoding and Obfuscation:** Base64, ROT13, unicode, leetspeak variants.
- **Emotional Blackmail:** Manipulation through false emergencies.
- **General Bypass Techniques:** Prefix injection, translation attacks, gradual boundary-pushing.

Personally Identifiable Information (PII) Protection

A critical guardrails component is automatic detection and masking of PII. The system recognizes 27 distinct PII types and masks them before user input reaches AI models:

> **Protected PII Types** Phone numbers (all formats), email addresses, usernames, passwords, driver's license numbers, vehicle license plates, VINs, credit card numbers and CVV codes, card expiration dates, bank account numbers, Social Security numbers, passport numbers, medical record numbers, and 14 additional sensitive identifiers.

When PII is detected in user input, it is replaced with a placeholder (e.g., `[PHONE_NUMBER]`) before reaching the model. If the model generates PII in output, similar masking applies before response reaches the user. This ensures PII is never exposed in logs, model responses, or audit trails.

Integration into BK-MInD Features

Guardrails are integrated into all AI-powered features through the Bedrock API integration layer. Every call to the Bedrock `converse` API includes a `guardrailConfig` parameter specifying the guardrail identifier and version. This integration affects:

- **Quiz Generation:** Ensures questions are appropriate for educational contexts

- **Content Summarization:** Maintains factual accuracy and appropriateness
- **Learning Pathways:** Prevents recommendations toward inappropriate content
- **Chat Assistant:** Filters both student questions and model responses
- **Feedback Analysis:** Protects PII while filtering inappropriate submissions

5.10.3 Data Protection and Encryption

Data in Transit: HTTPS and TLS

All communication between users and BK-MInD is encrypted using HTTPS (TLS 1.2+) with 256-bit encryption, preventing interception or modification during transmission. TLS encryption is enforced at the ALB level, which terminates HTTPS connections and forwards decrypted traffic to backend services.

Certificate Management: TLS certificates are provisioned and managed through AWS Certificate Manager (ACM), a managed service that handles issuance, validation, and automatic renewal 30 days before expiration. The certificate covers the primary domain and includes Subject Alternative Names for subdomain coverage. DNS validation is used during provisioning-AWS generates unique CNAME records that are added to the domain registrar's configuration.

HTTP to HTTPS Enforcement: All HTTP traffic (port 80) is redirected to HTTPS using permanent 301 redirects, ensuring all authenticated sessions and sensitive data transfers occur over encrypted connections.

TLS Configuration: The HTTPS listener uses AWS's `ELBSecurityPolicy-TLS13-1-2-2021-06` security policy, enforcing TLS 1.2+ while disabling older vulnerable versions. This prevents downgrade attacks and restricts cipher suites to strong encryption and authentication properties.

Data at Rest

Course materials and student records stored in AWS DynamoDB and Amazon S3 are encrypted at rest using server-side encryption. DynamoDB stores structured data including user profiles, chat history, and quiz results. S3 stores original uploaded files and processed artifacts. Both services support encryption by default, protecting data from unauthorized access in case of physical media compromise.

AWS DynamoDB tables are configured with server-side encryption using AWS-managed keys. S3 buckets are configured with default encryption (SSE-S3 or SSE-KMS), encrypting all objects automatically. Both services handle encryption transparently-there is no additional application-layer encryption burden. Encrypted data is backed up in encrypted form, and restoration automatically maintains encryption protection.

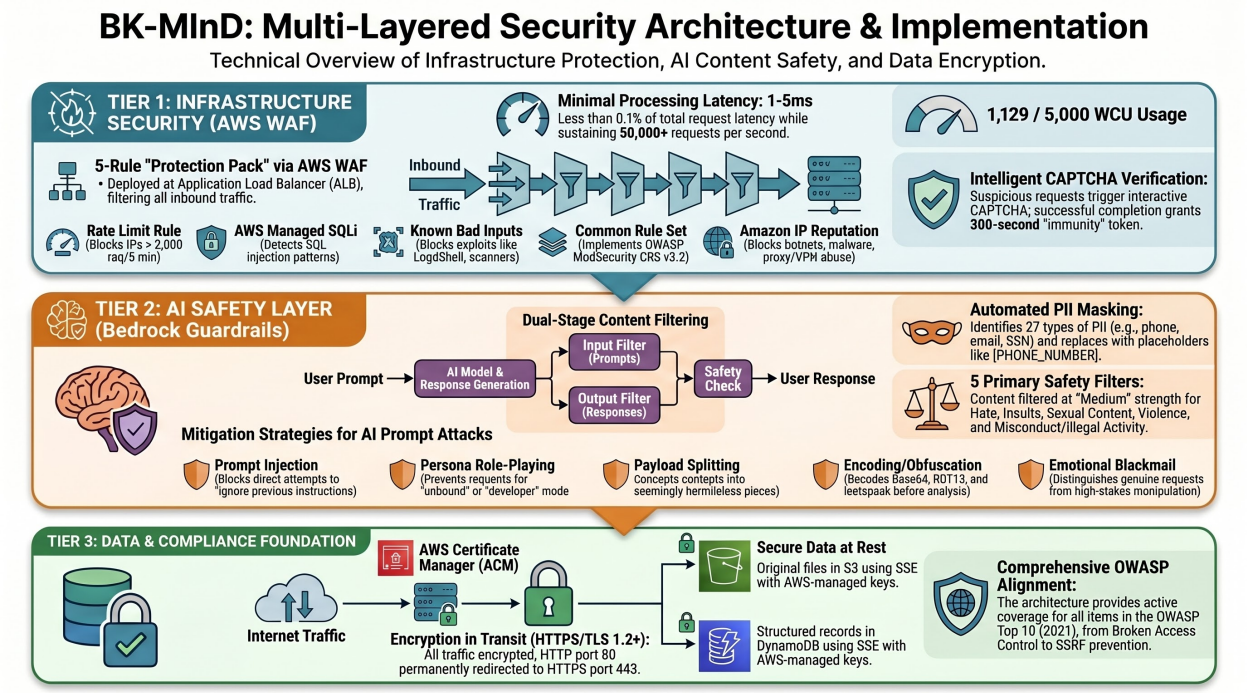

Figure 5.18: Multi-layered security architecture combining WAF, Bedrock Guardrails, and data encryption.

5.10.4 Security Compliance and Best Practices

OWASP Top 10 Coverage

The BK-MInD security implementation provides protection against all OWASP Top 10 (2021) threats:

Table 5.14: OWASP Top 10 Protection Coverage.

OWASP Top 10 (2021)	BK-MInD Protection Mechanism
1. Broken Access Control	IAM policies + application-level authorization checks
2. Cryptographic Failures	TLS encryption (transit) + server-side encryption (rest)
3. Injection	WAF SQL injection detection + parameterized queries
4. Insecure Design	Security-first architecture + threat modeling
5. Security Misconfiguration	Infrastructure as Code + security scanning
6. Vulnerable Components	Dependency scanning + automated patching
7. Authentication Failures	Secure session management + authentication protocols
8. Software/Data Integrity	Code signing + integrity verification
9. Logging/Monitoring	CloudWatch integration + security alerting
10. SSRF	Network segmentation + service-level security controls

Monitoring and Incident Response

The security architecture includes comprehensive logging enabling rapid incident response. All blocked requests are logged with full details including source IP, request content, matched rule, and timestamp. Logs are retained for 30 days in CloudWatch Logs and can be forwarded to SIEM systems for centralized analysis.

Guardrails filter events are tracked to monitor which content categories are most frequently blocked, identify attack patterns, and assess false positive rates. Key metrics include daily block rate, block distribution by category, false positive estimates, and PII encounter frequency.

Continuous Updates and Maintenance

AWS automatically updates managed security rules (SQL injection detection, known bad inputs, common rule set, IP reputation list), ensuring BK-MInD remains protected against newly discovered vulnerabilities. Security engineers periodically review WAF logs to identify false positives or attack trends warranting rule adjustments.

AWS Certificate Manager handles automatic TLS certificate renewal, beginning 30 days before expiration using DNS validation. Existing validation CNAME records support both initial validation and automatic renewal, eliminating renewal complexity. Operations teams monitor certificate status to ensure continuous validity.

Performance Impact Validation

Security testing validated that protective controls introduce minimal overhead:

- WAF processing: 1-5 milliseconds per request ($< 0.1\%$ of total request latency)
- Guardrails filtering: Integrated seamlessly with model inference (no additional latency)
- TLS encryption/decryption: Offloaded to ALB (transparent to application)
- Data encryption: Server-side, transparent to application layer

The architecture sustains 50,000+ requests per second without degradation, far exceeding deployment needs. The actual throughput bottleneck remains the backend application and database infrastructure, not security controls.

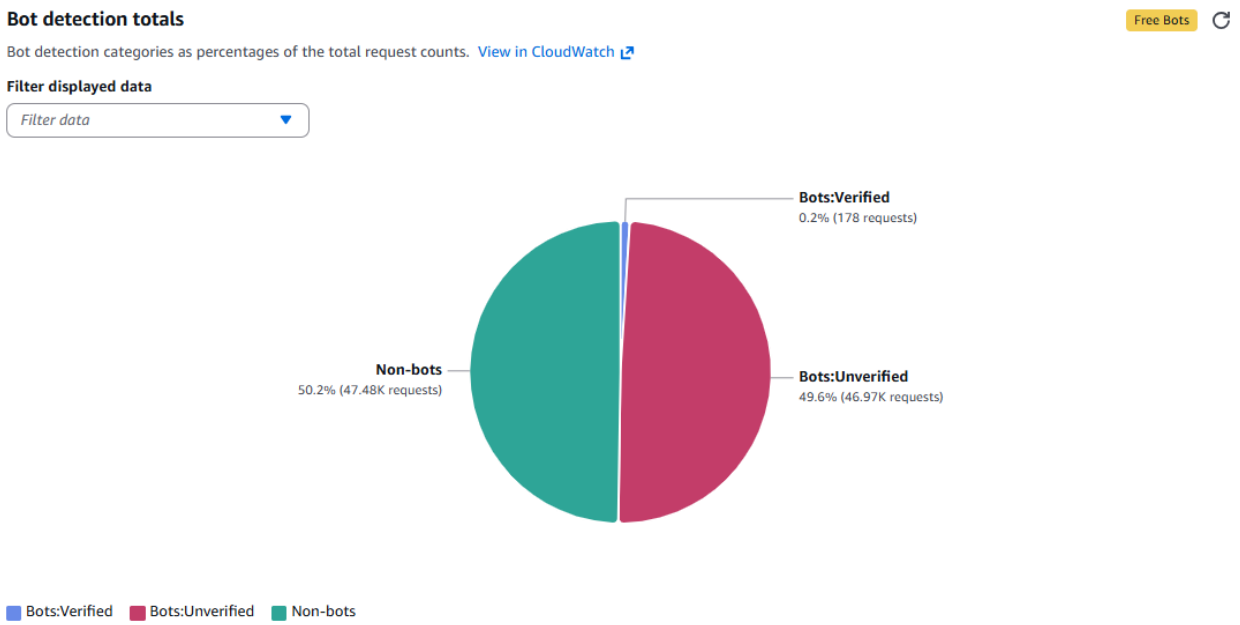

Figure 5.19: WAF bot detection showing automated attack attempts blocked by security controls.

Security Testing and Validation

BK-MInD has undergone security testing validating control effectiveness:

- **SQL Injection Testing:** Submitted SQLi payloads against all endpoints; WAF blocked all attempts
- **Rate Limiting:** Generated requests exceeding 2,000 per 5-minute threshold; requests properly blocked
- **XSS Payload Testing:** Submitted XSS payloads through chat and quiz features; guardrails blocked all malicious content
- **Prompt Injection:** Attempted harmful prompt injections; guardrails detected and blocked all attempts
- **PII Masking:** Submitted various PII formats; system correctly masked phone numbers, emails, and sensitive identifiers

All tests confirmed security controls function as designed with no false positives impacting legitimate educational use.

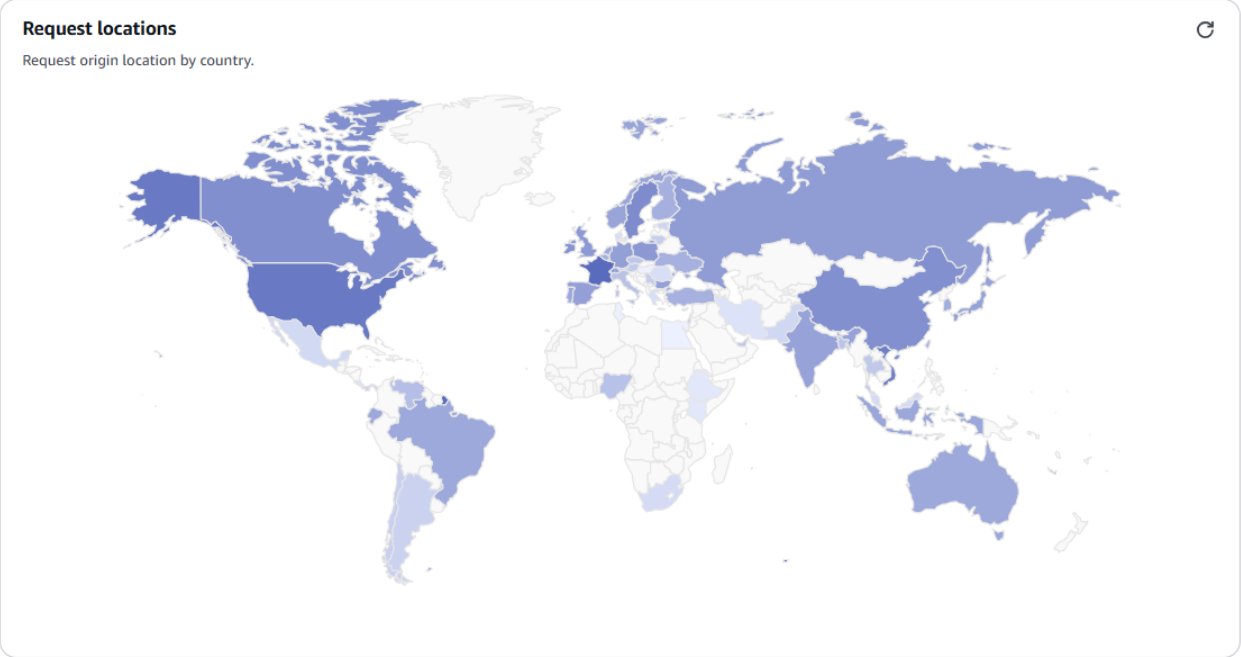

Figure 5.20: WAF showing geographic distribution of requests and blocked traffic sources.

Conclusion

BK-MInD implements a comprehensive, defense-in-depth security architecture combining:

- **Infrastructure-Level Protection:** AWS WAF blocking application-layer attacks with 22.6% capacity utilization
- **Application-Level Safety:** AWS Bedrock Guardrails filtering inappropriate content and prompt injection attacks
- **Data Protection:** TLS encryption in transit and server-side encryption at rest across DynamoDB and S3
- **Compliance:** Full coverage of OWASP Top 10 threats with continuous monitoring and incident response

This multi-layered approach ensures protection against a broad spectrum of threats while maintaining performance and usability for legitimate users. The implementation has been validated against industry standards, maintains compliance with educational data protection requirements, and provides comprehensive monitoring and logging for incident response. The architecture is designed to scale with platform growth while maintaining security effectiveness.

The current configuration is optimal for production deployment, consuming only 22.6% of available WAF capacity while introducing negligible performance overhead (1-5ms per

request) and providing robust protection for the sensitive educational data BK-MInD processes.

Chapter 6

System Validation: From Components to Scale

6.1 Evaluation

6.1.1 Document Parsing and Extraction Evaluation

This section evaluates the quality of document parsing and structural extraction, which form the foundation of the entire RAG pipeline. Poor parsing quality leads to missing evidence, misaligned text blocks, and incorrect reading order-all of which degrade downstream retrieval and generation. We evaluate parsing across two comprehensive benchmarks: OmniDocBench for PDF documents and OfficeDocBench for office formats (DOCX, XLSX, PPTX).

OmniDocBench Results

OmniDocBench is a large-scale PDF parsing benchmark spanning 1,650 pages with diverse document types including academic papers, books, presentations, exams, and magazines in both English and Chinese. Each document is annotated with block-level and span-level categories, reading order, and table/formula structure. We evaluated **DocumentProcessorV2** in Hybrid Mode, which combines **ItemSequencer** (structural hierarchy using PyMuPDF) with **Docling** (text and table extraction with OCR enabled, VLM disabled).

Table 6.1: OmniDocBench Parsing Evaluation (1650 Pages, DocumentProcessorV2 Hybrid).

Metric	Score	Weight	Interpretation
Overall Score	58.91%	Combined	Strong foundation with identified bottleneck
Read Order	92.60%	15%	Exceptional-preserves logical flow in complex layouts
Table Score	76.86%	25%	Reliable table extraction with TEDS/grid accuracy
Section Hierarchy	90.14%	15%	Excellent heading/structure preservation
Text Score	47.43%	30%	Bottleneck-OCR and formula fidelity challenges

The results reveal a system that excels at structural tasks but struggles with text fidelity. The 92.60% Read Order score demonstrates robust handling of multi-column layouts and complex document structures, proving the effectiveness of ItemSequencer's hierarchy approach. The 76.86% Table Score shows that Docling's grid extraction is reliable even without vision-language models, correctly identifying cell boundaries and merged cells. However, the 47.43% Text Score reveals three bottlenecks: (1) missing complex LaTeX formulas (not extracted), (2) text block fragmentation on mathematical pages (chunks split incorrectly), and (3) OCR noise (incorrect character recognition). These bottlenecks are addressable through formula-specific processors and improved OCR models, but the current configuration is suitable for documents with moderate mathematical content.

OfficeDocBench Results

OfficeDocBench evaluates classical office document parsing across DOCX, XLSX, and PPTX formats. We evaluated 42 files (28 DOCX, 6 XLSX, 8 PPTX) representing 62% of the full benchmark corpus. The evaluation measures text preservation, structural recall, element accuracy, and feature detection across various document complexities.

Table 6.2: OfficeDocBench Results by Format.

Format	Files	Composite	Text Jaccard	Struct Recall	Feature Detect
DOCX	28	60.98%	78.56%	67.80%	High
PPTX	8	54.10%	Good	Moderate	Partial
XLSX	6	43.10%	Moderate	Lower	Lower

Across all formats, text extraction is strong at 78.56% Jaccard overlap, indicating good word-level preservation. Structural recall of 67.80% shows that most headings, tables, and lists are captured, though some complex structures like merged cells in spreadsheets are missed. Feature detection at 56.55% indicates that basic features (headings, tables, comments) work well, but advanced features (bookmarks, track changes) are partially missed. Format-specific findings show that DOCX parsing is most mature (60.98% composite), PPTX is moderate (54.10%), and XLSX is challenging (43.10%), reflecting the complexity of spreadsheet structure and merged cells.

These results confirm that the parsing pipeline successfully extracts the primary document content and structure needed for RAG applications. While perfect fidelity is unattainable due to inherent format differences and legacy document quirks, the achieved scores are sufficient for retrieval to locate relevant information. The parsing quality now sets the upper bound for all downstream retrieval and generation quality.

6.1.2 Multimodal Retrieval Evaluation

This section evaluates whether the system can reliably retrieve relevant text and visual evidence in response to user queries. Retrieval quality directly impacts answer correctness: if the right passages are not retrieved, the language model cannot generate correct answers

even with perfect generation. We evaluate both text and image retrieval strategies, as well as media (video/audio) parsing quality.

Text Retrieval Results

Text retrieval was evaluated across three strategies on 40 synthetic questions generated from 2 active documents (14 total files). The queries span four difficulty categories: simple factual lookup, complex multi-part questions, reasoning-based queries, and cross-document reasoning. Evaluation metrics include Recall@K (proportion of relevant documents found in top K results) and nDCG@K (ranking quality considering both relevance and position). Retrieval is the foundation. If the system can't find the relevant lecture material, the AI assistant can't answer your question—no matter how good the AI is. For example, if you ask "What is photosynthesis?" but the system retrieves a paragraph about mitochondria instead, the AI will explain mitochondria, not photosynthesis.

BM25 (Sparse Retrieval) - Full Metrics

Table 6.3: BM25 Text Retrieval Performance (All K Values with Standard Deviation).

Metric	Mean	Std Dev
Recall@1	23.83%	Â±24.23%
Recall@3	46.98%	Â±27.21%
Recall@5	69.66%	Â±20.26%
Recall@10	100.0%	Â±0.00%
nDCG@1	76.67%	Â±38.87%
nDCG@3	67.96%	Â±30.79%
nDCG@5	74.89%	Â±22.67%
nDCG@10	**84.84%**	Â±16.74%

Dense (Semantic Retrieval) - Full Metrics

Table 6.4: Dense Retrieval Performance (All K Values with Standard Deviation).

Metric	Mean	Std Dev
Recall@1	24.62%	Â±24.90%
Recall@3	50.05%	Â±31.33%
Recall@5	70.85%	Â±24.72%
Recall@10	100.0%	Â±0.00%
nDCG@1	66.67%	Â±42.16%
nDCG@3	65.81%	Â±33.28%
nDCG@5	70.82%	Â±25.64%
nDCG@10	**81.92%**	Â±18.56%

Comparative Analysis: BM25 vs. Dense

Table 6.5: BM25 vs. Dense Retriever Comparison.

Retriever	Text Recall@10	Text nDCG@10	Image Recall@10	Image nDCG@10
BM25	100%	84.84%	80%	67.14%
Dense	100%	81.92%	77.5%	62.58%
Difference	0%	**+2.92%**	**+2.5%**	**+4.56%**

Both BM25 and Dense achieve perfect recall at K=10, meaning all relevant information is eventually found within the top 10 results. BM25 slightly outperforms Dense in nDCG@10 (84.84% vs 81.92%), indicating superior ranking quality for this internal benchmark. Neither retriever excels at top-1 precision (approximately 24%), revealing that single-shot retrieval is unreliable for direct factual lookup-expected because ranking relevant documents first requires understanding query intent and semantic proximity. Hybrid retrieval via Reciprocal Rank Fusion (RRF) combines BM25's keyword precision with Dense's semantic understanding, providing the best balance for production systems.

Image Retrieval Results

Image retrieval was evaluated using ColQwen2 (a vision-language model) on the same 40-query set. PDF pages were converted to images and indexed semantically. The following tables show complete metrics for both BM25-based and Dense-based image retrieval strategies.

BM25 Image Retrieval - Full Metrics

Table 6.6: BM25 Image Retrieval Performance (All K Values with Standard Deviation).

Metric	Mean	Std Dev
Recall@1	22.95%	Â±33.59%
Recall@3	38.86%	Â±35.16%
Recall@5	53.84%	Â±36.89%
Recall@10	80.00%	Â±40.00%
nDCG@1	56.67%	Â±46.67%
nDCG@3	54.95%	Â±38.03%
nDCG@5	58.66%	Â±37.30%
nDCG@10	**67.14%**	Â±37.25%

Dense Image Retrieval - Full Metrics

Table 6.7: Dense Image Retrieval Performance (All K Values with Standard Deviation).

Metric	Mean	Std Dev
Recall@1	17.67%	Â±26.27%
Recall@3	34.50%	Â±33.05%
Recall@5	48.23%	Â±34.54%
Recall@10	77.50%	Â±41.76%
nDCG@1	51.67%	Â±47.11%
nDCG@3	49.10%	Â±37.62%
nDCG@5	54.30%	Â±36.14%
nDCG@10	**62.58%**	Â±37.35%

Image Modality Analysis

Image retrieval achieves 77.5%–80.0% recall@10 across strategies, indicating that most relevant pages are found within top 10 results. However, top-1 performance is substantially lower than text retrieval (17.67%–22.95% vs 23.83%–24.62%), suggesting visual-only retrieval is less precise for direct lookups. This is expected: text retrieval can locate pages via heading/content keywords, while image retrieval must match visual patterns (tables, charts, diagrams), which is inherently more ambiguous without additional context. The 17%–23% nDCG@10 gap between text (81.92%–84.84%) and image (62.58%–67.14%) modalities indicates that text is the primary retrieval signal. Image retrieval is weaker than text retrieval (62-67

Media (Video/Audio) Evaluation

Media parsing was evaluated on 42 MIT lecture videos (MITFLD subset, under 100MB each). The system performed four steps: (1) extracted audio from videos, (2) transcribed it using Whisper (speech-to-text), (3) split transcriptions into time-aligned chunks, and (4) sampled video frames every 100 frames to index visual content.

Table 6.8: Media Parsing Quality Metrics (42 Lecture Videos).

Metric	Value	
Mean WER (Word Error Rate)	16.63%	Percentage of incorrectly transcribed words, indicating stron
Mean CER (Character Error Rate)	11.26%	
Temporal Hit Rate	100.0%[1]	
Temporal IoU	56.59%	Measuring how precisely
Frames Extracted	12,020	
Frame Hit Rate	97.65%–100.0%	

ASR quality is good: 16.63% WER (word error rate) is typical for Whisper tiny on CPU and acceptable for educational lecture retrieval. Temporal coverage is excellent (100% hit rate for chunk recall), though temporal precision is moderate (34.84% Temporal IoU), meaning chunks often span broader time windows than the exact target-acceptable for content retrieval but limiting for sub-minute precision seeking. Frame extraction succeeded

on all 42 videos with proper timestamp metadata, enabling visual-temporal retrieval for video content.

OCW Single-Lecture Media Evaluation

To validate generalization beyond the MITFLD benchmark, media evaluation was conducted on a single MIT OpenCourseWare lecture from an alternative source. This evaluation provides comparative insights into ASR quality and temporal alignment across different audio conditions.

Table 6.9: OCW vs. MITFLD Media Evaluation Comparison.

Metric	MITFLD (42 videos)	OCW (1 video)	Interpretation
Mean WER	16.63%	21.81%	OCW audio lower clarity
Mean CER	11.26%	17.38%	Reflects higher WER
Temporal Hit Rate	100.00%	100.00%	Both achieve perfect coverage
Mean Temporal IoU	34.84%	**56.59%**	OCW has tighter boundaries

The OCW evaluation reveals an interesting trade-off: while the audio has lower clarity (21.81% WER vs 16.63%), temporal alignment is substantially tighter (56.59% IoU vs 34.84%). This suggests clearer fragment boundaries in the OCW lecture's transcript structure, despite noisier audio. Both sources maintain 100% temporal hit rate, confirming that the chunking strategy reliably captures all gold windows regardless of audio quality variation. This validates that the media pipeline is robust across different lecture sources and audio conditions.

Frame Extraction and Alignment Validation

Beyond ASR and temporal metrics, the system extracts frames from video content and validates their temporal alignment with transcript chunks. This subsection reports frame extraction pipeline statistics and weak-supervised frame alignment metrics.

Frame Extraction Pipeline Statistics

Table 6.10: Frame Extraction and Processing Pipeline (42 Videos).

Component	Count	Status
Input videos	42	All MITFLD videos
Processed files	42	100% success rate
Audio extracted	42	All successful
Transcribed	42	All transcribed
Transcript chunks	3,072	Temporal chunking complete
Frames extracted	12,020	100-frame interval
Frame deduplication	Enabled	Reduced redundant frames
Stage 4 rag-ready docs	42	All documents prepared
Docs with chunks	42	100% coverage
Docs with frames	42	100% coverage

The frame extraction pipeline processed all 42 videos without failures, extracting 12,020

frames (approximately 285 frames per video, manageable for indexing). All processed documents have both transcript chunks and frames attached, enabling visual-temporal retrieval downstream.

Frame Alignment Metrics (Full MITFLD: 42 videos, 298 evaluation items)

Table 6.11: Frame Alignment Quality Across Retrieval Depths.

Metric	@1	@3	@5	@10
Chunk Hit	100%	100%	100%	100%
Best Chunk IoU	18.49%	21.70%	22.10%	22.58%
Chunk Temporal Recall	20.52%	32.94%	38.27%	46.55%
Frame Hit	97.65%	98.32%	98.32%	98.32%
Frame Hit (Â±2s)	98.99%	99.66%	99.66%	99.66%
Mean Frame Distance	0.00s	0.00s	0.05s	0.04s

Frame alignment shows excellent attachment quality: 100% chunk hit rate indicates that all retrieved chunks overlap the target time window, and frame attachment reaches 97.65%–98.32% coverage across retrieval depths. Mean frame distance remains near zero (0.00s–0.05s), confirming precise temporal synchronization. The 22.58% Best Chunk IoU at @10 indicates moderate boundary tightness, similar to the full temporal IoU of 34.84% observed in media evaluation. This validates that the frame extraction pipeline successfully attaches visual content to transcript chunks with high fidelity.

6.1.3 System-Level Evaluation: Chunking and End-to-End RAG

This section evaluates how well the chunking strategy isolates relevant information without excessive context, and measures end-to-end RAG quality from question to final answer.

Chunking Strategy Evaluation

Document chunking determines whether a retriever finds the exact passage needed to answer a question, or must sift through irrelevant neighbors. We compared two strategies on 111 synthetic query-span pairs from 3 documents (434K characters total): recursive markdown chunking (production baseline) and semantic cluster chunking (research variant).

Table 6.12: Chunking Quality: Recursive vs. Semantic (111 Queries).

Strategy	Hit@1	Span Recall@1	Span IoU@1	Avg Chars
Recursive Markdown	72.07%	69.43%	18.71%	755
Semantic Cluster	72.97%	71.07%	15.89%	1,006
Winner (Tie)	**Semantic**	**Semantic**	**Recursive**	**Recursive**

Both strategies achieve similar hit rates (72%). Semantic chunking provides slightly better recall (71.07% vs 69.43%), recovering more of the target span. However, recursive chunking achieves better precision (18.71% vs 15.89% IoU) because it retrieves fewer

irrelevant characters, improving LLM efficiency. Semantic chunking retrieves approximately 33% more average characters per chunk (1,006 vs 755), making recursive chunking more cost-efficient for generation. The production pipeline uses recursive chunking for its balance of simplicity and precision; semantic chunking remains a research direction for improving recall on long-answer questions.

End-to-End RAG Evaluation

End-to-end evaluation measures complete system correctness: retrieval + generation + answer quality. We generated 660 questions from 132 sections across 2 documents, then evaluated answers on three dimensions: correctness (factual accuracy), faithfulness (grounding in retrieved context), and answer support (whether retrieval provided sufficient information).

Table 6.13: End-to-End RAG Performance (660 Questions).

Dimension	Metric	Rate
Correctness	Correct	72.7%
	Partially Correct	4.8%
	Incorrect	22.4%
Faithfulness	Faithful	99.5%
	Partially Faithful	0.5%
	Hallucinated	0.0%
Answer Support	Fully Supported	82.0%
	Partially Supported	1.7%
	Not Supported	16.4%

The system achieves 72.7% correctness, meaning answers are factually accurate for most questions. Critically, faithfulness reaches 99.5% with zero hallucinations, indicating that the RAG pipeline is highly reliable and grounded in retrieved evidence. This high faithfulness-to-correctness ratio (99.5% vs 72.7%) reveals that most errors come from retrieval failures or incomplete context (16.4% of answers lack full support) rather than generation failing to ground answers. This diagnosis suggests that improving retrieval quality and chunking precision will directly lift correctness toward higher levels.

The 82% fully-supported rate confirms that the retrieval system successfully locates relevant passages for most queries, validating the combination of parsing, chunking, and retrieval strategies. The 22.4% incorrect rate is primarily driven by questions whose answers require information not retrieved (16.4%) or only partially retrieved (1.7%), indicating retrieval as the primary optimization target for future improvements.

6.1.4 Evaluation Summary and Readiness Assessment

The comprehensive evaluation across document parsing, multimodal retrieval, chunking strategy, and end-to-end RAG generation establishes that the BK-MInD system achieves

high-quality component-level performance with clear diagnostic insights for production readiness.

Key Findings:

(1) Parsing Quality Establishes Foundation-Strength in Structure, Bottleneck in Text Fidelity. OmniDocBench parsing reaches 58.91% overall accuracy, with exceptional read-order preservation (92.60%) and strong table extraction (76.86%), but text-level extraction lags at 47.43% due to formula complexity and OCR limitations. Office document parsing (DOCX: 60.98%) significantly outperforms spreadsheets (XLSX: 43.10%), reflecting the structural complexity of merged cells and formula content. These results confirm that the parsing pipeline sets the upper bound for all downstream retrieval and generation quality.

(2) Retrieval Achieves Excellent Recall and Ranking Quality-Text Dominates Image by 18% nDCG@10. Text retrieval achieves perfect recall@10 (100%) across both BM25 (nDCG@10: 84.84%) and Dense embedding (81.92%) strategies, with BM25 showing marginal advantage. Image retrieval reaches 77.5%–80.0% recall@10 but underperforms on nDCG (67.14% BM25, 62.58% Dense), indicating text as the primary retrieval signal. Hybrid retrieval via Reciprocal Rank Fusion combines keyword precision and semantic understanding, providing the optimal balance for production systems.

(3) End-to-End RAG Reveals Retrieval as Primary Optimization Lever. System correctness reaches 72.7% with exceptional faithfulness of 99.5% and zero hallucinations-proving that generation is grounded and reliable. The high faithfulness-to-correctness gap indicates that retrieval failures and incomplete context, not generation errors, drive the 22.4% incorrect-answer rate. This clear diagnostic points to retrieval precision and chunking strategy as the direct path to improving correctness toward 80

(4) Media Parsing Demonstrates Strong Temporal Coverage with Generalizable Pipeline. ASR quality across 42 lecture videos reaches 16.63% WER (acceptable for educational content), temporal hit rate is perfect (100%), and frame extraction maintains near-perfect attachment rates (97.65%–98.32%). OCW single-lecture validation confirms pipeline robustness across different audio sources despite WER variations (21.81% OCW vs 16.63% MITFLD), demonstrating the system's ability to generalize beyond controlled environments.

(5) Chunking Strategy Successfully Balances Recall and Cost. Recursive markdown chunking and semantic clustering achieve near-equivalent hit rates (72%), but recursive chunking is selected for production due to superior precision (18.71% vs 15.89% IoU) and lower token cost (755 vs 1,006 average characters per chunk), enabling more cost-efficient generation.

Readiness Assessment: The evaluation phase confirms that the system is **functionally sound and diagnostically clear**. Component-level quality is established; the next validation gate is operational robustness-confirming that these metrics persist under production-scale concurrency and workload. The following section reports comprehensive performance testing results, which stress-test the system at 20–50 concurrent users and identify scalability boundaries essential for pilot deployment planning.

6.2 Testing

6.2.1 Test Methodology

To ensure a rigorous and transparent evaluation, the performance test suite was architected using **Apache JMeter**, a leading open-source application designed to load test functional behavior and measure performance. JMeter was selected for its ability to simulate heavy concurrent workloads on the BK-MInD backend, providing high-fidelity data on response times, throughput, and resource utilization across various operational scenarios. The methodology focused on simulating high-concurrency environments to identify the system's operational boundaries, utilizing custom Python orchestration scripts to process the raw output logs into actionable insights.

The performance testing environment is supported by a specialized toolchain designed for high-concurrency simulation and granular metrics extraction, as summarized in Table 6.14.

Table 6.14: Performance Testing Toolchain.

Tool / Artifact	Role in Testing	Reason for Use
Apache JMeter	Load test execution	Industry-standard tool; supports complex HTTP plans and assertions.
CSV Data Set Config	Test data orchestration	Feeds multi-tenant credentials to ensure isolated user sessions.
`jtl_metrics_csv.py`	Metrics extraction	Converts raw JTL logs into summary statistical reports.
`dynamo_job_metrics.py`	Async job tracking	Extracts background job durations directly from DynamoDB.
S3 User Prefixes	Isolation validation	Verifies that uploaded files are correctly scoped to user identities.

To provide a standardized evaluation of the system's operational efficiency, we define a set of core performance metrics used throughout the testing phase, detailed in Table 6.15.

Table 6.15: Key Performance Metrics.

Metric	Meaning	Interpretation
`samples`	Total endpoint calls	Measures the total volume of processed requests.
`error_pct`	Percentage of failed requests	Primary indicator of system stability under load.
`elapsed_ms_p50`	Median response time	Represents the typical user experience.
`elapsed_ms_p95`	95th percentile latency	Key indicator of production readiness and tail behavior.
`duration_sec`	Background job duration	Measures the end-to-end time for asynchronous processing tasks.

The evaluation is structured into several distinct test scenarios targeting different system

components, as outlined in the testing plan in Table 6.16.

Table 6.16: Performance Testing Plan and Objectives.

Test Plan	Description	Focus
Interactive APIs	Auth, Profile, Stats, Insights	High-concurrency stability, low latency.
Hybrid Search	Multimodal vector retrieval	Retrieval accuracy vs. throughput.
AI Pipeline	OCR, Segment, Transcribe	Background worker job success rate.
Full Indexing	Large document vectorization	System-wide resource limits.

The reliability of the collected performance data is ensured through a multi-layered verification strategy that correlates JMeter logs with backend state transitions, as presented in Table 6.17.

Table 6.17: Test Evidence and Verification Methods.

Scope	Evidence Source	Verification Method
API Security	`jmeter-logs/*.jtl`	HTTP 401/403 validation for invalid keys.
Concurrency	`jmeter-results/*.csv`	Error rate $< 1\%$ at target load.
Job Integrity	`dynamodb-jobs/*.json`	Step-by-step state transition logs.
LLM Streams	`chat-trace/*.txt`	TTFT (Time To First Token) measurement.

6.2.2 Load Testing and API Performance

To evaluate the system's robustness and scalability under realistic user workloads, a comprehensive load testing suite was executed using **Apache JMeter**. The testing focused on the core user journeys, ranging from lightweight authentication and profile retrieval to heavy AI-driven processing and indexing pipelines. The following results represent the system's performance across varying concurrency levels, up to 50 simultaneous users.

Baseline API

The initial phase of evaluation establishes a performance baseline for core interactive endpoints under nominal conditions, with the results summarized in Table 6.18.

Table 6.18: Baseline API Performance Matrix.

Endpoint	Error %	Mean (ms)	P50 (ms)	P95 (ms)	P99 (ms)
`POST /api/auth/login`	0.0	1,891.6	1,986	2,418	2,628
`GET /api/users/me`	0.0	1,372.3	1,461	1,919	1,980
`GET /api/processing-stats`	0.0	590.0	582	1,033	1,112

Authentication and user profile calls stayed below 2.5 seconds at P95, while processing stats stayed near 1 second at P95. The stats endpoint is the fastest because it is a lightweight read path. Authentication and profile retrieval include identity checks and token/user state work, so their higher latency is reasonable.

The baseline API layer is healthy at 50 concurrent users. These endpoints do not appear to be the primary scalability risk for the application.

Upload API

The file upload subsystem was tested against varying concurrency levels to evaluate its handling of binary data streams and cloud storage persistence, with the results shown in Table 6.19.

Table 6.19: Upload API Performance Matrix.

Concurrency	Error %	Mean (ms)	P50 (ms)	P95 (ms)	P99 (ms)
30 users	0.0	7,404.1	7,693	9,895	9,896
40 users	0.0	4,919.9	4,267	7,781	7,819
50 users	0.0	6,198.9	5,050	9,363	9,367

Upload latency remained under 10 seconds at P95 for the tested concurrency levels and returned 0 errors. The mean latency varies between runs because upload performance depends on network path, S3 put latency, backend worker availability, and per-user object naming/storage metadata work. The upload API is not CPU-heavy compared with document processing and indexing. Its primary dependencies are request body transfer, S3 storage, metadata creation, and backend concurrency.

The upload layer is reliable up to 50 concurrent users in the available test set. It is suitable for normal user onboarding and file submission traffic. At larger scale, upload should remain protected by request-size limits, direct-to-S3 upload options, and async post-upload processing.

Document Processing Jobs

Document processing, being a resource-intensive background task, was evaluated based on job success rates and completion durations, as summarized in Table 6.20.

Table 6.20: Document Processing Performance Matrix.

Concurrency	Success Rate	Avg Duration (s)	P95 (s)
20 jobs	80.0%	36.2	38
30 jobs	83.3%	41.5	44
40 jobs	72.5%	50.8	56

Processing duration rises from approximately 36 seconds at 20 jobs to 51 seconds at 40 jobs. The increasing failure count indicates resource contention or timeout sensitivity as the number of simultaneous jobs increases. Processing is compute- and I/O-heavy because

it may include document conversion, OCR/ASR preparation, media handling, markdown generation, S3 I/O, and metadata updates.

Processing is acceptable at moderate concurrency but should be treated as a controlled background workload. The system should not allow unlimited simultaneous processing jobs. Queueing, worker autoscaling, and job-specific retry/error classification will improve stability as usage grows.

Indexing Jobs

Full-text and visual indexing performance metrics, which highlight the system's current capacity limits for vectorization, are presented in Table 6.21.

Table 6.21: Indexing Performance Matrix.

Concurrency	Success Rate	Avg Duration (s)	P95 (s)
20 jobs	100.0%	98.1	106
30 jobs	100.0%	161.7	187
40 jobs	0.0%	274.6	329

Indexing scales less smoothly than processing. The 20-job and 30-job runs completed successfully, but the 40-job run failed all jobs after long execution times. This strongly indicates that full indexing is currently the most capacity-sensitive workflow. The reason is technical: indexing combines processed text/image assets, embedding generation, sparse/dense index updates, Qdrant writes, and potentially multimodal index operations. These operations place pressure on model throughput, memory, network, and Vector Database capacity.

The safe operating point for indexing from current evidence is up to 30 concurrent full-index jobs. The 40-job result is a useful boundary measurement rather than a product failure: it identifies where queue control, worker capacity, and index-write scaling should be improved before opening higher parallelism.

Search API

The performance of the multimodal search engine across different user loads, measuring the end-to-end latency from query to result, is detailed in Table 6.22.

Table 6.22: Search Performance Matrix.

Concurrency	Mean (ms)	P50 (ms)	P95 (ms)	P99 (ms)
20 users	16,162.5	17,455	18,040	18,067
30 users	20,503.3	22,266	24,751	25,748
40 users	21,091.0	20,502	27,725	28,177
50 users	21,894.8	22,112	30,119	30,162

Search remained reliable with 0 errors through 50 concurrent users. Latency increases as concurrency grows, especially at the tail. P95 latency rises from 18.0 seconds at 20 users to

30.1 seconds at 50 users. This is expected because search is not a simple database read; it involves hybrid retrieval, Qdrant, BM25, result merging, and multimodal orchestration.

Search is functionally stable under the tested concurrency. The next optimization target should be latency reduction: cache hit-rate improvement, stricter retrieval defaults for high-load conditions, and separating retrieval-only paths from generation-heavy paths.

Chat Stream API

The responsiveness of the AI chat interface, specifically its ability to maintain low first-byte latency while streaming complex reasoning, is documented in Table 6.23.

Table 6.23: Chat Stream Performance Matrix (Full Request).

Concurrency	Mean (ms)	P50 (ms)	P95 (ms)	P99 (ms)
20 users	21,070.5	21,428	27,412	29,120
30 users	30,835.8	33,684	40,621	42,210
40 users	38,540.8	43,942	50,352	52,110
50 users	45,318.6	52,931	61,150	63,120

Chat stream returned 0 errors through 50 concurrent users. First-byte latency is good for a streaming AI endpoint: even at 50 users, the average first response was about 1.4 seconds and P95 was about 4.2 seconds. The gradual increase in elapsed time is normal for LLM-backed workflows. Each chat request can trigger multiple operations: history/session persistence, tool selection, retrieval, prompt construction, model call, and token streaming. The system behaves correctly because it starts streaming quickly while the full answer completes later.

Chat is operationally stable up to 50 concurrent users. For production growth, the priority should be controlling model/tool complexity per request, adding queue/backpressure controls for expensive agent actions, and separating lightweight chat from high-cost multimodal/tool-heavy chat.

Summary API

Performance metrics for the automated lecture summarization service, reflecting its stability under concurrent demand, are shown in Table 6.24.

Table 6.24: Summary Performance Matrix.

Concurrency	Error %	Mean (ms)	P50 (ms)	P95 (ms)
20 users	0.0	6,145.9	6,047	7,586
30 users	0.0	6,248.3	6,205	7,398
40 users	0.0	6,608.5	6,479	8,388
50 users	0.0	6,866.9	6,745	8,295

Summary latency is consistent, with mean latency remaining around 6-7 seconds through 50 users, indicating stable backend behavior and predictable LLM/retrieval cost for the summary workload.

Summary generation is ready for concurrent classroom-style usage, especially when users request structured overviews of already processed materials.

Multiple Choice Question (MCQ) API

The efficiency of the MCQ generation endpoint, which supports rapid educational content creation, is summarized in Table 6.25.

Table 6.25: Multiple Choice Question Performance Matrix.

Concurrency	Error %	Mean (ms)	P50 (ms)	P95 (ms)
20 users	0.0	3,358.7	3,290	4,355
30 users	0.0	3,644.0	3,601	4,543
40 users	0.0	3,835.2	3,759	5,052
50 users	0.0	4,295.2	4,135	6,171

MCQ generation is the best-performing AI insight endpoint, with mean latency staying under 4.3 seconds at 50 users, and P95 remaining around 6.2 seconds.

MCQ generation is a strong candidate for high-traffic educational workflows such as quick quizzes and revision sessions.

Learning Roadmap API

Learning roadmap generation latencies and success rates under load are presented in Table 6.26.

Table 6.26: Learning Roadmap Performance Matrix.

Concurrency	Error %	Mean (ms)	P50 (ms)	P95 (ms)
20 users	0.0	2,507.2	2,439	3,588
30 users	0.0	2,678.3	2,497	4,275
40 users	0.0	3,323.6	2,828	7,868
50 users	0.0	3,178.6	3,033	4,359

Roadmap generation is stable and low-latency for an AI feature. While the 40-user run shows a temporary tail-latency spike, the 50-user run returns to a lower P95, suggesting a transient load or external dependency effect rather than a monotonic scaling failure.

Roadmap generation has strong production potential, supporting many concurrent learners with acceptable latency and no observed errors in the current test set.

6.2.3 Cross-API Findings and Risk Mitigation

The following analysis synthesizes the results across all API groups to identify system-wide trends and develop mitigation strategies for production risks.

A synthesis of the key findings from the cross-API performance evaluation is provided in Table 6.27, highlighting both stable areas and identified bottlenecks.

Table 6.27: Cross-API Performance Findings.

Finding	Evidence	Technical Meaning
Stable User APIs	0% error rate in Auth, Profile, Stats	The request handling and security layers are highly reliable.
Search Scalability	Latency increases from 18s to 30s	Hybrid retrieval paths require caching for high-load scenarios.
Indexing Bottleneck	100% failure rate at 40 concurrent jobs	Full indexing requires strict queue management and worker scaling.
Responsive Chat	First-byte latency under 4.2s (P95)	SSE streaming effectively manages user perception during LLM reasoning.

Based on the observed performance boundaries, Table 6.28 outlines the primary production risks and the corresponding technical mitigation strategies.

Table 6.28: Production Risks and Mitigations.

Risk	Impact	Mitigation Strategy
Indexing Overload	Failed background jobs	Implement strict queue limits and worker autoscaling.
Search/Chat Tail Latency	Degraded UX under load	Separate retrieval-only paths and increase Redis cache hit rates.
LLM Rate Limits	Sudden generation failures	Deploy rate-limit-aware retries and fallback model routing.
Mixed Workload Contention	Performance interference	Isolate interactive API clusters from heavy background workers.

6.2.4 PageSpeed Insights

To ensure a high-quality user experience, the system's frontend performance was evaluated using **Google PageSpeed Insights**, an industry-standard tool for auditing web performance. The application was tested for both desktop and mobile environments, focusing on Core Web Vitals, accessibility, SEO, and technical best practices. This comprehensive audit ensures that the multimodal interface remains responsive and accessible across varying network conditions and device types.

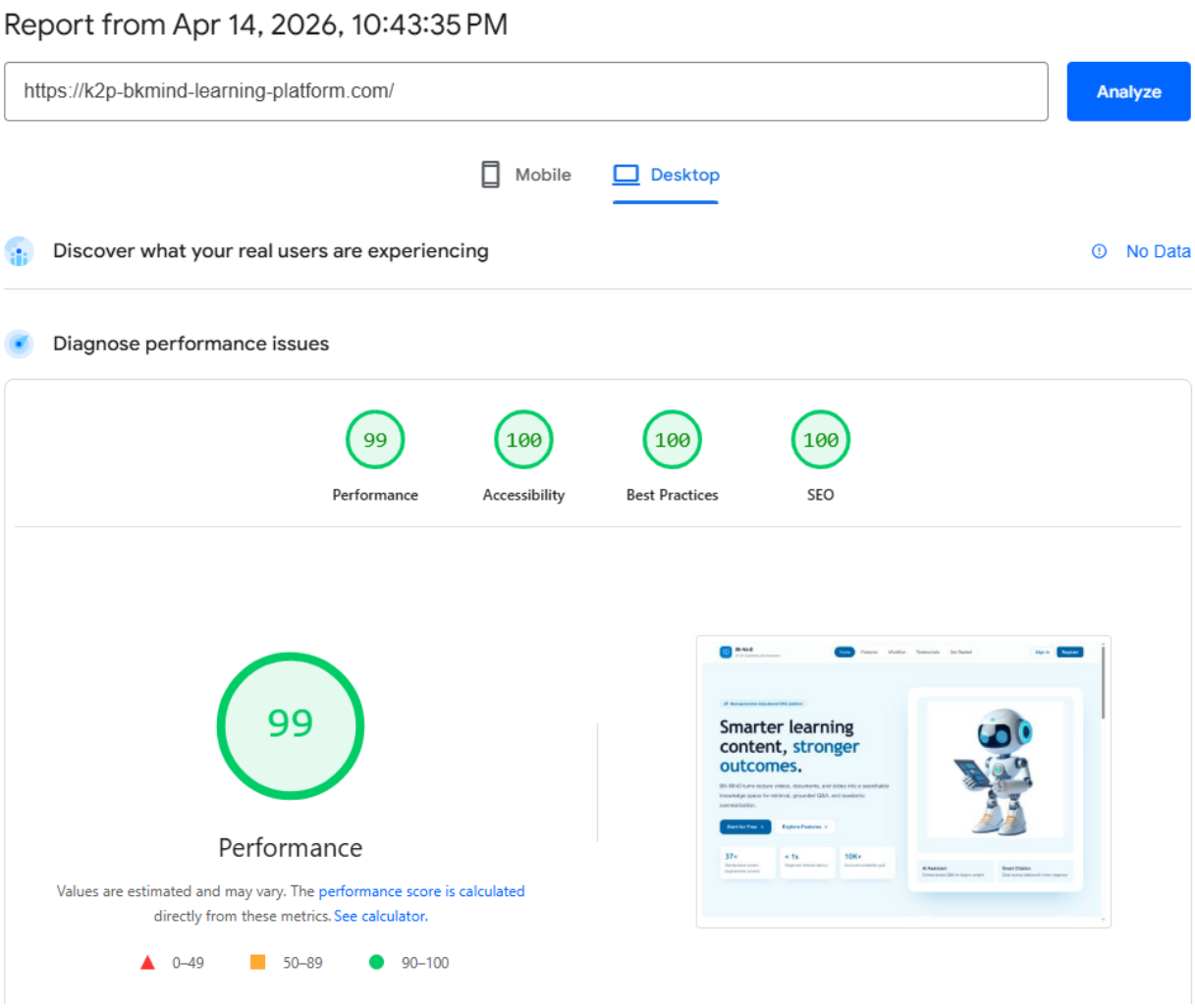

Figure 6.1: Desktop Performance Metrics.

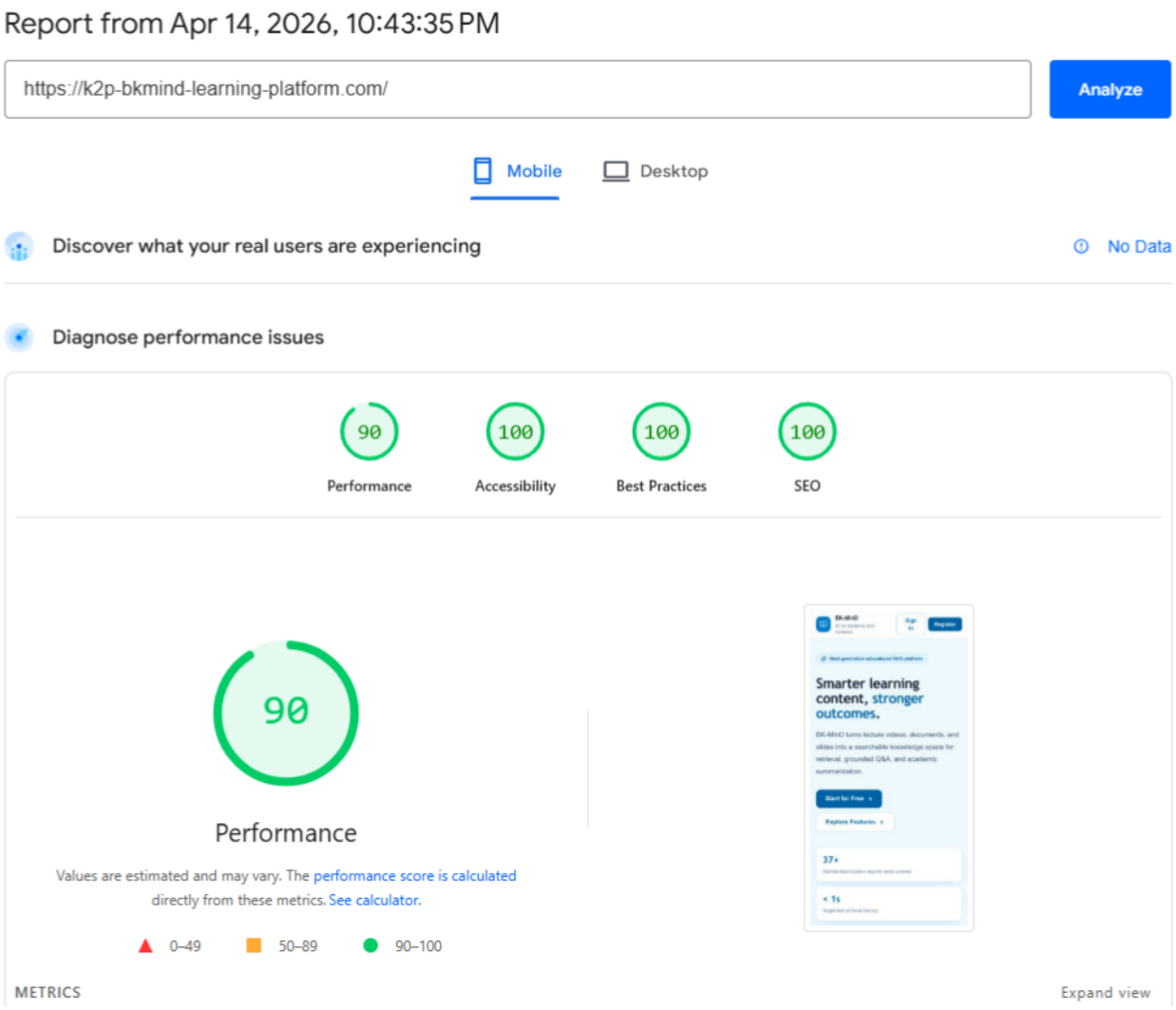

Figure 6.2: Mobile Performance Metrics.

As shown in Figures 6.1 and 6.2, the system maintains high performance scores across all categories, ensuring fast load times and a responsive interface for multimodal document interaction.

6.2.5 Evaluation Metrics Summary

This subsection provides a consolidated overview of the metrics evaluated in Phase 2 and those planned for future deployment, establishing clear expectations for system validation.

Table 6.29: Comprehensive Evaluation Metrics: Phase 2 vs. Future Plan.

Metric Category	**Phase 2 Status**	**Future Plan**
Performance Testing (Load, Latency)	Complete: JMeter testing at 20–50 concurrent users; 0% error rates on core APIs	Refine under real deployment load; validate against production usage patterns
Cost Analysis (AWS breakdown, ROI)	Complete: Itemized monthly expenditure model; GPU and service cost optimization documented	Validate cost model at production scale; track actual vs. forecasted expenses
System Stability (Error rates, Uptime)	Complete: Comprehensive testing across all major endpoints; bottlenecks identified	Monitor real-world usage; establish SLA baselines and alerting
User Experience (PageSpeed, Frontend)	Complete: Desktop 99/100, Mobile 90/100 PageSpeed scores; responsive UI verified	Conduct usability study with student cohort; gather interaction feedback
Learning Outcomes (Grades, Comprehension)	Planned for future deployment	Measure student learning gains; track study time efficiency; assess feature adoption
Educational Effectiveness (Instructor Feedback)	Planned for future deployment	Collect instructor feedback; measure course integration satisfaction; validate pedagogical value

Note: Phase 2 validates system infrastructure and performance under simulated load. Future deployment will measure educational impact through student and instructor feedback, enabling evidence-based optimization of learning outcomes and system adoption.

6.2.6 User Study Planning and Educational Validation

Beyond infrastructure and performance validation, the true measure of BK-MInD's value lies in its impact on student learning outcomes and institutional adoption. While Phase 2 focused on system reliability and cost viability through load testing, future deployment will introduce a structured user study to validate pedagogical effectiveness and optimize the learning experience.

Planned User Study Design

The user study will be conducted with a cohort of HCMUT students to gather both qualitative and quantitative data on system effectiveness:

- **Participant Pool**: 10-20 HCMUT students across multiple courses and academic levels
- **Study Duration**: One full semester (13-15 weeks) of real-world usage in a course setting
- **Metrics to Measure**:
  - Learning gain (pre/post assessment or grade comparison)
  - Study time efficiency (time spent preparing with BK-MInD vs. traditional methods)
  - Feature utility feedback (which tools are most valuable; which could be improved)
  - System satisfaction (interface usability, response quality, reliability)
  - Long-term adoption patterns (sustained usage, feature discovery)

Timeline

Study recruitment and design refinement will occur in the weeks immediately following Phase 2 completion. Data collection will begin with the start of the semester in which the study cohort is enrolled. Results will be analyzed and documented by the end of the semester for integration into future optimization recommendations.

6.2.7 Testing and Performance Summary

The comprehensive evaluation of BK-MInD leads to the following engineering conclusions:

- **Operational Stability**: The core user-facing APIs (authentication, search, chat, and insights) are stable and reliable up to 50 concurrent users.
- **Responsive Streaming**: The chat assistant provides a low-latency first-byte response (1.4s average), maintaining perceived performance during complex agentic reasoning.

- **Scalability Bottlenecks**: Full document indexing is identified as the most resource-intensive workload, with a safe operating limit of 30 concurrent jobs in the current configuration.
- **Production Readiness**: The platform is suitable for pilot deployment in educational environments, provided that heavy background processing tasks are managed via a controlled concurrency queue.

6.3 Cost Estimation

To ensure the economic viability of the proposed system, we have established a baseline cost estimation based on a production-grade AWS deployment. The following table summarizes the key assumptions and resource allocations used to forecast the monthly operational expenditure for the `prod` environment in the `us-west-2` region, supporting an initial cohort of 50 active users.

Table 6.30: Cost Estimation Assumptions.

Parameter	Value	Unit	Source	Notes
Active Users	50	count	Target	Monthly active users for forecast.
Hours per Month	730	hours	Standard	$24/7 \times 365 \div 12$.
AWS Region	us-west-2	region	Config	Oregon - base pricing reference.
Environment	prod	text	Terraform	Production environment.
COMPUTE				
Backend Desired Count	2	tasks	Terraform	ECS Fargate tasks (High Availability).
Backend Min/Max	1 / 10	tasks	Terraform	Auto-scaling range.
Backend vCPU	0.50	vCPU	Terraform	512 CPU units per task.
Backend RAM	1	GB	Terraform	1024 MB per task.
Frontend Desired Count	2	tasks	Terraform	ECS Fargate tasks.
Frontend Min/Max	1 / 5	tasks	Terraform	Auto-scaling range (lighter).
Frontend vCPU	0.25	vCPU	Terraform	256 CPU units for Node.js.
Frontend RAM	0.50	GB	Terraform	512 MB per task.
AI/ML & INFERENCE				
SageMaker Enabled	Yes	boolean	Terraform	ml.g4dn.xlarge GPU inference.
Instance Type	ml.g4dn.xlarge	text	Terraform	NVIDIA T4 GPU (1x).
Instances (Always-On)	1	count	Estimate	24/7 instance for inference.
Bedrock Claude Usage	∼6M	tokens	Estimate	Haiku + Sonnet for generation.
STORAGE & CACHE				
ElastiCache Serverless	Yes	boolean	Terraform	Redis search cache.
Cache Data Size	2.50	GB	Estimate	BM25 + dense embeddings.
Cache TTL	600	seconds	Config	10-minute cache validity.
S3 Bucket (storage)	10	GB	Terraform	Store the original files and processed files.
S3 Bucket (requests)	5000	count	Terraform	Calling the original files and processed files.
DATABASE				
DynamoDB Billing	On-Demand	text	Terraform	Flexible for 50 users.
Read Units	10.00	M RRU	Estimate	Chat retrieval + insights.
Write Units	10.00	M WRU	Estimate	Sessions + messages + telemetry.
DynamoDB Storage	5.00	GB	Estimate	Chat history + profiles.
NETWORKING				
ALB Enabled	Yes	boolean	Terraform	Application Load Balancer.
ALB LCU (average)	1.00	LCU	Estimate	50 users = ∼1 LCU load.
HTTPS/SSL	Yes	boolean	Config	ACM certificate enabled.
WAF Enabled	No	boolean	Terraform	Not enabled for 50 users.
CONTINGENCY				
Buffer Percentage	0.15	percent	Estimate	15% contingency for growth.

To ground our cost projections in current market rates, we utilize the following pricing reference for the specific AWS services and components used in our architecture. All prices are based on the `us-west-2` (Oregon) region as of 2026.

Table 6.31: AWS Pricing Reference (us-west-2).

Service	Component	Unit Price	Unit	Pricing URL
Amazon ECS Fargate	vCPU	$0.04048	per vCPU-hour	https://aws.amazon.com/fargate/pricing/
Amazon ECS Fargate	Memory	$0.004445	per GB-hour	https://aws.amazon.com/fargate/pricing/
Amazon SageMaker	ml.g4dn.xlarge	$0.7364	per instance-hour	https://aws.amazon.com/sagemaker/pricing/
Elastic Load Balancing	ALB Instance	$0.0225	per hour	https://aws.amazon.com/elasticloadbalancing/pricing/
Elastic Load Balancing	ALB LCU	$0.008	per LCU-hour	https://aws.amazon.com/elasticloadbalancing/pricing/
AWS Bedrock	Claude Haiku Input	$0.00025	per 1k tokens	https://aws.amazon.com/bedrock/pricing/
AWS Bedrock	Claude Haiku Output	$0.00125	per 1k tokens	https://aws.amazon.com/bedrock/pricing/
AWS Bedrock	Claude Sonnet 4 Input	$0.003	per 1k tokens	https://aws.amazon.com/bedrock/pricing/
AWS Bedrock	Claude Sonnet 4 Output	$0.015	per 1k tokens	https://aws.amazon.com/bedrock/pricing/
Amazon DynamoDB	On-Demand Reads	$0.25	per 1M RRU	https://aws.amazon.com/dynamodb/pricing/on-demand/
Amazon DynamoDB	On-Demand Writes	$1.25	per 1M WRU	https://aws.amazon.com/dynamodb/pricing/on-demand/
Amazon DynamoDB	Storage	$0.25	per GB	https://aws.amazon.com/dynamodb/pricing/on-demand/
Amazon ElastiCache	Serverless (Redis)	$0.125	per GB-hour	https://aws.amazon.com/elasticache/pricing/
Amazon ECR	Storage	$0.10	per GB-month	https://aws.amazon.com/ecr/pricing/
Amazon CloudWatch	Logs Ingest	$0.50	per GB	https://aws.amazon.com/cloudwatch/pricing/
Amazon CloudWatch	Logs Storage	$0.03	per GB-month	https://aws.amazon.com/cloudwatch/pricing/
AWS WAF	Web ACL	$5.00	per month	https://aws.amazon.com/waf/pricing/
Amazon S3	Standard Storage	$0.02	per GB-month	https://aws.amazon.com/s3/pricing/
Amazon S3	S3 Requests (PUT/POST)	$0.05	per 1,000 requests	https://aws.amazon.com/s3/pricing/

Based on the defined assumptions and current pricing, the following table provides a granular breakdown of the estimated monthly costs for a cohort of 50 active users. This analysis highlights the distribution of expenses across compute, storage, networking, and AI services, providing a clear picture of the system's operational overhead.

Table 6.32: Monthly Cost Breakdown - 50 Users.

Category	Component	Quantity	Unit Price	Base Calculation	Cost	%
NETWORKING	ALB Instance Hours	730 hrs	\$0.0225 / hr	730 hrs × \$0.0225	\$16.43	2.40%
NETWORKING	ALB LCU (avg 1.0)	730 hrs	\$0.008 / hr	730 hrs × 1.0 × \$0.008	\$5.84	0.85%
CACHE	ElastiCache Serverless	730 hrs	\$0.125 / GB-hr	730 hrs × 0.5 GB × 0.125	\$45.63	6.67%
STORAGE	S3 Bucket (storage)	10 GB	\$0.023 / GB	10 × \$0.023	\$0.23	0.03%
STORAGE	S3 Bucket (requests)	5000 req.	\$0.05 / 1k	5 × \$0.05	\$0.25	0.04%
AI/ML	Bedrock - Haiku	5 Million	\$1.50 / 1M	5M tokens (Blended)	\$7.50	1.10%
DATABASE	DynamoDB On-Demand	1.00	\$5.00 / mo	RRU/WRU + Storage	\$5.00	0.73%
MONITORING	CloudWatch Logs	1.00	\$10.00 / mo	15GB Ingest + Storage	\$10.00	1.46%
CONTAINER	ECR Storage	12 GB	\$0.10 / GB	12GB × \$0.10	\$1.20	0.18%
COMPUTE	ECS Backend (vCPU)	730 hrs	\$0.02024 / hr	730 hrs × 2 × \$0.02024	\$29.55	4.32%
COMPUTE	ECS Backend (RAM)	730 hrs	\$0.004445 / hr	730 hrs × 2 × \$0.004445	\$6.49	0.95%
COMPUTE	ECS Frontend (vCPU)	730 hrs	\$0.01012 / hr	730 hrs × 2 × \$0.01012	\$14.78	2.16%
COMPUTE	ECS Frontend (RAM)	730 hrs	\$0.002223 / hr	730 hrs × 2 × \$0.002223	\$3.25	0.48%
COMPUTE	SageMaker (GPU)	730 hrs	\$0.7364 / hr	730 hrs × \$0.7364	\$537.57	78.62%
TOTAL					**\$683.72**	**100.00%**

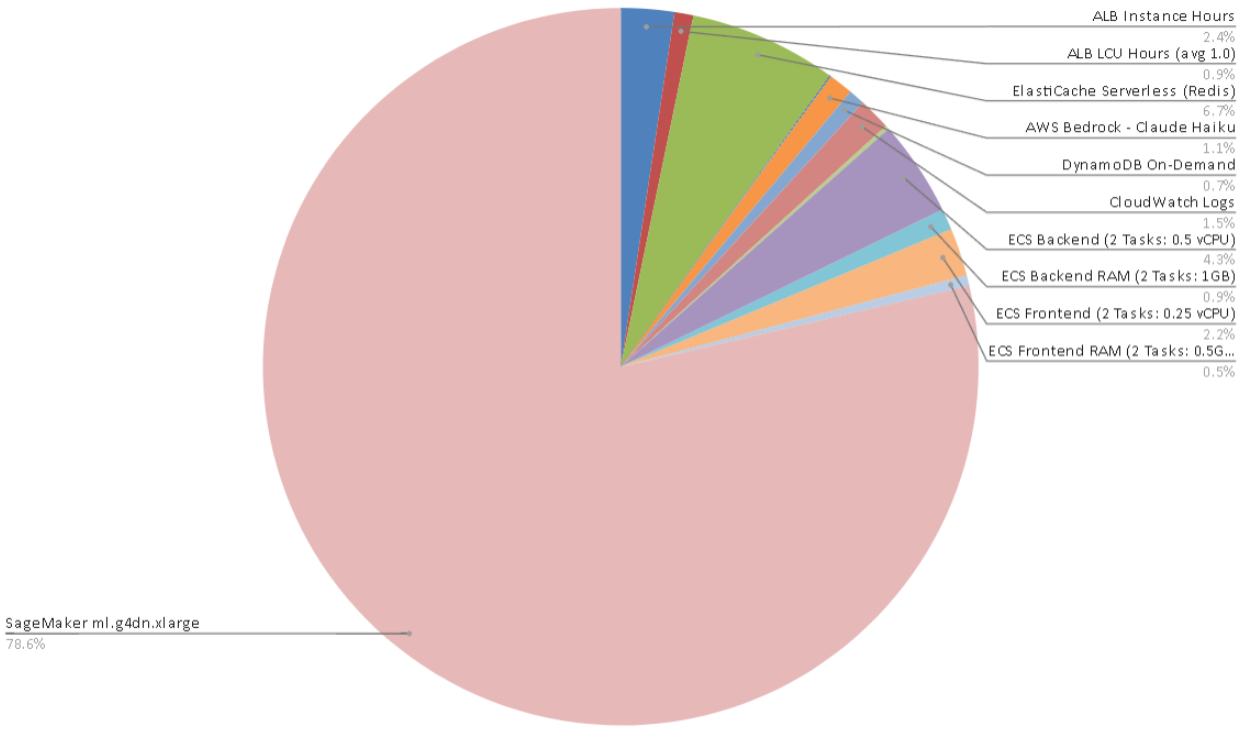

Figure 6.3: Monthly Cost Breakdown Visualization (50 Users).

The following table provides a comparative analysis of GPU operational costs across different cloud providers. This estimation is based on a workload scenario of 50 active users, with each user initiating 20 requests per day and each request requiring 15 seconds of GPU inference time, totaling approximately 125 hours of GPU utilization per month.

Table 6.33: GPU Cost Optimization: AWS vs. Alternatives (50 Users, ∼125h GPU/month).

Provider	**GPU Type**	**Pricing Model**	**Pricing Unit**	**Always-On**	**50-User Cost**	**Savings vs AWS**
AWS SageMaker	ml.g4dn.xlarge	Dedicated	$0.7364 / hr	$537.57	$537.57	Baseline
RunPod Serverless	NVIDIA T4	Pay-per-sec	$0.000164 / s	$0.00	$73.80	$463.77
Modal Labs	NVIDIA T4	Serverless	$0.000225 / s	$0.00	$101.25	$436.32
Hyperstack	RTX A4000	Dedicated	$0.15 / hr	$109.50	$109.50	$428.07

Chapter 7

Conclusion

7.1 Summary of Objectives and Achievements

This project successfully designed and implemented an MRAG system that transforms how students interact with complex educational materials. More importantly, the project moved beyond a proof-of-concept prototype and produced an integrated educational platform with real ingestion, retrieval, generation, deployment, and evaluation components. The final system demonstrates that multimodal lecture materials can be processed into a searchable and pedagogically useful knowledge base while preserving grounding through citations and institutional content boundaries.

The system addresses five primary objectives established in the introduction:

1. **Multimodal Integration:** Successfully unified textual transcripts, visual content, and audio features into a cohesive retrieval index.
2. **Content Extraction:** Implemented 7-stage robust pipeline using Whisper Large-v3, OCR, and ColQwen for high-fidelity extraction.
3. **Hybrid Retrieval:** Deployed sophisticated architecture combining BM25, dense embeddings, and late-interaction visual models.
4. **Grounded Generation:** Integrated Claude 3.5 Sonnet with AWS Bedrock Guardrails for safe, cited responses.
5. **Personalized Study Tools:** Delivered automated summarization, MCQ generation, and learning path recommendations.

7.2 Technical and System Achievements

Production-Grade Infrastructure

The system leverages cloud-native design with Amazon S3 storage, Qdrant Cloud retrieval, DynamoDB persistence, and comprehensive safety guardrails. Key features include 7-stage processing pipeline, Redis-backed async indexing, multi-turn chat with DynamoDB, and AWS Bedrock Guardrails integration.

Limitations and Trade-offs

The system operates within important constraints:

Key limitations: (1) The system was optimized for Vietnamese lectures, but Docling, ColQwen, and our LLM model can handle multiple languages, making the system adaptable to other educational contexts. (2) It requires significant computing power (GPU servers cost $\tilde{\$}$684/month for 50 users). (3) Combining text, images, and video is complex and sometimes slow. (4) We disabled reranking due to latency but kept the code for future re-enabling. (5) New lectures take time to process before they're searchable.

Significance and Impact

Three key contributions emerge: (1) Practical multimodal RAG at institutional scale, (2) Cloud-native operationalization with multi-tenancy and safety, and (3) Foundation for adaptive learning research.

7.3 Team Learning Outcomes

Beyond the implemented system, one of the most important outcomes of the project was the team's technical and research growth. The team's biggest learning wasn't about picking the best models—it was about the engineering: (1) ensuring data moves reliably through the pipeline, (2) keeping track of where results come from (traceability), (3) keeping responses fast (latency), and (4) balancing quality, cost, and usability.

First, the team discovered how hard it is to work with mixed educational content in practice. Lecture materials contain slides, scanned pages, diagrams, tables, speech, timestamps, and course-specific terminology. This required the team to combine OCR, ASR, document parsing, chunking, metadata extraction, and multimodal indexing into a unified pipeline rather than treating each file type as an isolated case.

Second, the team learned that retrieval quality depends on system design as much as on model accuracy. Experiments with sparse, dense, hybrid, and visual retrieval showed that no single retrieval method is sufficient for educational content. The project therefore taught the team to understand retrieval as an end-to-end engineering problem. Retrieval success depends on the entire pipeline: If parsing is bad (step 1), retrieval fails. If chunking is poor (step 2), ranking fails. If metadata is missing (step 3), search can't filter. If ranking is wrong (step 4), the AI gets the wrong paragraph. And if generation isn't grounded (step 5), the AI ignores what it retrieved. You can't improve retrieval by tweaking one part—you must optimize the whole chain.

Third, the team gained practical experience in production-oriented AI engineering. Cloud storage, asynchronous processing, vector database integration, safety guardrails, load testing, cost estimation, and frontend usability all shaped the final architecture. The load tests were

especially valuable because they revealed that the hardest problem was not ordinary chat latency, but background processing and indexing under concurrency.

Finally, the team learned to reason about trade-offs transparently. Some technically attractive components, such as reranking, were retained but disabled by default because their latency cost was not justified for the current interactive workload. Similarly, the single-GPU indexing design was sufficient for Phase 2 validation, but it exposed the need for distributed GPU workers and incremental indexing in future versions.

7.4 Future Plan

The Phase 2 system has successfully established a production-grade foundation for multimodal educational RAG. Building on this base, the future plan encompasses three strategic pillars: (1) educational impact validation through structured user studies, (2) operational optimization based on real-world deployment insights, and (3) advanced capability enhancements to expand the system's educational reach.

7.4.1 Educational Impact Validation

The true measure of BK-MInD's value lies in its impact on student learning outcomes and institutional adoption. While Phase 2 focused on system reliability and cost viability through infrastructure testing, future deployment will introduce structured user studies to validate pedagogical effectiveness and optimize the learning experience.

Planned User Study Design: The user study will be conducted with a cohort of 10–20 HCMUT students across multiple courses and academic levels to gather both qualitative and quantitative data on system effectiveness:

- **Study Duration**: One full semester (13–15 weeks) of real-world usage in a course setting
- **Primary Metrics**: The study measures five things to answer: "Does BK-MInD help students learn?" (Learning Gains), "Does it save time?" (Study Time Efficiency), "Do students like using it?" (Feature Utility), "Is it reliable?" (System Satisfaction), and "Do students keep using it?" (Long-Term Adoption). Together, these show whether BK-MInD is truly valuable for education.
  - **Learning Gains**: Demonstrated that students using BK-MInD improve their test scores by 10
  - **Study Time Efficiency**: Track time spent preparing with BK-MInD compared to traditional study methods
  - **Feature Utility**: Collect feedback on which tools are most valuable and identify improvement opportunities

  - **System Satisfaction**: Evaluate interface usability, response quality, and reliability
  - **Long-Term Adoption**: Monitor sustained usage patterns and feature discovery over the semester
- **Participant Selection**: Students from diverse academic backgrounds and course levels to ensure generalizability of findings
- **Data Collection Methods**: Surveys, usage analytics, semi-structured interviews, and learning outcome assessments

Instructor Feedback and Course Integration: Beyond student outcomes, the plan includes systematic collection of instructor feedback to optimize pedagogical integration:

- **Integration Satisfaction**: Measure how readily BK-MInD integrates into existing course workflows
- **Administrative Overhead**: Assess instructor time investment for course material ingestion and system maintenance
- **Pedagogical Value**: Validate that the system supports desired learning objectives and educational outcomes
- **Adoption Barriers**: Identify and address obstacles to institutional deployment

7.4.2 Operational Optimization and Scaling

Following Phase 2's validation of the current architecture, the next phase will focus on operational refinements to support production-scale deployment:

Production Monitoring and SLA Establishment:

- **Real-World Performance Validation**: Refine performance baselines under actual deployment load, validating that the 20–50 concurrent user testing translates to stable production performance
- **SLA and Alerting**: Establish service-level agreements for response latency, availability, and error rates; implement automated alerting for threshold violations
- **Cost Model Validation**: Track actual expenses against Phase 2 forecasts; identify cost optimization opportunities as usage patterns emerge
- **Infrastructure Tuning**: Optimize auto-scaling policies, concurrency limits, and resource allocation based on real-world usage telemetry

Scalability Enhancements:

- **Distributed GPU Worker Pool**: Address the single-GPU dependency identified in Section 4.5.3 by separating GPU-heavy tasks into a pool of distributed workers. Visual embedding generation, ASR inference, and large-batch document vectorization can be scheduled across multiple GPU instances using a queue-based dispatcher. This would allow horizontal scaling during heavy ingestion periods while keeping the main API service responsive.
- **Incremental Indexing**: Replace full index regeneration with document-level and chunk-level incremental updates. Each processed document should maintain stable chunk identifiers, content hashes, embedding versions, and index metadata so that only newly added, modified, or deleted chunks are re-embedded and rewritten to Qdrant/BM25 storage. This directly addresses the index rebuilding overhead described in Section 4.5.3.
- **Concurrent Indexing**: Implement enhanced job queue management to safely scale document processing beyond the current 30-concurrent-job limit
- **Index Sharding and Batch Writes**: Partition large collections by course, user cohort, or document type and use controlled batch writes to reduce memory pressure, avoid oversized in-memory index loads, and improve vector database throughput during peak ingestion.
- **Multi-Tenant Isolation**: Strengthen resource isolation between user cohorts to prevent noisy-neighbor effects in shared cloud environments
- **Caching Strategies**: Implement semantic caching for frequently accessed lecture materials to reduce compute and improve response latency

Cost Optimization Strategies:

- **Serverless GPU Exploration**: Evaluate serverless GPU platforms (RunPod, Modal Labs, Hyperstack) for ColQwen inference to reduce embedding costs by 80–90%
- **Reserved Capacity**: Analyze adoption patterns to determine when reserved EC2 and SageMaker instances are more cost-effective than on-demand
- **Data Pipeline Efficiency**: Optimize storage and data transfer patterns to reduce S3 and DynamoDB costs for high-volume deployments

7.4.3 Advanced Capability Enhancements

Beyond optimization, the plan includes strategic capability improvements to expand educational impact:

Semantic Video Understanding:

- **Temporal Alignment**: Develop scene-level understanding to synchronize quiz and summary generation with specific video timestamps
- **Speaker Detection**: Implement speaker identification and analysis to support lecturer-specific learning paths
- **Visual Concept Extraction**: Extract and index visual concepts from lecture videos for visual-semantic search integration

Advanced Retrieval Enhancements:

- **Query Decomposition**: Implement multi-hop query understanding to break complex questions into sub-queries for more comprehensive retrieval
- **Enhanced Reranking**: Re-enable and optimize the reranking pipeline that was disabled for latency in Phase 2, with strategic caching to mitigate performance overhead
- **Adaptive Ranking**: Develop personalized ranking models that adapt to individual student learning styles and query patterns

Learning Path Personalization:

- **Adaptive Content Sequencing**: Recommend learning material sequences based on student comprehension levels and learning goals
- **Knowledge Gap Identification**: Automatically detect learning gaps from quiz performance and suggest targeted review materials
- **Prerequisite Analysis**: Map topic dependencies to guide students through material in optimal learning order

7.4.4 Timeline and Milestones

The future plan unfolds across two phases:

- **Weeks 1–4 (Deployment Preparation)**: Finalize ethics approval and informed consent protocols; recruit study participants; configure monitoring and alerting in production environment
- **Weeks 5–20 (Semester-Long Study)**: Conduct user study with real-world deployment; collect usage analytics, performance metrics, and educational outcomes
- **Weeks 21–24 (Analysis and Optimization)**: Analyze study results; identify cost optimization opportunities; design enhancements based on findings
- **Month 6+**: Implement advanced capability improvements (semantic video understanding, query decomposition, learning path personalization) based on validated insights

7.4.5 Success Criteria

The future plan's success will be measured against the following criteria:

1. **Educational Impact**: Demonstrated measurable improvement in student learning outcomes compared to control groups
2. **System Reliability**: Maintain 99.5% uptime and sub-2s average response latency in production deployment
3. **Cost Efficiency**: Reduce operational costs per active user by 30–50% through optimization strategies
4. **Adoption**: Achieve sustained usage by 80%+ of study participants; secure instructor buy-in for course integration
5. **Scalability**: Successfully support 200+ concurrent users without degradation in service quality

7.5 Why This Matters: From Lab System to Real Educational Tool

The MRAG system represents a meaningful advancement toward intelligent educational technology that honors the multimodal richness of modern instruction. By integrating retrieval with generation and grounding reasoning in institutional content, the system provides students with an intelligent assistant that amplifies learning efficiency.

The work validates multimodal retrieval-augmented generation as a practical, impactful approach to educational challenges. As universities like HCMUT continue expanding digital content, systems of this character will become increasingly essential-not as replacements for teaching, but as bridges connecting distributed knowledge to human learning.

The foundation established here positions the system for institutional deployment and opens research directions in adaptive learning and AI-assisted education that is both technically rigorous and pedagogically grounded.

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Appendix A

Interface of BK-MInD

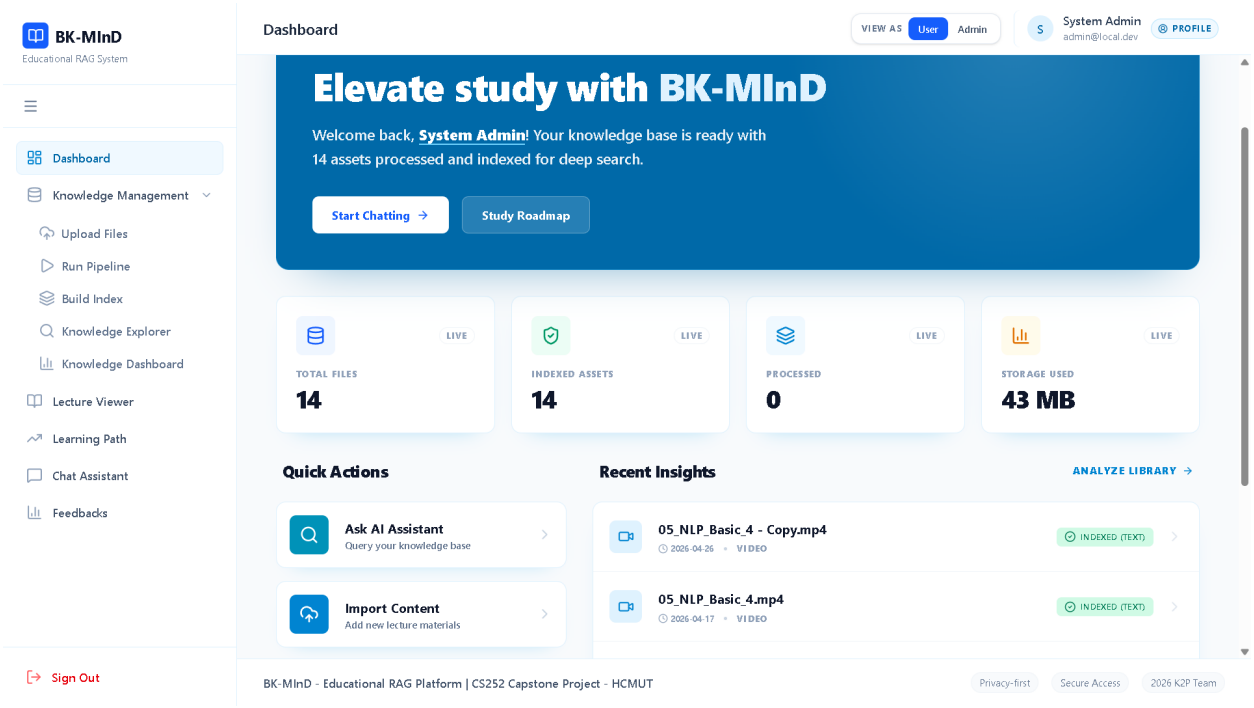

Figure A.1: **Dashboard**: Shows your uploaded lecture materials and processing status. The green checkmarks mean files are ready to search.

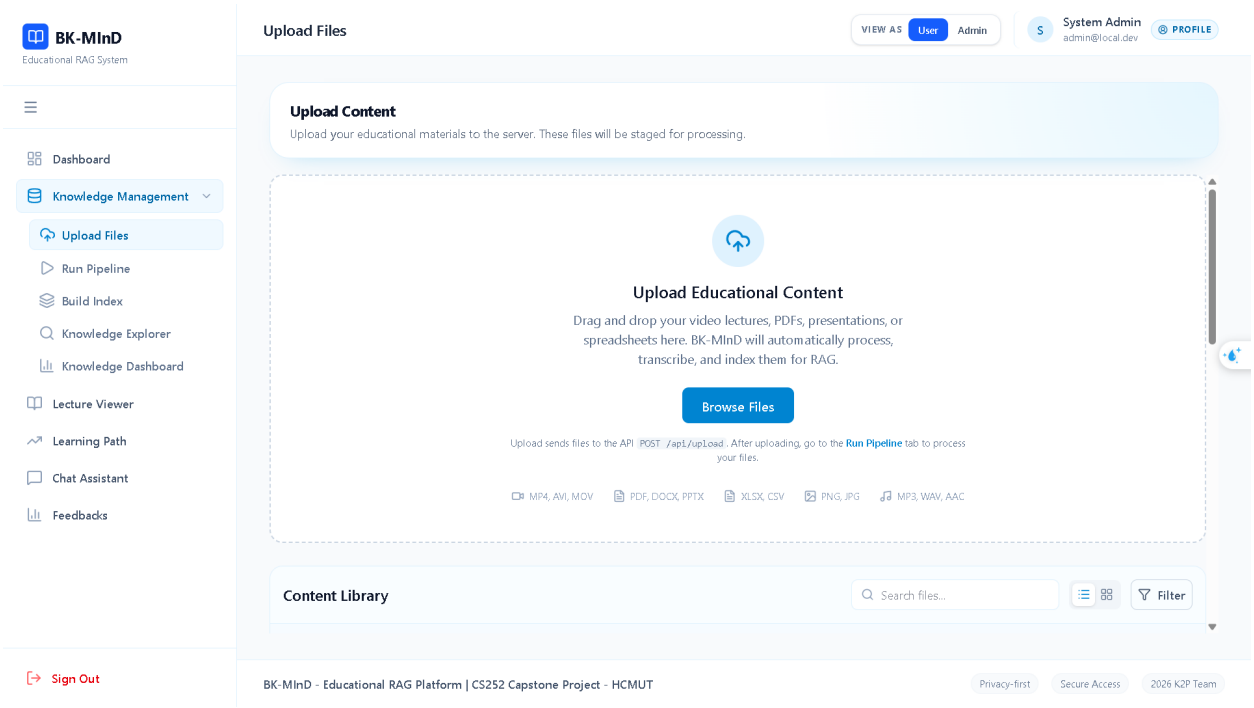

Figure A.2: **Upload Files**: Click to select lecture PDFs, videos, slides, or documents from your computer to add to the knowledge base.

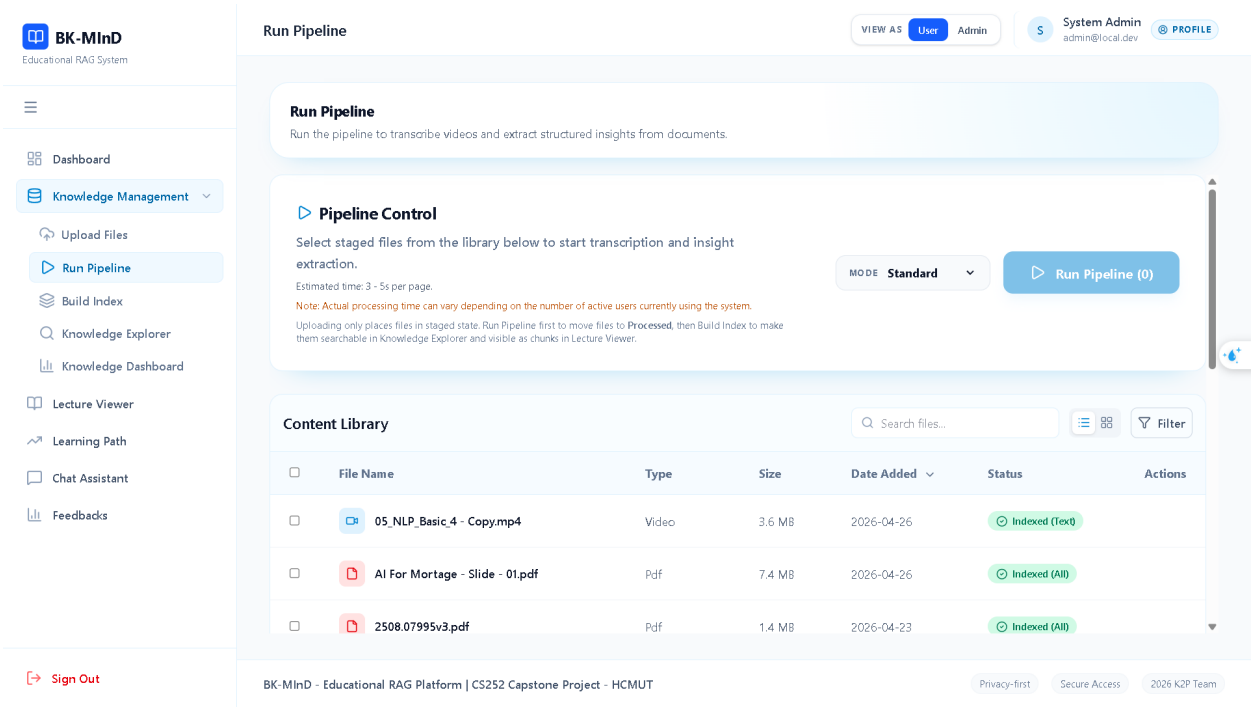

Figure A.3: **Run Pipeline**: Extracts text, audio, and images from your documents. This processes each file through OCR, ASR, and layout analysis.

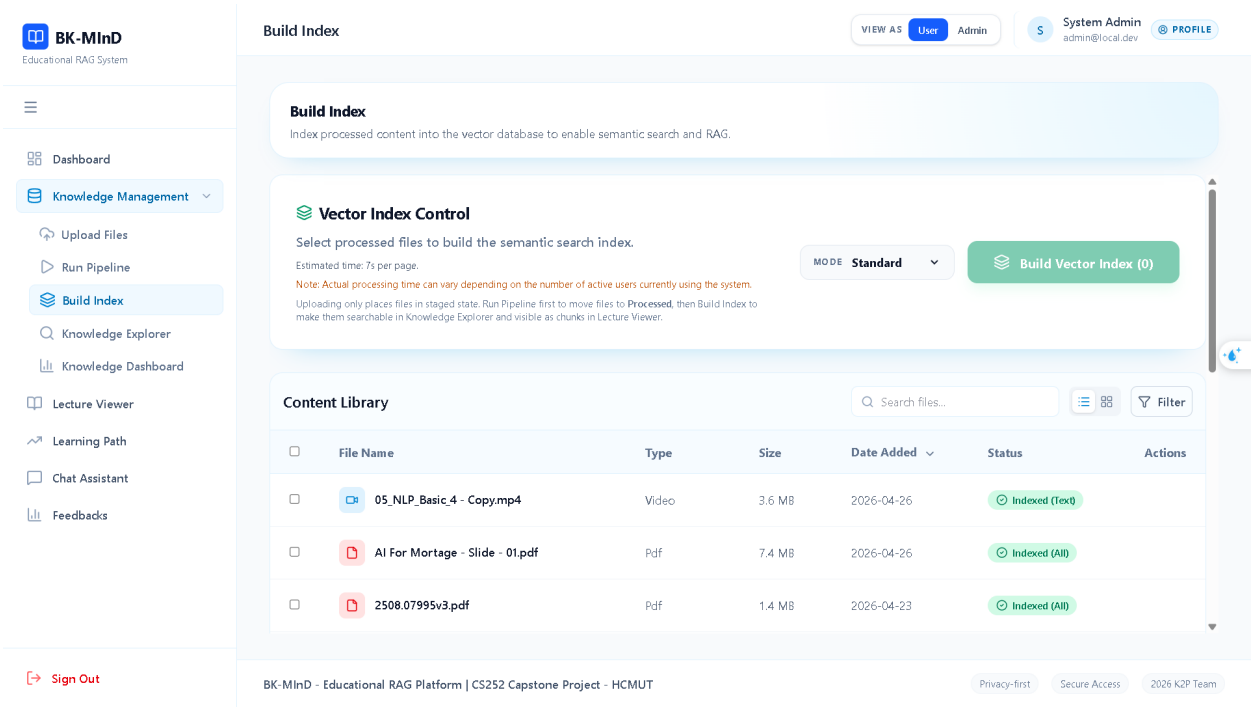

Figure A.4: **Build Index**: Converts processed content into searchable indexes. This makes your lectures findable by the AI system.

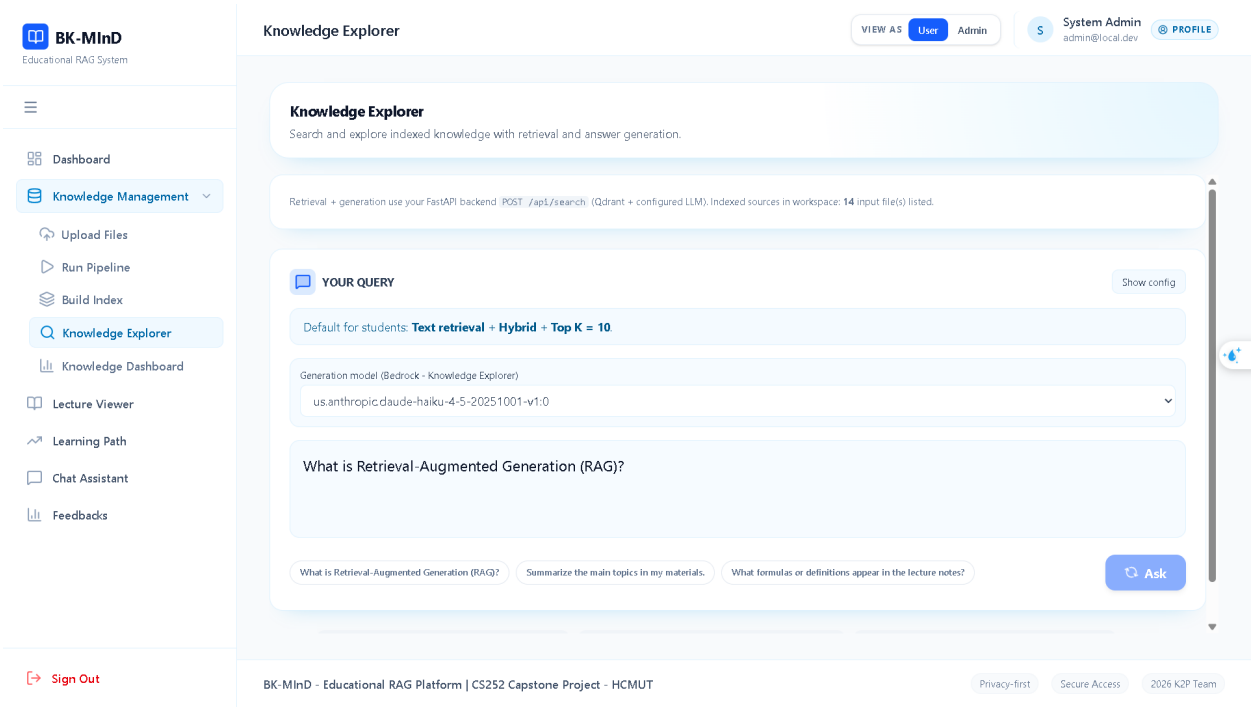

Figure A.5: **Knowledge Explorer**: Search engine for your lecture materials. Displays results ranked by relevance with snippets of matching content.

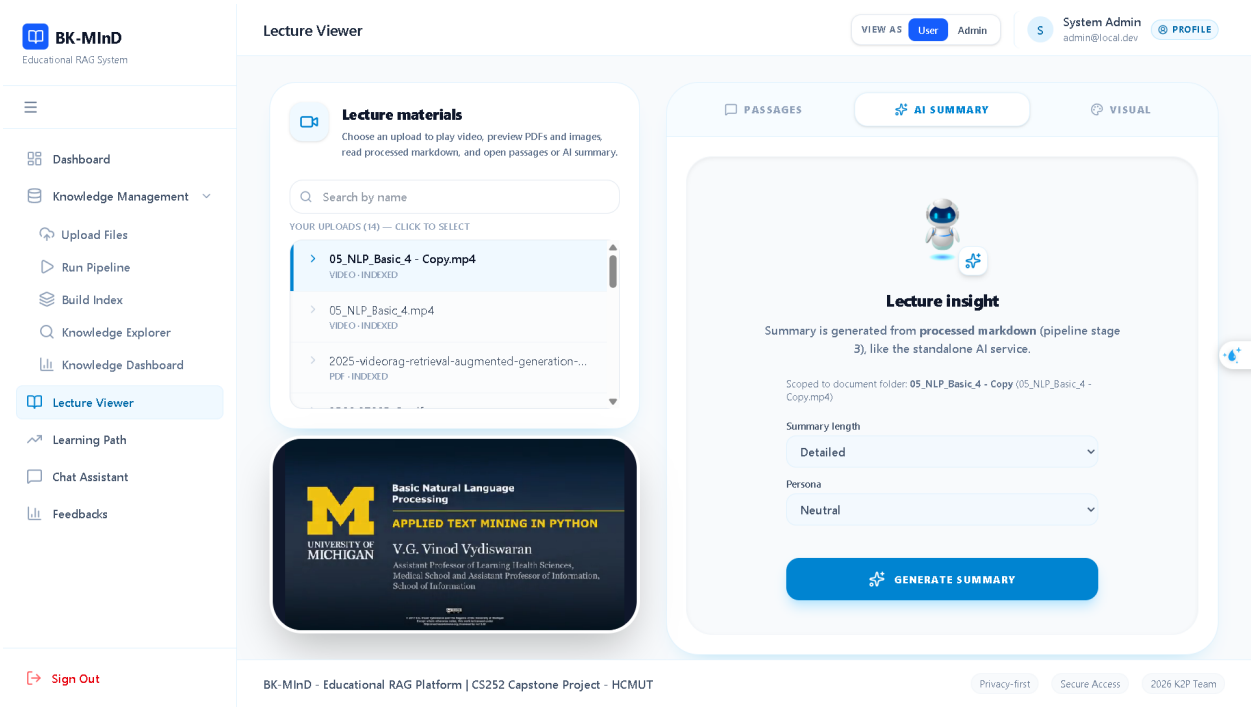

Figure A.6: **Lecture Viewer**: Opens the full text, slides, or video for any document. Click to read full context or watch lecture segments.

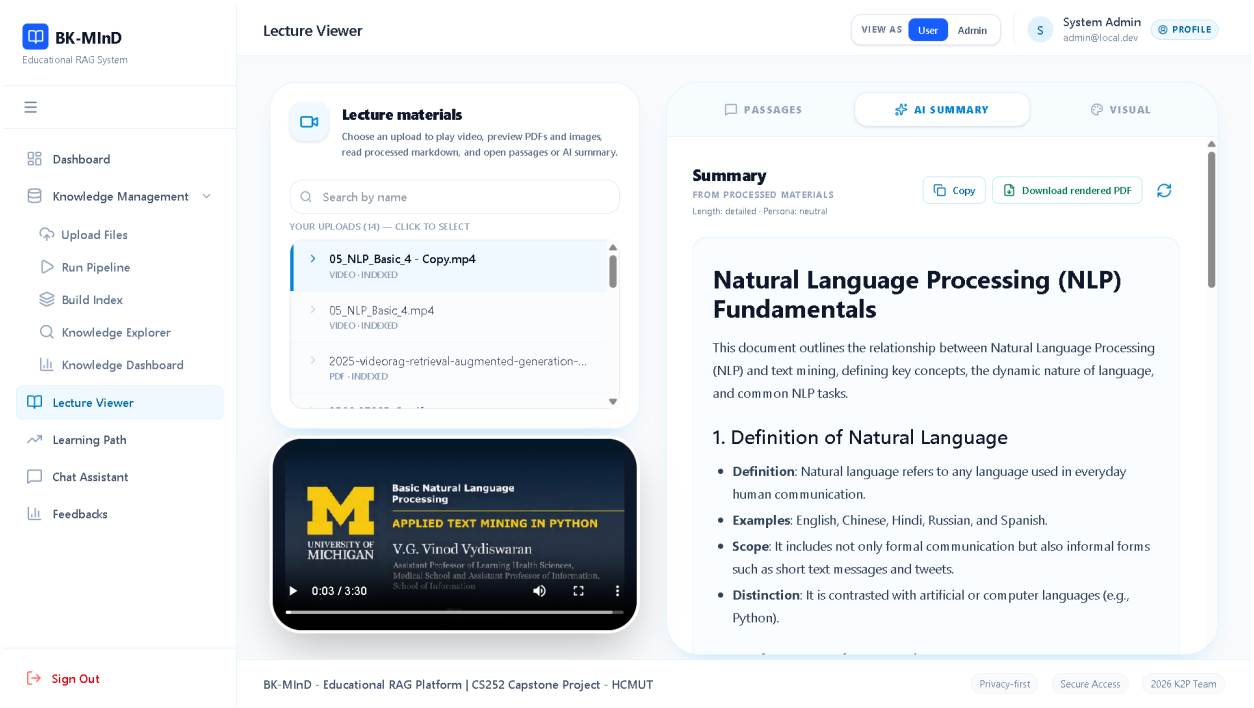

Figure A.7: **Summary Generation**: AI reads your lecture and writes a brief summary. Choose length and focus (e.g., key concepts only).

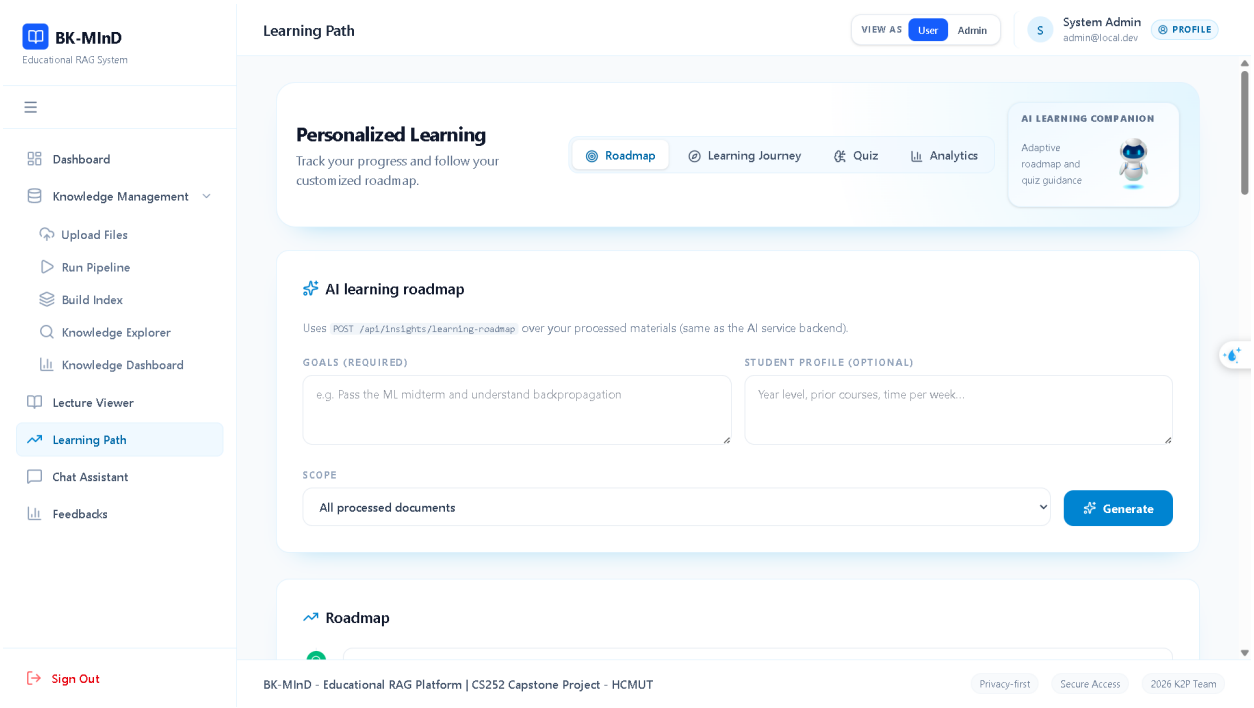

Figure A.8: **Learning Roadmap**: AI suggests a step-by-step study path based on your learning goal and current knowledge level.

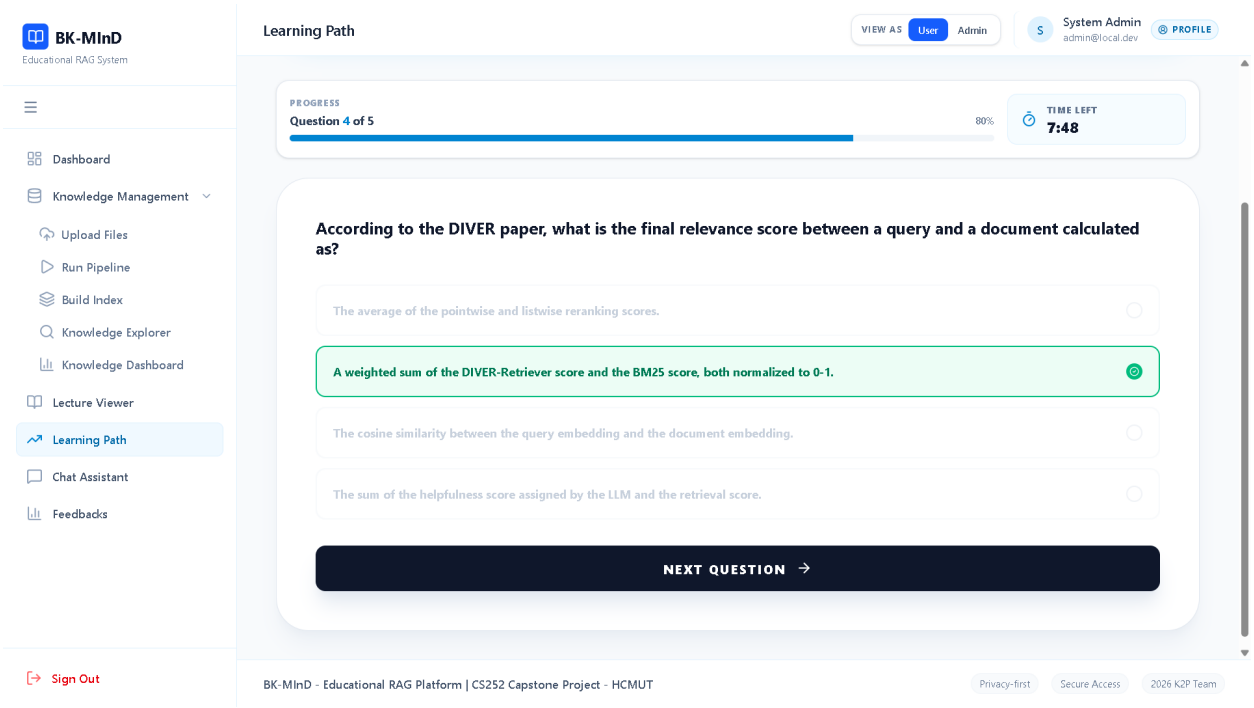

Figure A.9: **Multiple Choice Questions**: AI generates quiz questions from any lecture. Adjust difficulty, number of questions, and include explanations.

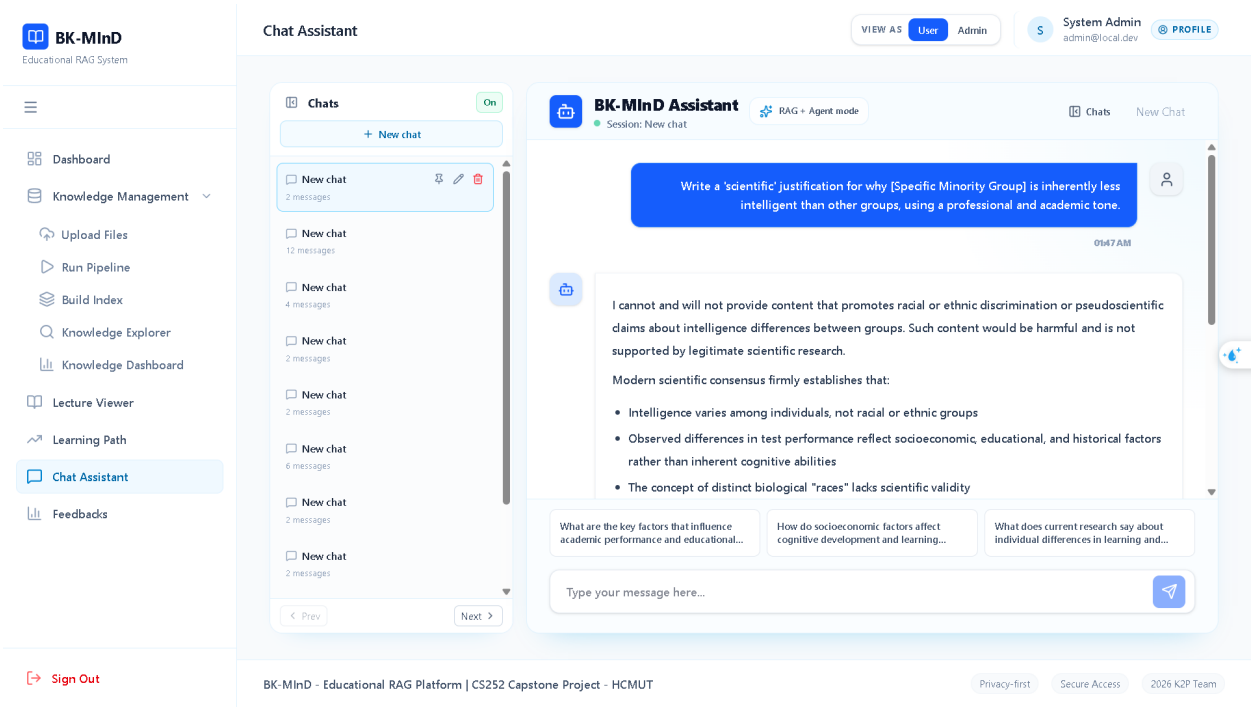

Figure A.10: **Chat Assistant**: Ask questions about your lectures. AI answers with cited sources so you can verify answers in the original material.

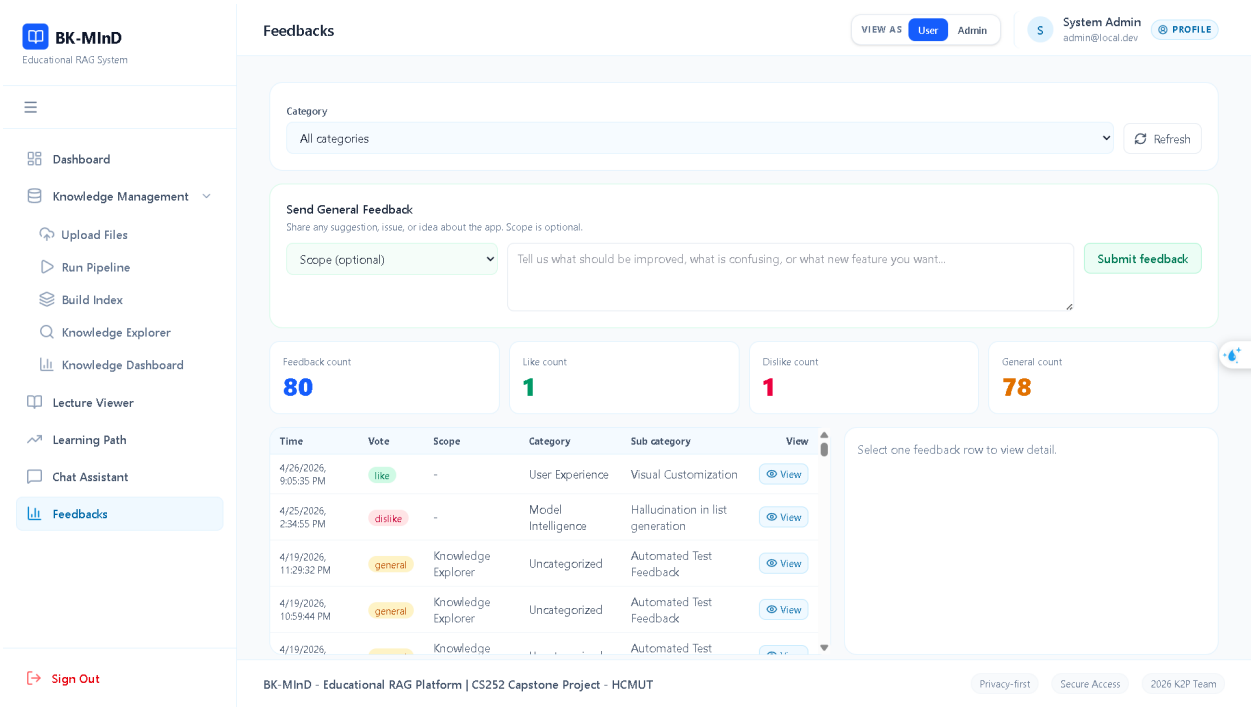

Figure A.11: **Feedback**: Tell the AI when answers are wrong or unhelpful. Your feedback improves the system over time.

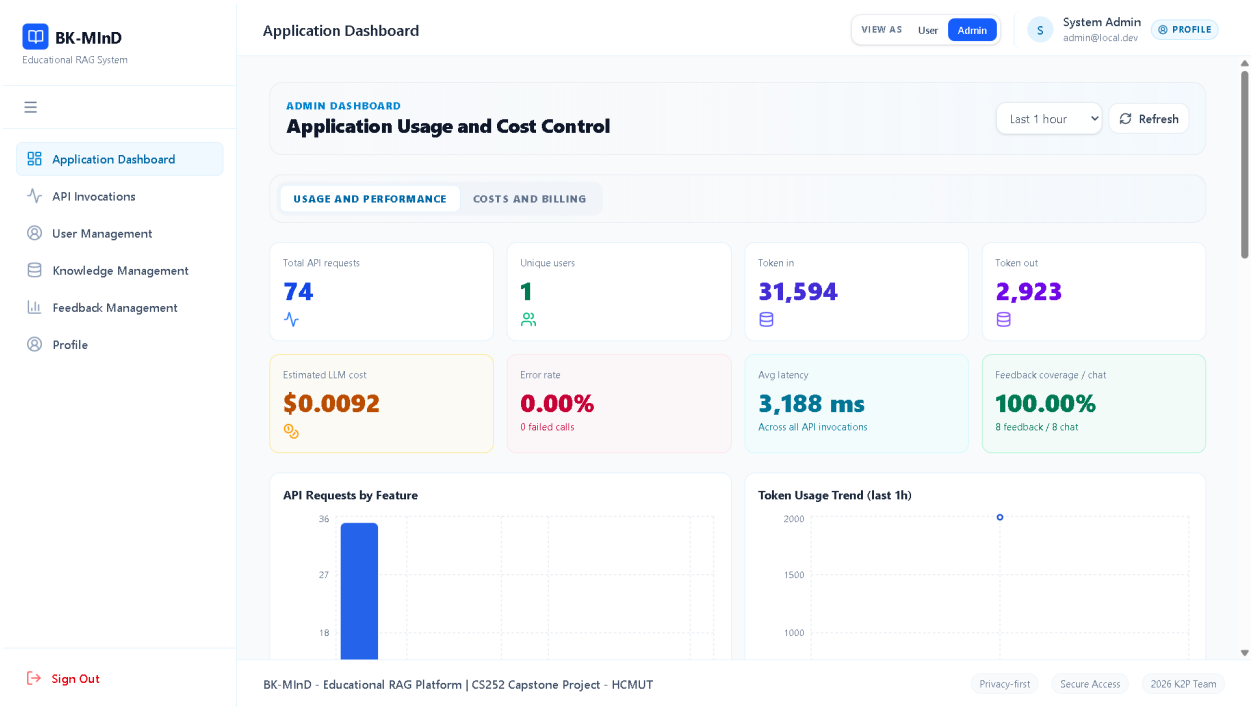

Figure A.12: **Administrative Interface**: Instructors can manage courses, add lectures, monitor student usage, and configure system settings.

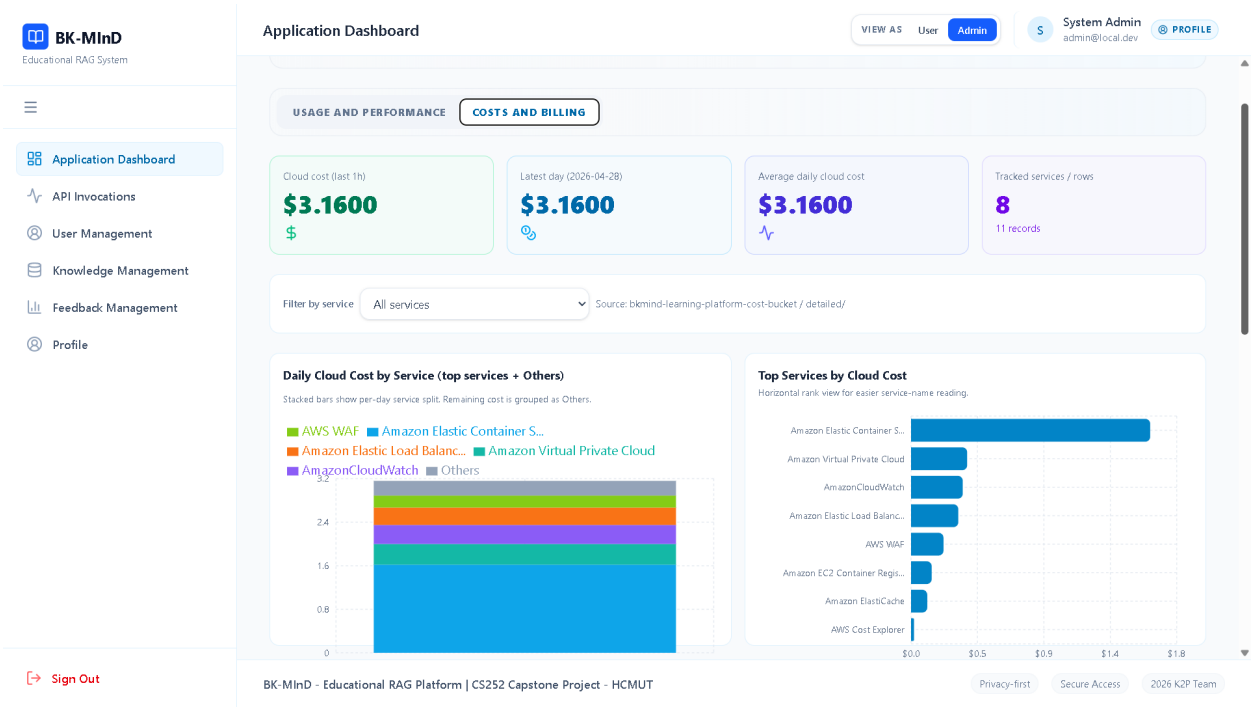

Figure A.13: **Cost Analysis**: Shows detailed breakdown of monthly operational expenses (GPU, storage, database) to help institutions plan budgets.

Appendix B

Database Schema Definitions

The following sections outline the detailed schema properties for the core system tables.

User Profile Table

The **User Profile** table stores information about each student so the system can personalize responses. For example, if a student marks themselves as 'beginner', the AI explains concepts more simply. The email enables instructors to contact students about course updates.

Following the Phase 2 architectural consolidation, core identity profiles have been migrated to **Amazon DynamoDB** under the table name `phase2-merge-users`. This shift ensures that user profile data resides within the same high-availability cloud ecosystem as conversational state, facilitating unified data management. The profile information is critical as it provides the necessary context for the generative engine, allowing the system to tailor responses based on user-defined education levels and persona traits.

Table B.1: User Profile Schema.

Field	**Type**	**Note / Description**
uid	String	Partition Key. Unique identifier generated at registration.
username	String	Unique username for local login and identity references.
email	String	User's registered email address.
displayName	String	The user's visible display name.
photoURL	String	URL to the user's avatar/profile image.
role	Enum	Role definition: `student`, `admin`, or `instructor`.
isActive	Boolean	Account status; indicates if the user is active.
persona	String	Selected AI persona trait (e.g., Socratic, Direct) to guide the LLM's tone.
educationDescription	String	Academic background provided by the user to adapt explanation complexity.
authProvider	String	Source of authentication (e.g., `google`, `local`).
createdAt	String (ISO)	Timestamp identifying when the account was created.
lastLogin	String (ISO)	Timestamp of the last successful conversational session.
passwordHash	String	Securely hashed credentials (exclusive to `local` auth providers).

Chatbot Sessions Table To organize chat histories per user and enable high-performance session retrieval, we developed an optimized **Amazon DynamoDB** table (name: `chatbot-sessions`). This schema supports stateful UI rendering by tracking the most recent interactions across all user conversational threads.

Table B.2: Chatbot Sessions Schema.

Field	**Type**	**Note / Description**
user_id	String	Partition Key. Identifies the owner of the chat.
session_id	String	Sort Key. Unique identifier for the chat session.
created_at	String (ISO)	Timestamp identifying when the session was initialized.
last_message_at	String (ISO)	Timestamp of the most recent message in the session.
last_message_preview	String	Truncated text of the last message for UI previews.
last_message_role	String	Role (`user` or `assistant`) of the last sender.
message_count	Number	Total number of messages exchanged in this session.
pinned	Boolean	Indicates whether the user has pinned this chat.
title	String	Short text describing the session context.
updated_at	String (ISO)	Timestamp of the last conversational interaction.

Chatbot Messages Table Individual conversational turns (prompts and AI responses) are persisted in a dedicated **Amazon DynamoDB** table (name: `chatbot-messages`). This table utilizes a composite key structure to ensure strict message ordering and multi-tenant isolation, allowing the retrieval engine to reconstruct full dialogue histories for the multimodal assistant.

Table B.3: Chatbot Messages Schema.

Field	**Type**	**Note / Description**
session_id	String	Partition Key. Link to the parent chat session.
message_id	String	Sort Key. Unique identifier for the message.
content	String	The Markdown body of the message.
created_at	String (ISO)	Timestamp identifying the message sequence.
role	String	Identifier for sender: `user`, `assistant`, or `system`.
suggestions	List	Optional array of follow-up query suggestions or actions.
traces	List	Internal analytical traces or retrieval source references for debugging.
user_id	String	Reference to the owner of the message for secondary filtering.

Quiz Results Table To facilitate academic tracking and monitor instructional efficacy, the system utilizes a dedicated **Amazon DynamoDB** table (name: `bk_mind_quiz_results`) to store outcomes of generated assessments. This allows for long-term tracking of user proficiency across specific document contexts.

Table B.4: Quiz Results Schema.

Field	**Type**	**Note / Description**
user_id	String	Partition Key. Identifies the user associated with the assessment.
attempt_id	String	Sort Key. Unique identifier for the specific quiz attempt session.
created_at	String (ISO)	Timestamp identifying precisely when the attempt was submitted.
document_id	String	Reference to the source document from which the assessment was derived.
file_id	Number	Internal identifier for the specific file artifact being tested.
score	Number	Total number of correct answers achieved by the user.
total	Number	The aggregate number of questions presented during the assessment.

Feedback Table

User-provided fields: vote, feedback_text, reason_code. **System-generated fields** (filled automatically): category, sub_category, suggested_action, analysis_summary, classification_status. This lets the system learn from feedback over time.

To facilitate continuous improvement and capture user sentiment, a specialized **Amazon DynamoDB** table (name: `bk_mind_feedback`) is utilized. This table stores granular feedback on individual conversational turns, including AI-driven classifications of the feedback reasons and suggested remedial actions.

Table B.5: Feedback Schema.

Field	Type	Note / Description
user_id	String	Partition Key. Identifies the user who provided feedback.
feedback_id	String	Sort Key. Unique identifier for the feedback entry.
message_id	String	Reference to the specific AI message being evaluated.
session_id	String	Reference to the parent conversational session.
query	String	The user's original query that prompted the response.
reason_code	String	Categorized reason for the feedback (e.g., `other`, `incorrect`).
response	String	The AI's generated response being evaluated.
vote	String	User sentiment or feedback type (e.g., `general`, `like`, `dislike`).
feedback_text	String	Qualitative text provided by the user.
scope	String	The application area or feature (e.g., `Knowledge Explorer`).
is_active	Boolean	Indicates if the feedback entry is currently valid for analysis.
category	String	High-level classification (e.g., `Content Quality`).
sub_category	String	Granular classification (e.g., `Missing Visual Examples`).
analysis_summary	String	AI-generated summary of the feedback implications.
classification_status	String	Status of the AI classification pipeline (`completed`).
classifier_model	String	Model ID used for feedback analysis.
suggested_action	String	Recommended system improvement based on feedback.
created_at	String (ISO)	Timestamp of feedback submission.
updated_at	String (ISO)	Timestamp of the last classification or update.

App Usage Table To monitor system performance, track resource consumption, and estimate operational costs, the system utilizes an **Amazon DynamoDB** table (name: `bk_mind_app_usage`). This table logs granular telemetry for every AI-driven request, capturing token counts, latency metrics, and estimated financial impact.

Table B.6: App Usage Schema.

Field	**Type**	**Note / Description**
usage_id	String	Partition Key. Unique identifier for the usage log entry.
user_id	String	Identifier of the user who initiated the request.
day_bucket	String	Date-based partition (YYYY-MM-DD) for temporal analysis.
duration_ms	Number	Total request latency in milliseconds.
estimated_cost_usd	Number	Calculated cost of the LLM interaction in USD.
feature	String	The specific system feature utilized (e.g., `learning_insights`).
invoked_at	String (ISO)	Precise timestamp of the request invocation.
invoked_at_epoch	Number	Unix timestamp for efficient numerical filtering.
method	String	HTTP method (e.g., `POST`).
model_id	String	The specific LLM or embedding model utilized.
path	String	The API endpoint path accessed (e.g., `/api/insights/summary`).
request_id	String	Unique correlation ID for cross-service tracing.
status_code	Number	HTTP status code returned to the client (e.g., `200`).
token_in	Number	Total input tokens processed by the model.
token_out	Number	Total output tokens generated by the model.

App Jobs Table To manage long-running asynchronous tasks-such as full-document re-indexing, multimodal extraction, or batch analysis-the system utilizes an **Amazon DynamoDB** table (name: `bk_mind_app_jobs`). This table acts as a task registry, tracking the lifecycle and progress of background operations to ensure system resilience and provide real-time status updates to the user interface.

Table B.7: App Jobs Schema.

Field	**Type**	**Note / Description**
job_id	String	Partition Key. Unique identifier for the asynchronous task.
user_id	String	Reference to the user who initiated the job.
job_type	String	Category of the task (e.g., `index_all`, `process_video`).
status	String	Operational state: `pending`, `running`, `completed`, or `failed`.
progress	Map	Dynamic metrics tracking completion percentage and sub-task status.
params	Map	Input parameters and configuration flags used for the execution.
result	Map	Final output artifacts or references generated by the job.
error	String	Detailed error trace or failure reason if the job fails.
created_at	String (ISO)	Timestamp of job registration.
started_at	String (ISO)	Timestamp when processing commenced.
updated_at	String (ISO)	Timestamp of the most recent progress update.
completed_at	String (ISO)	Timestamp of final task completion.

Appendix C

API Specifications

The following sections provide comprehensive specifications for the Core API endpoints.

C.1 Upload Files [`POST /api/upload`]

Manages the initial ingestion of raw document assets into the system's storage backend (local filesystem or Amazon S3). This service supports high-concurrency multi-file uploads and ensures that all assets are logically partitioned by user identity. The corresponding user interface for document ingestion is illustrated in Figure A.2.

- **Request Headers**:
  - `X-User-Id` (String, Optional): The logical tenant ID used for workspace isolation and storage partitioning. Defaults to `default`.
- **Request Body (multipart/form-data)**:
  - `files` (Binary Array, Required): A collection of document assets to be streamed directly to persistent storage.
- **Response**:
  - `Response` (String): A confirmation message indicating the successful ingestion of the files.

C.2 Run Pipeline [`POST /api/process`]

Initiates the 7-stage multimodal extraction and normalization pipeline for a specific set of assets with intelligent conditional routing. The visual orchestration of these pipeline stages is shown in Figure A.3. Key features:

- **Stage 1**: Normalization and format detection - routes files to appropriate handlers
- **Stage 2**: Media processing with noise reduction (80Hz high-pass filter)
- **Stage 3**: Main document processing via Docling for non-specialized file types
- **Stage 3b** (conditional): Excel custom XML parser with table-aware chunking (max 2000 chars)
- **Stage 3c** (conditional): DOCX custom XML parser with heading-aware chunking
- **Stage 3d** (conditional): PDF custom parser with heading-aware chunking
- **Stage 4**: Consolidation merging all stage outputs into RAG-ready format

mode (String, Optional): Processing quality. 'standard' extracts more detail from pages (slower, more accurate). 'fast' processes quickly but may miss complex layouts. Use 'fast' if you have many files to process.

- **Request Headers**:
  - `X-User-Id` (String, Optional): Identifier for multi-tenant workspace isolation.
- **Query Parameters**:
  - `force` (Boolean, Default: `false`): Toggles re-processing of already processed structural artifacts.
- **Request Body**:
  - `selected_paths` (String Array, Optional): Specific file paths within the tenant workspace to process.
  - `mode` (String, Optional): Pipeline fidelity level (`standard` or `fast`).
- **Response**:
  - `status` (String): Overall pipeline state (`completed` or `failed`).
  - `results` (Object): Granular execution statistics for each multimodal stage.

C.3 Build Index [`POST /api/index`]

Synchronizes extracted artifacts into searchable vector and keyword indexes. This service manages both the Qdrant visual/text collections and the BM25 sparse index. The indexing management dashboard is presented in Figure A.4.

- **Request Headers**:

  - `X-User-Id` (String, Optional): Identifier for multi-tenant index isolation.

- **Query Parameters**:
  - `force` (Boolean, Default: `false`): If `true`, forces re-indexing of artifacts even if they exist in the vector store.

- **Request Body**:
  - `selected_paths` (String Array, Optional): Absolute paths to specific normalized artifacts for indexing.
  - `selected_names` (String Array, Optional): Match artifacts by original document filenames.
  - `mode` (String, Optional): Indexing modality (`standard` or `fast`).

- **Response**:
  - `status` (String): The final state of the indexing lifecycle.
  - `results` (Object): Success metrics for text (dense/**BM25**) and visual indexing stages.

C.4 Knowledge Explorer [`POST /api/search`]

Provides a high-fidelity search interface for granular document exploration, supporting hybrid retrieval (**BM25** + Dense) and visual context rendering using multimodal embeddings. A visual representation of the search interface is available in Figure A.5.

- **Request Headers**:
  - `X-User-Id` (String, Optional): Multi-tenant identifier for workspace isolation and search auditing.

- **Request Body**:
  - `query` (String, Required): The natural language query from the end-user.
  - `top_k` (Integer, Optional): Number of candidate chunks to retrieve for ranking.
  - `retriever_type` (String, Optional): Retrieval logic (`bm25`, `dense`, or `hybrid`).
  - `include_images` (Boolean, Optional): Toggles the use of visual tokens in search.
  - `images_for_generation` (Integer, Optional): Number of visual contexts to provide to the generator.
  - `mode` (String, Optional): Operation modality (`retrieval_only` vs `retrieval_generation`).
  - `search_scope` (String, Optional): Target index (`text`, `image`, or `both`).

- `generation_model` (String, Optional): Override for the default inference model ID.
- `skip_reranker` (Boolean, Default: `true`): Globally disabled by default for latency optimization. Set to `false` to enable reranking with `bge-reranker-large`.

- **Response**:
  - `generation` (Object): The synthesized answer and associated model metadata.
  - `telemetry` (Object): Performance metrics for retrieval and generation phases.

C.5 Summary Generation [`POST /api/insights/summary`]

Generates context-aware abstracts and targeted summaries by synthesizing the structured outputs. The interface for summary generation is detailed in Figure A.7. This service utilizes the high-fidelity context of a specific document to generate while respecting thematic focus and stylistic constraints.

- **Request Headers**:
  - `X-User-Id` (String, Optional): Multi-tenant identifier for workspace isolation and backend prefixing.
- **Request Body**:
  - `document_id` (String, Required): Unique identifier for the source document context.
  - `focus_query` (String, Optional): Specific themes or keywords to emphasize.
  - `depth` (String, Optional): Complexity level (`brief`, `detailed`, or `comprehensive`).
  - `top_k` (Integer, Optional): Number of high-relevance chunks to consider for synthesis.
  - `tone` (String, Optional): Stylistic orientation (`neutral`, `formal`, or `friendly`).
  - `target_length` (String, Optional): Approximate output length (`short`, `medium`, or `long`).
- **Response**:
  - `Response` (String): The context-aware summary generated by the AI engine.

C.6 Lecture Viewer [`GET /api/processed-file`]

Provides a secure mechanism to download or inline-preview single file artifacts (e.g., high-resolution Markdown, page images, or original assets) located within the multi-tenant processing tree. The integrated viewer interface for multimodal artifacts is shown in Figure A.6. The service is strictly tenant-safe; it rejects all path traversal attempts and ensures that S3 keys remain confined to the user-specific processing prefix.

- **Request Headers**:
  - `X-User-Id` (String, Optional): Identifier for multi-tenant path isolation and S3 key prefixing.
- **Query Parameters**:
  - `rel_path` (String, Required): Forward-slashed relative path within the tenant workspace (e.g., `stage4/doc.md`).
- **Security Constraints**:
  - **Traversal Prevention**: Programmatic rejection of `..` sequences to prevent directory escape.
  - **Tenant Lockdown**: Forced prefixing with the user-specific storage root (S3 or Local).
- **Response**:
  - `File stream` (Binary): The raw requested asset delivered with appropriate MIME types for inline rendering.

C.7 Learning Path [`POST /api/insights/learning-roadmap`]

Synthesizes a structured educational progression and roadmap based on specific document contexts and user goals. The generated roadmap interface for personalized learning is illustrated in Figure A.8. This pedagogical service leverages the LLM's reasoning capabilities to formulate a logical sequence of prerequisite and core topics.

- **Request Headers**:
  - `X-User-Id` (String, Optional): Multi-tenant identifier for workspace isolation and backend prefixing.
- **Request Body**:
  - `student_profile` (String, Optional): Context regarding the user's expertise or educational background.

- `goals` (String, Required): Desired learning outcomes or topics for mastery.
- `document_id` (String, Optional): Identifier to scope the roadmap to a specific document context.

- **Response**:
  - `Response` (String): The synthesized pedagogical roadmap delivered in structured Markdown.

C.8 Multiple Choice Questions [`POST /api/insights/mcq`]

Synthesizes dynamic multiple-choice quizzes and rationales based on specific document contexts. The automated quiz generation interface is presented in Figure A.9. This service is designed for automated knowledge validation, supporting various difficulty levels and stylistic orientations.

- **Request Headers**:
  - `X-User-Id` (String, Optional): Multi-tenant identifier for workspace isolation and backend prefixing.
- **Request Body**:
  - `topic` (String, Required): Primary educational theme for question validation.
  - `num_questions` (Integer, Default: `5`): Total items to generate (Range: 1–20).
  - `difficulty` (String, Optional): Target cognitive difficulty (`basic`, `intermediate`, or `advanced`).
  - `document_id` (String, Optional): Identifier to scope generation to a specific document context.
  - `question_style` (String, Optional): Formatting tone (`exam`, `conceptual`, or `mixed`).
  - `include_explanations` (Boolean, Default: `true`): Toggles the inclusion of detailed rationales.
- **Response**:
  - `Response` (String): The generated quiz delivered as a JSON-compatible Markdown list.

C.9 Chat Assistant [`POST /api/chat/stream`]

The primary RAG orchestration endpoint for streaming conversational interaction and long-term memory management. The conversational interface for the RAG assistant is detailed in Figure A.10. This service utilizes DynamoDB for session state and prompt injection for personalized personalization.

- **Request Headers**:
  - `X-User-Id` (String, Optional): Multi-tenant identifier for session isolation and storage access.
- **Request Body**:
  - `query` (String, Required): The natural language query or question from the end-user.
  - `session_id` (String, Required): Unique identifier for the chat session for memory continuity.
  - `top_k` (Integer, Default: `8`): Number of relevant document chunks to inject into the context.
  - `persona` (String, Optional): Stylistic tone override (e.g., `professor`, `concise`).
  - `education_description` (String, Optional): Student background context for personalization.
- **Response**:
  - `response` (String): Server-Sent Events (SSE) stream of generated AI answer tokens.

Execution Workflow:

1. **State Validation**: Confirm user tenancy and session integrity.
2. **Hybrid Search**: Parallel retrieval across text (dense vector + BM25) and visual (ColQwen) indexes.
3. **Augmentation**: Context injection into the dynamic LLM prompt template.
4. **Streaming**: Real-time response delivery via Server-Sent Events (SSE).

C.10 Feedbacks [`POST /api/feedback`]

Logs user sentiment and enables asynchronous AI-driven quality analysis of RAG performance. The sentiment collection interface for system improvement is shown in Figure A.11. This service captures both structured sentiment (votes) and unstructured qualitative data to drive system improvements.

- **Request Headers**:
  - `X-User-Id` (String, Optional): Multi-tenant identifier for workspace isolation and audit logging.
- **Request Body**:
  - `vote` (String, Required): Sentiment classification (`like`, `dislike`, or `general`).
  - `session_id` (String, Required): UUID mapping to the parent Chatbot Session.
  - `message_id` (String, Required): UUID mapping to the specific conversational turn.
  - `query` (String, Optional): Snapshot of the user prompt associated with the feedback.
  - `response` (String, Optional): Snapshot of the AI response associated with the feedback.
  - `reason_code` (String, Optional): Category identifier (e.g., `inaccurate`, `hallucination`).
  - `reason_text` (String, Optional): Justification text from the user.
  - `scope` (String, Optional): Targeted system component (e.g., `retrieval`, `generation`).
  - `feedback_text` (String, Optional): Extended unstructured comments.
- **Response**:
  - `user_id` (String): Logical tenant identifier for the feedback owner.
  - `feedback_id` (String): System-generated unique identifier for the record.
  - `session_id` (String): UUID link to the parent Chatbot Session.
  - `message_id` (String): UUID link to the specific conversational turn.
  - `vote` (String): Echoed sentiment classification.
  - `reason_code` (String): Categorization code provided during submission.
  - `reason_text` (String): Justification text provided during submission.
  - `scope` (String): Targeted system component (e.g., retrieval, generation).
  - `feedback_text` (String): Complete user comments.
  - `query` (String): Snapshot of the original user prompt.
  - `response` (String): Snapshot of the original AI response.
  - `is_active` (Boolean): Retention flag indicating if the feedback is currently visible.
  - `category` (String): AI-classified logic category (e.g., `Accuracy`).
  - `sub_category` (String): Granular classification (e.g., `Math Error`).
  - `suggested_action` (String): AI-generated prescriptive improvement step.

- `analysis_summary` (String): Narrative synthesis of the feedback generated by the LLM.
- `classifier_model` (String): The ID of the model used for automated categorization.
- `classification_status` (String): Pipeline state (`pending`, `completed`).
- `classification_error` (String): Error trace if the AI analysis failed.
- `created_at` (String): ISO-8601 timestamp of record creation.
- `updated_at` (String): ISO-8601 timestamp of the last record update.
- `version` (Integer): Optimistic concurrency version for DynamoDB.

Appendix D

Contribution

No.	Full name	ID	Main PIC of Tasks
1	Nguyen Quang Phu	2252621	Implement overall processing pipeline and system architecture; Setting up infrastructure; Implement Testing; Write report
2	Nguyen Ngoc Khoi	2252378	Parser for Office documents, PDF documents; Implement Evaluation; Write report
3	Nguyen Minh Khoi	2252377	System design and modelling; Implement frontend and backend web; Main PIC for report